Environmental, Health and Safety | 2026

Environmental, Health & Safety Trade Partner Manual

DISCLAIMER: The information contained in this Manual is not intended to serve as a substitute for the exercise of good engineering judgment by the engineers nor as a substitute for determinations made by Trade Partners of appropriate manner and methods of operations and safety aspects of work under their control. This Manual is also not intended to be all inclusive or replace a Trade Partner's or Trade Partner's corporate or site-specific safety program and is not intended to, nor shall it, supersede any more stringent federal, state, local and other applicable laws, codes, rules, regulations and/or practices. All Trade Partner's and Trade Partner's site-specific safety programs shall meet or exceed the requirements of the Whiting-Turner EH&S program, the contract documents and all federal, state, local and other applicable laws, codes, rules, regulations and/or practices. In the event of any conflicts between the material contained therein and any more stringent laws, codes, rules, regulations and/or practices, the more stringent laws, codes, rules, regulations and/or practices shall govern.

This Manual and all information contained therein is confidential and intended for the sole use of Trade Partners and Trade Partners on Whiting-Turner projects. Trade Partners are prohibited from distributing this information to any third parties. The Whiting-Turner Contracting Company expressly disclaims warranties for the information contained in this Manual and makes no representations to third parties regarding the reliability, suitability, correctness, or completeness of such information. Whiting-Turner assumes no responsibility or liability and shall not be responsible and/or liable for damages or losses of any kind, including but not limited to direct, indirect, incidental, exemplary, special or consequential damages or losses, arising from, attributable to and/or resulting from the use of such information by third parties however caused and on any theory of liability, whether in contract, strict liability or tort.

Rev. No.	Date	Reason
0	November 2018	Initial
1	January 2026	Added revision block, added Sections 2 and 3, reformatted Section 7 to match new EH&S Manual.

Table of Contents

1. Trade Partner EH&S Requirements

Policy Statement

The following establishes Whiting-Turner's minimal EH&S requirements for all Trade Partners on a Whiting-Turner project. This Manual is not meant to be all inclusive and it does not, in any way, excuse the Trade Partner from adhering to the entirety of the Owner's requirements and OSHA's 29 CFR 1926 Safety and Health Guidelines for Construction nor is it a substitute for the Trade Partner's own site-specific safety program. The Trade Partner's own site-specific safety program shall meet or exceed the requirements of the Whiting-Turner Trade Partner EH&S Manual, the Contract documents and the most current federal, state, local or other applicable codes and regulations.

1. **General Requirements**
    - 1.1. Each Trade Partner shall identify and submit the qualifications of a safety representative/competent person to Whiting-Turner as the primary, on-site contact for safety related issues.
    - 1.2. The safety representative may be a supervisor and they shall have, at a minimum, the OSHA 30-hour Outreach Training Program for Construction.
    - 1.3. The Trade Partner will provide a first aid/CPR trained person when one or more of the Trade Partner's employees are working
    - 1.4. The Trade Partner's supervisor(s) and safety representative shall make frequent and regular inspections of their work areas and activities.
    - 1.5. Hazards identified that are under their control shall be corrected immediately and all other identified hazards shall be reported to the Whiting-Turner Project Superintendent.
    - 1.6. One documented inspection shall be conducted each week and made available at Whiting-Turner's request.
    - 1.7. The Trade Partner's on-site supervisor and the Trade Partner's designated on-site safety representative shall schedule and attend a pre-construction safety meeting with the Whiting-Turner Superintendent and Site Safety Manager to discuss the Trade Partner safety requirements.
    - 1.8. The pre-construction safety meeting should take place at least five (5) working days before startup to allow for review of required documentation.
    - 1.9. The Trade Partner shall provide a translator whenever there are non-English speaking workers on site. Trade Partner supervisors shall be able to communicate in English.
    - 1.10. Trade Partners, who in turn contract out parts of their work, have sole responsibility to see that their lower-tier Trade Partners comply with project safety requirements. Additionally, Whiting-Turner's Project Manager and/or Whiting-Turner's Superintendent shall be notified that the lower-tired Trade Partners are arriving at least five (5) days before work starts. The Trade Partners will be held directly accountable for all lower tier

Trade Partners. Trade Partners shall provide a competent person onsite full-time to oversee and direct lower-tier Trade Partners' while actively performing work.

1.11. The Trade Partner's superintendent(s) and/or designated safety representative shall attend the weekly coordination meeting where safety issues will be addressed. Emergencies should be handled through the Whiting-Turner Field Office according to the posted Emergency Action Plan established prior to the start of work.
1.12. Each Trade Partner shall have the appropriate licenses, registrations and insurance to complete their work.
1.13. Prior to the start of work the Trade Partner will establish clear lines of communication.
1.14. Prior to the start of work the Trade Partner shall define clear roles and responsibilities.
1.15. Each Trade Partner with chemicals on site shall have a spill kit on the project with the appropriate materials for adequate and prompt clean-up. In addition, their site-specific safety plan shall address spill prevention and containment measures.
1.16. Only communication radios are permitted on Whiting-Turner projects.

2. Activity Hazard Analysis.

2.1. For each phase or major type of work/definable feature of work an AHA will be completed to identify the following:
  2.1.1. Health and safety considerations
  2.1.2. Description of steps to be performed
  2.1.3. Hazards associated with each step
  2.1.4. Required action to eliminate or control the hazard
  2.1.5. Focus four hazards and controls
  2.1.6. Trade Partner supervision sign-off
2.2. **The AHA is required to be submitted to the Whiting-Turner project team ten (10) days prior to start of the task.** The Trade Partner shall not be released to work until the AHA is received by the Whiting-Turner Superintendent or their designee.

3. Pre-task Planning.

3.1. This daily plan is designed to take place at the start of each work shift. Trade Partner supervisors shall meet with their crews to discuss the tasks to be accomplished and the steps that need to take place to work safely. All tradespersons should review and sign the relevant PTP for their assigned work. The main components of the Pre-task Plan will include the following:
  3.1.1. Evaluating the work area
  3.1.2. Potential hazard checklist
  3.1.3. Required actions to eliminate or control the hazard
  3.1.4. Crew sign-off
  3.1.5. A copy of the PTP shall be kept near the work location.
3.2. The information the supervisors are relaying to the tradespersons is the same that was developed in the AHA however, the PTP will more greatly define the plan for that phase of work as it applies to each crew performing the work.

4. **Aerial and Scissor Lifts**
   4.1. Trade Partner employees shall comply with their company's or project requirements for fall protection when working from a scissor lift. Climbing above the platform of any lift is prohibited; workers observed committing this unsafe act will be removed from the project for a minimum of 3 days— prior to returning to work they shall be retrained by their employer. Manufacturer instructions shall be followed for movement of any lift.

5. **Chemicals and Other Potentially Hazardous Materials**
   5.1. A copy of the Trade Partner's site-specific Chemical Inventory List and site specific SDS shall be submitted to the Whiting-Turner project team and updated as applicable. The chemical inventory list shall be reviewed on a monthly basis by a qualified person. A Project Hazar Communication Station [containing a hard copy of all safety data sheets applicable to the site] will be established and maintained in the Whiting-Turner Project Office. The Whiting-Turner Hazard Communication Station does not eliminate the need or requirement for the Trade Partner to establish and maintain a site-specific chemical inventory.
   5.2. In the event unknown and/or potentially hazardous materials are encountered during construction, that portion of the work will stop and Whiting-Turner will be notified immediately. Work will not resume until the Whiting-Turner Project Manager, Superintendent, EH&S Manager or third-party administrator authorizes it.

6. **Nitrogen Awareness**
   6.1. Workers who have the potential of being exposed shall be trained in the hazards associated with nitrogen. Potential hazards:
      6.1.1. Asphyxiation
      6.1.2. Cold burns
      6.1.3. Fire in oxygen-enriched atmosphere
      6.1.4. Pressure build-up/Ice plug formation/Explosion
   6.2. A site assessment will be conducted for potential nitrogen hazard exposure prior to starting work and the assessment shall be documented.
   6.3. When a potential hazard exists, appropriate signage will be utilized and adhered to.
   6.4. Appropriate barricades shall be utilized if determined by and in accordance with the site assessment.
   6.5. Barricades will provide a safe zone of 3' in diameter or greater if determined by oxygen monitoring results.
   6.6. All nitrogen cylinders should contain an identifying label.
   6.7. All nitrogen cylinders shall be properly secured at all times to prevent tipping, falling or rolling. They can be secured with straps or chains connected to a wall bracket or other fixed surface, or by use of a cylinder stand.
   6.8. Protective cap shall be in place when the cylinder is not in use.
   6.9. Nitrogen shall not be used to power pneumatic tools or blowers.

7. Communication and Compliance

7.1. Trade Partner management personnel is responsible for ensuring that all EH&S principles, policies and procedures are clearly communicated and understood by their respective employees. Managers and supervisors are expected to enforce all requirements uniformly.

7.2. All persons working on a Whiting-Turner project are responsible for using safe work practices. In addition, it is expected that each person follows all procedural safeguards and assists in maintaining a safe work environment.

8. Disciplinary Action

8.1. When safety policies and procedural safeguards are violated or frequent involvement in accidents or infractions is revealed, disciplinary action shall be considered. The intent of any disciplinary action taken is to convey the severity or potential severity of the infraction and bring about desired safe work practices.

8.2. When a person is observed committing an unsafe act, they shall be informed by means of a written safety notice and he/she shall attend the next scheduled jobsite safety orientation. In addition, employees observed committing unsafe acts while performing a task that exposes them to fall, electrical, caught-in/between or struck-by hazards shall be retrained by their employer prior to attending the jobsite orientation and returning to work. At the discretion of Whiting-Turner management, depending on the severity of the violation by a Trade Partner employee, the person may be suspended from the project for an allotted time or removed from the project permanently.

8.3. When a Trade Partner implements their own company procedures for safety infractions committed by their employees on a Whiting-Turner job site, the Trade Partner shall notify Whiting-Turner and provide a copy of the notice of infraction and related disciplinary actions to Whiting-Turner.

9. Drug & Alcohol Testing

9.1. All Trade Partners are required to perform testing as required to comply with all owner requirements and Whiting-Turner's policy for a drug free workplace.

10. Employee Safety Training

10.1. A site-specific safety orientation shall be conducted prior to allowing any workers access to the field. All employees completing the orientation will be issued a numbered orientation sticker or badge to be displayed on their person.

10.2. Additional OSHA Construction Standards that require specific training include, but are not limited to:

10.2.1. Hazard Communication Training - 29 CFR 1926.1200.
10.2.2. Stairway and Ladder Safety Training - 29 CFR 1926.1050.
10.2.3. Fall Protection Training - 29 CFR 1926. 503
10.2.4. Personal Protective Equipment - 29 CFR 1926.95
10.2.5. Scaffold Training - 29 CFR 1926.450

10.3. Trade Partners are responsible for ensuring their employees receive proper training before authorizing them to work on site.

11. Weekly Safety Meetings

11.1. Project workers shall attend at least one safety meeting each week. Copies of meeting minutes shall be submitted to the Whiting-Turner project team with the Trade Partner Daily Progress Report for the day the meeting is held. Meeting minutes shall indicate the name of the Trade Partner and date of the meeting. The supervisor(s) and the attendees shall sign minutes.

12. Quarterly Safety Focus Participation

12.1. All Trade Partners shall participate in the Quarterly Safety Focus meetings; these meetings will take place at 11:30am local time on the first Wednesday of the month in February, May, August and November.

2. Roles and Responsibilities

2.1. Trade Partner's Safety Representative.

2.1.1. The Trade Partner's safety representative and job foreman shall have the authority to act to prevent physical harm or significant property damage.

2.1.2. Is required, at a minimum, to have completed an OSHA 30-hour training program for the construction industry and shall meet the definition of a competent person for the given work activity as defined by OSHA standard 1926.32(f).

2.1.3. Assists in the design of their company's safety process and ensures conformity to Whiting-Turner's EH&S Program.

2.1.4. Assists in promoting continuity of the site-specific safety program.

2.1.5. Supports implementation of the process and program.

2.1.6. Supports training of their company's personnel regarding the Whiting-Turner safety process and the requirements in this program.

2.1.7. Continually evaluates their company's safety process.

2.1.8. Communicates the requirements of this program to their company's employees.

2.1.9. Assists the Project Safety Manager in all safety activities with regards to their company.

2.1.10. Disciplines and takes prompt corrective actions when directed by Whiting-Turner management personnel, or when conditions warrant such actions.

2.1.11. Ensures their company's employees follow all aspects of this program.

2.1.12. Makes daily safety inspections of the work being performed by their company and make necessary immediate corrective action to eliminate unsafe acts and conditions.

2.1.13. Documentation of inspections is to be submitted weekly to Whiting-Turner if requested.

2.1.14. Assures the OSHA 300 Injury and Illness Log Form Report is properly completed.

2.1.15. Reviews incident reports and initiate immediate corrective action.

2.1.16. Provides job foreman with appropriate material for use in conducting weekly "toolbox" safety meetings.

2.1.17. Attends foreman "toolbox" safety meetings and evaluate their effectiveness.

2.1.18. Assists in the preparation of the incident investigation and reporting procedures.

2.1.19. Encourages programs for recognition of individual employee's safety efforts and their contribution toward improved work procedures.

2.1.20. Controls and ensures the availability of the necessary safety equipment, including employee's personal protective equipment.

2.1.21. Coordinates activities with those of the other Trade Partners and the Whiting-Turner Project Safety Manager or Superintendent.

2.1.22. Attends safety meetings monthly or more frequently, as required.

2.1.23. Shares experiences, questions and problems at safety meetings.

2.1.24. Attends the weekly project safety committee meeting.

2.2. **Trade Partner Employees.**

2.2.1. Shall attend the project safety orientation and complete the acknowledgement form prior to beginning Whiting-Turner Projects.
2.2.2. Shall perform work in a safe manner for prevention of incidents to themselves, fellow tradespersons, the public and property.
2.2.3. Shall attend weekly toolbox talks.
2.2.4. Shall alert supervisor of hazards and unsafe acts.
2.2.5. Shall notify supervisor immediately of any incident.
2.2.6. Shall comply with their company's Safety Program, Whiting-Turner's EH&S Program and all applicable federal, state and local codes and regulations.
2.2.7. Shall use Stop Work Authority, if needed, to help prevent incidents.

3. Pre-Mobilization Safety Submittals & Ongoing Safety Management

1. Each Trade Partner shall identify (on the competent person acknowledgement form) and submit the qualifications of a competent person to Whiting-Turner prior to the start of work.
2. Trade Partner shall submit a site-specific safety plan (SSSP) prior to start of work.
    - 2.1. Activity hazard analyses (AHA) for major phases of work, submitted with company safety program may be accepted in lieu of SSSP—at the discretion of the Whiting-Turner project team.
    - 2.2. Trade Partner supervisor is responsible to maintain a copy of these plans at the worksite and reviewing with their employees for each new task or phase.
3. Site specific safety data sheets (SDS) and chemical inventories are to be provided to Whiting-Turner prior to start of work.
    - 3.1. SDS shall be submitted for all hazardous chemicals/products to be used on the project.
    - 3.2. Trade Partner supervisor is responsible to maintain copies of SDS and chemical inventory at the worksite.
    - 3.3. Trade Partner supervisor is responsible for updating Whiting-Turner each time a new chemical is brought on site; the update shall be reflected in their SDS binder and chemical inventory list.
4. Each Trade Partner is required to designate a site safety representative (SSR). SSR shall be on site at all times and shall have the knowledge and authority of the competent person. SSR shall be able to conduct site walks with Whiting-Turner personnel to ensure the safety of Trade Partner's/Trade Partner's workers on the project. Manpower totals below include all tiered Trade Partner employees. Proof of training shall be submitted prior to mobilization or at orientation. The qualifications for the SSR are as follows:
    - 4.1. Minimum requirement proof of OSHA 30 hour submitted
    - 4.2. Trade Partners with (30) or more workers on site will be evaluated by the Whiting-Turner's management team along with Whiting-Turner's EH&S Manager regarding the Trade Partner's/Trade Partner's site-specific safety performance. If the Trade Partner's/Trade Partner's past or current site safety performance indicates improved safe work practices and conditions are needed to help ensure the safety of the Trade Partner crews and others, Whiting-Turner at its discretion, may require the Trade Partner to provide a full-time Site Safety Representative to be present onsite with no other collateral duties.
5. Certification for operators of powered industrial trucks and cranes is required to be submitted to the Whiting-Turner project team prior to the operation of said equipment.
6. All on site personnel, (Trade Partners, tiered Trade Partners and their employees) are required to participate in a mandatory safety orientation session prior to commencing with any work on site. Trade Partner shall provide a translator for any non-English speaking employees during orientation and any job wide meetings/stand-downs. Employees may be asked to attend orientation again for repeat violations or deficiencies.
7. Trade Partner shall conduct weekly safety meetings for all workers (employed or contracted). Whiting-Turner Superintendent and/or other Whiting-Turner personnel may attend this safety meeting.

    7.1. Provide a legible sign-in sheet and meeting agenda to the Whiting-Turner field office at the conclusion of the meeting.
    7.2. Trade Partner is also required to submit daily work reports to the Superintendent, on Whiting-Turner's form.
8. In coordination with the Trade Partner's/Trade Partner's AHA, complete a Pre-task Plan (PTP) at least once per day, per crew/task.
    8.1. PTPs shall be updated as hazards or conditions change. If necessary, a new PTP may be completed and reviews with the crew.
    8.2. Copies of PTP(s) shall be submitted to the Whiting-Turner project team upon request.
    8.3. Tasks should be reviewed in the area where work is to be performed and crew members should be involved in the development of the plan (safety "huddle").
9. Documented pre-shift inspections are to be completed for mobile equipment, heavy equipment and cranes.
    9.1. Equipment is to be taken out of service for deficiencies noted.
    9.2. These inspections are to be maintained in the work area for review.

4. Incident Investigation and Reporting

4.1. All work-related injuries, regardless of severity, shall be reported to Whiting-Turner immediately. Incidents may include:

4.1.1. Spills
4.1.2. Releases to the environment
4.1.3. Property damage
4.1.4. Incidents
4.1.5. Injuries
4.1.6. Near misses

4.2. An incident investigation report shall be completed by the appropriate Trade Partner supervisor and submitted to the Whiting-Turner project team within 24 hours of the incident. A copy of an initial incident report shall include the following:

4.2.1. Basics facts
4.2.2. Preliminary severity classification
4.2.3. Immediate and corrective actions

3. Incidents involving the general public, regardless of severity, shall be reported to Whiting-Turner immediately. An incident investigation report shall be completed by the appropriate Trade Partner supervisor and submitted to the Whiting-Turner project team within 24 hours of the incident.
4. All fatal incidents shall be reported to OSHA within 8 hours and all in-patient hospitalization, amputation, or eye loss shall be reported within 24 hours.
5. All recordable and lost-time incidents shall be reported to the client (host facility) within 24 hours.

5. Compliance Enforcement

5.1. To ensure compliance with this program and all other established OSHA standards, the Whiting-Turner Contracting Company has established this procedure of compliance enforcement for all Trade Partners. This is established to promote safety excellence and eliminate offenders. This program may be used or may be superseded with more severe discipline based on the degree of the infraction(s). In any case, Whiting-Turner has sole authority in what type of discipline is issued relating to the project, up to and including removal from the project. Trade Partners are required to provide Whiting-Turner with any written disciplinary notices issued to site employees by their management.

5.2. To assist in our efforts to provide a safe workplace, the following disciplinary plan shall be used on each Whiting-Turner project.

- 5.2.1. First infraction: written warning; coach the employee.
- 5.2.2. Second infraction: written warning and a meeting shall be held between the employee, his/her supervisor and the Whiting-Turner Superintendent—or their designee.
- 5.2.3. A copy of the written warning is sent to the employee's company's office.
- 5.2.4. A written warning requires the Trade Partner's supervisor to assure the employee has satisfactorily completed an appropriate training session related to the safety policy violated.
- 5.2.5. This training shall be completed within five (5) working days from issuance of the written warning. Until that training has occurred the employees shall not resume the same work activity.
- 5.2.6. Proof of training or retraining shall be provided to Whiting-Turner.

5.3. Third infraction the worker is removed from the project, indefinitely.

- 5.3.1. If repeat occurrences with other crew members are found the supervisor of said crew members shall be subject to removal from the project.

5.4. All warnings, verbal or otherwise, shall be documented in a log to be used as a reference point by the project team to track unsafe trends and to act as a tool to enhance safety.

6. Safety Incentive Program Guidelines.

6.1. The implementation of a project safety incentive program is at the discretion the Whiting-Turner project team. The project's safety awards and incentives will be based primarily upon the data collected during safety observations. All superintendents, project managers and safety managers are required to perform inspections.

6.2. There are two important aspects of safety award programs that need to be kept in mind:

6.2.1. Provide safety awards for safe behavior and for activities related to maintaining a safe work environment.

6.2.2. Make safety awards and incentives significant enough to support compliance but not significant enough to generate false reporting of safety data.

7. Safety Practices

E.1 Sanitation Guidelines

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart D - Occupational Health and Environmental Controls; employers shall establish and maintain basic sanitation provisions for all employees in all places of employment as specified in the following paragraphs.

Procedures

1. **Drinking water supply.**
   - 1.1. Each employer shall provide their employees with an adequate supply of potable water.
   - 1.2. The employer shall take one or more of the following steps to ensure every employee has access to drinking water:
     - 1.2.1. Supply single-service cups. Where single service cups are supplied, a sanitary container for the unused cups and a receptacle for disposing of the used cups shall be provided.
     - 1.2.2. Supply sealed-one time use water containers. Where sealed one-time use water containers are supplied, a receptable for disposing of the used containers shall be provided.
     - 1.2.3. Portable containers used to dispense drinking water to more than one person shall be equipped with a faucet or drinking fountain. Drinking water containers shall be capable of being closed tightly and shall be otherwise designed, constructed and services so that sanitary conditions are maintained. Water shall not be dipped from containers.
     - 1.2.4. Any container used to store or dispense drinking water shall be clearly marked as to the nature of its contents and shall not be used for any other purpose.

2. **Non-potable water.**
   - 2.1. On outlets dispensing non-potable water, signs cautioning that the water is unsafe for drinking, washing, or cooking shall be conspicuously posted.

3. **Toilets**.
   - 3.1. When sanitary sewers are not available, one of the following facilities, unless prohibited by local codes, shall be provided: chemical toilets, recirculating toilets, or other toilet systems as approved by state/local governments.
   - 3.2. Each toilet facility shall be equipped with a toilet seat and toilet seat cover. Each toilet facility - except those specifically designed and designated for females - shall be equipped with a metal, plastic, or porcelain urinal trough. All shall be provided with an adequate supply of toilet paper and a holder for each seat.
   - 3.3. Toilet facilities shall be so constructed that the occupants shall be protected against weather and falling objects; all cracks shall be sealed and the door shall be tight-fitting, self-closing and capable of being latched.

3.4. One toilet facility shall be provided for every ten (10) persons on site.
3.5. Separate toilet facilities shall be provided for each gender.
3.6. Provisions for routinely servicing and cleaning all toilets and disposing of the sewage shall be established before placing toilet facilities into operation.
3.7. Cost shall not be a factor in maintaining toilet facilities.

4. **Washing facilities.**
   4.1. Washing facilities shall be provided at toilet facilities and as needed to maintain healthful and sanitary conditions.
   4.2. Each washing facility shall be maintained in a sanitary condition and provided with water (either hot and cold running water or tepid running water), soap and individual means of drying. However, where it is not practical to provide running water, hand sanitizers may be used as a substitute.
   4.3. Whenever employees are required by a standard to wear protective clothing, change rooms with storage facilities for street clothes and separate storage facilities for protective clothing shall be provided.

5. **Vermin control.**
   5.1. Enclosed workplaces shall be constructed and maintained, as far as practical, to prevent the entrance or harborage of rodents, insects and other vermin. An effective extermination program shall be instituted where the presence of such vermin is detected.

E.2 Medical and First-Aid Requirements

Introduction

Each Trade Partner shall have one person on site with a valid certificate in first aid, cardiopulmonary resuscitation (CPR) and automated external defibrillator (AED) while actively engaged in work activities.

Procedures

1. In addition to the first aid kit Whiting-Turner supplies, each Trade Partner's first aid kit shall be following Whiting-Turner's Bloodborne Pathogen Policy.

2. The contents of the first aid kit shall be placed in a weatherproof container with individual sealed packages for each type of item and shall be inspected at least weekly.

3. Emergency telephone numbers shall be conspicuously posted.

4. Where the eyes or body of any person may be exposed to injurious corrosive materials (e.g. during concrete placement), suitable facilities for quick drenching or flushing of the eyes and body shall be provided within the work area for immediate emergency use.

5. Prior to the commencement of construction activities, each employer shall prepare for the best transportation method to the nearest health care facility in the event of an incident.

E.3 Temporary Facilities

Introduction

Trade Partner providing temporary construction buildings, facilities, fencing and access routes and anchoring systems for temporary structures shall be submitted to Whiting-Turner for review.

Procedures

1. The design and construction of temporary structures shall consider the following loadings:
   1.1. Dead and live loads
   1.2. Soil and hydrostatic pressures
   1.3. Wind loads
   1.4. Rain and snow loads
   1.5. Seismic forces

NOTE: Local jurisdiction inspections and engineering may be required for temporary structures.

NOTE: Temporary structures built within another building or structure shall meet the requirements of 29CFR1926.151(b).

2. Trailers and other temporary structures used as field offices, to house personnel, or for storage shall be anchored with rods and cables or by steel straps to ground anchors. The anchor system shall be designed to withstand winds and shall meet applicable State or local standards for anchoring mobile trailer homes.

3. Fencing and warning signs
   3.1. Temporary project fencing shall be installed per Whiting-Turner's fencing policy particularly on all projects located in areas of active use by members of the public. More careful consideration will also be given to those areas proximate to family housing areas and/or school facilities.
   3.2. Signs warning of the presence of construction hazards and requiring unauthorized persons to keep out of the construction area shall be posted on the fencing. At the minimum, posting shall be on all fenced sides of the project and spaced one sign every 100 feet.

E.4 Personal Protective Equipment

Introduction

All Trade Partner employees on Whiting-Turner projects shall wear all the personal protective equipment necessary to complete their jobs safely. Mandatory personal protective equipment required on the project site includes, but is not limited to, hardhat/helmet, safety glasses, gloves, high visibility vests and sturdy leather work boots. A competent person onsite will determine additional necessary equipment. Each Trade Partner working on a Whiting-Turner project site will comply with 29 CFR 1926, Construction Industry Regulations, Subpart E – Personal Protective and Lifesaving Equipment; in addition to the following guidelines.

Procedures

1. All Trade Partner employees and visitors to project sites are required to wear safety glasses that comply with ANSI Z87.1.
   - 1.1. Dark lenses are not to be worn inside of buildings, in enclosed areas or at night.
   - 1.2. Tinted lenses shall be worn on light colored membrane roofs.
   - 1.3. Prescription eyeglasses and sunglasses that do not comply with ANSI Z87.1 are prohibited.
   - 1.4. A pair of fully enclosed safety goggles or face shield is required for grinding concrete.
   - 1.5. Face shields are required for: welding, burning and cutting; using abrasive wheels; chop saws; portable grinders or files, chippings concrete, stone, or metal; drilling or working under dusty conditions; using explosive actuated fastening or nailing tools; overhead work; work with hazardous liquids or gases.
   - 1.6. All face shields shall be compatible with a hardhat/helmet.

NOTE: Face shields are to be worn, in addition to safety glasses.

2. All Trade Partner employees and visitors to project sites are required to wear hardhats/helmets that comply with ANSI Z89.1.
   - 2.1. Aluminum hardhats and bump caps are not permitted on Whiting-Turner projects.
   - 2.2. For security and identification purposes, all hardhats/helmets shall display the Trade Partner name and/or decal indicating for whom the employee works as well as the employee's name.
   - 2.3. Employees exposed to electrical voltages of 600V or greater shall wear hardhats/helmets that meet the requirements of ANSI Z89.2 type hardhats/helmets.
3. All Whiting-Turner employees, Trade Partner employees and visitors to project sites are required to wear substantial leather work boots.
   - 3.1. Employees working with jackhammers, tampers and similar equipment are required to utilize metatarsal guards over their work boots.
4. Where necessary, each employee shall use equipment with filter lenses that have a shade number appropriate for the work being performed for protection from injurious light radiation.
5. All Trade Partner employees and visitors to project sites are required to have hand protection that is determined to be the appropriate level of protection by the competent person.
6. Workers exposed to roofing tar shall wear long sleeved shirts and gloves. Workers who are directly exposed to hot tar shall also wear a full apron and face shield.

7. Where an employee could be exposed to noise more than 85 dba TWA, their employer shall provide hearing protection which will reduce the noise to an acceptable level. If the noise levels are determined to cause an 8-hour TWA exposure greater than 85 dba, the Trade Partner shall be required to submit a detailed hearing conservation program to Whiting-Turner. This program shall be accepted prior to beginning work.
8. Personal protective equipment labeled "WT" or Whiting-Turner" is reserved for the use of Whiting-Turner employees only.

9. **Roles and Responsibilities**
   - 9.1. Trade Partner management shall:
     - 9.1.1. provide necessary PPE and training,
     - 9.1.2. monitor use of PPE,
     - 9.1.3. provide replacement PPE when needed,
     - 9.1.4. identify any new hazards that would require the use of PPE; and
     - 9.1.5. be responsible for the assurances of PPE adequacy, maintenance and sanitation.
   - 9.2. Trade Partner employees shall:
     - 9.2.1. properly use and care for assigned PPE and
     - 9.2.2. immediately inform supervisor if PPE is damaged or not effective.

10. **Additional Information, Requirements and Considerations**
    - 10.1. Retraining in the use of PPE shall be performed as necessary.
    - 10.2. PPE training shall be documented.
    - 10.3. All PPE shall be maintained in a sanitary/clean condition.
    - 10.4. To ensure proper maintenance, Whiting-Turner does not allow employees to provide their own PPE.
    - 10.5. All PPE shall be the proper fit and size for the wearer.
    - 10.6. Defective PPE shall immediately be discarded or removed from service.

E.5 Hearing Conservation

Introduction

Whiting-Turner recognizes that excessive noise can cause permanent hearing loss if appropriate administrative or engineering controls or personal protective equipment is not used. Limiting exposure to excessive noise through engineering controls is Whiting-Turner's preferred method of control.

Procedures

Permissible Noise Exposures

Duration per day, hours	Sound level dba, slow response
8	85
6	92
4	95
3	97
2	100
1 ½	102
1	105
½	110
¼ or less	115

1. Protection against the effects of noise exposure shall be provided when the sound levels exceed those shown in the table above. The measurement shall be observed on the A-scale of a sound level meter at slow response.
2. When employees are subjected to sound levels exceeding those shown above, feasible administrative or engineering controls shall be utilized.
3. If such controls fail to reduce sound levels within the levels shown above, personal protective equipment shall be provided and used to reduce the noise exposure.
4. In all cases where all feasible engineering and administrative controls have been put in place and the sound levels exceed the values shown in the table above, a continuing, effective hearing conservation program shall be administered.
5. If the noise levels are determined to cause an 8-hour TWA exposure greater than 85 dba, the Trade Partner shall provide a comprehensive hearing conservation program before continuing work. At a minimum this program shall include:
   5.1. Noise survey data for typical work they perform.
   5.2. Noise dosimetry data for typical exposures from the work they perform.
   5.3. Training records for employees working on the Whiting-Turner Project.

Additional Information, Requirements and Considerations

6. Workers who are exposed to hazardous noise shall be provided annual training on the hearing conservation program.
7. Personal noise exposure monitoring procedures (typically with a dosimeter) may be used when exposure limits may exceed 85 dBA, 8-hour time-weighted average. If conducting a noise survey, it is recommended to use a sound level meter (SLM) to produce a noise map of an area or integrated sound level meter (ISLM) for highly variable noise.

8. Audiometric testing shall be performed annually for individuals exposed to noise equal to or greater than 85 dBA, 8-hour time-weighted average.
9. A baseline audiogram shall be established within the first 6 months of exposure; at least 14 hours without exposure to workplace noise is required prior to establishing a baseline audiogram.
10. Workers shall be notified in writing of a standard threshold shift within 21 days of determination.
11. Hearing protection shall be re-evaluated in the event of a standard threshold shift.
12. Hearing protection is evaluated for the specific noise environments in which the hearing protection will be used.
13. Employee exposure records shall be kept for at least 30 years, except for
    13.1. Background data related to environmental, or workplace, monitoring or measuring — such as laboratory reports and worksheets — shall only be retained for 1 year, so long as you preserve certain interpretive documents relevant to the interpretation of the data for 30 years.

E.6 Hazard Communication

Introduction

Hazard communication is a system utilized by employers to ensure that all affected employees are kept informed of the types of chemicals in their workplace, the hazardous properties of the chemicals with which they will work and both safe handling procedures and measures to take to protect themselves to work safely with these chemicals.

Trade Partners are required to develop their own written site-specific hazard communication program. Each Trade Partner is required to supply the Whiting-Turner project team with hard copies of the site-specific chemical inventory list and all Safety Data Sheets (SDS) for any hazardous material used on site, prior to beginning work on any Whiting-Turner project.

Procedures

1. **General Requirements.**
    - 1.1. To meet the requirements as a general/managing Trade Partner all Whiting-Turner projects shall follow the below listed procedures:
    - 1.2. Implement and maintain a central repository in the jobsite trailer/office that contains all Trade Partner site specific chemical inventories and SDS.
    - 1.3. The project safety orientation should include information on where the central repository for all SDS and chemical inventories will be kept at the project.
    - 1.4. Require a representative from all Trade Partners to update the chemical inventory and SDS monthly to ensure that an accurate account of hazardous materials on site is maintained.
    - 1.5. Develop and maintain a written site-specific hazard communication program for the duration of the project.
    - 1.6. Provide training to all project team members that may be using a hazardous material during the project. The assigned Area EH&S Manager shall be contacted to help with this requirement.
    - 1.7. Containers that are not labeled with the contents and associated hazards shall not be allowed on any Whiting-Turner project.
    - 1.8. If employees are wearing any kind of respirator, ensure that all the proper sections of 29 CFR 1910.134 Respiratory Protection are being followed. Consult assigned Area EH&S Manager, if necessary.
    - 1.9. Hazardous materials that can create large areas of disturbance shall be considered for off-hours application, to prevent exposure to the entire jobsite workforce. An example of this may be a two-part epoxy floor install.
    - 1.10. Ensure that Trade Partners utilize the hierarchy of controls to be highest extent possible.

2. **Nitrogen Awareness.**
    - 2.1. Workers who have the potential of being exposed shall be trained in the hazards associated with nitrogen. Potential hazards:
        - 2.1.1. Asphyxiation
        - 2.1.2. Cold burns
        - 2.1.3. Fire in oxygen-enriched atmosphere

2.1.4. Pressure build-up/Ice plug formation/Explosion

2.2. A site assessment will be conducted for potential nitrogen hazard exposure prior to starting work and the assessment shall be documented.

2.3. When a potential hazard exists, appropriate signage will be utilized and adhered to.

2.4. Appropriate barricades shall be utilized if determined by and in accordance with the site assessment.

2.5. Barricades will provide a safe zone of 3' in diameter or greater if determined by oxygen monitoring results.

2.6. All nitrogen cylinders should contain an identifying label.

2.7. All nitrogen cylinders shall be properly secured at all times to prevent tipping, falling or rolling. They can be secured with straps or chains connected to a wall bracket or other fixed surface, or by use of a cylinder stand.

2.8. The protective cap shall be in place when the cylinder is not in use.

2.9. Nitrogen shall not be used to power pneumatic tools or blowers.

3. **Hydrogen Sulfide (H2S).**

3.1. Hydrogen sulfide is a toxic, colorless gas, with the odor of rotten eggs at low concentrations; it is soluble in water and is flammable.

3.2. Whiting-Turner employees typically do not perform work in locations that would overexpose them to Hydrogen Sulfide. However, a site risk assessment prior to the start of would reveal such exposure. In general, working in the following areas and conditions increases a worker’s risk of overexposure to hydrogen sulfide

3.2.1. Confined spaces (for example pits, manholes, tunnels, wells) where hydrogen sulfide can build up to dangerous levels.

3.2.2. Windless or low-lying areas that increase the potential for pockets of hydrogen sulfide to form.

3.2.3. Marshy landscapes where bacteria break down organic matter to form hydrogen sulfide.

3.2.4. Hot weather that speeds up rotting of manure and other organic materials and increases the hydrogen sulfide vapor pressure.

3.3. If during a site assessment it is determined that there is a possibility of coming into contact with hydrogen sulfide, the methods of detecting H2S will be by the use of fixed or portable monitors and will alarm at the appropriate permissible exposure limits 10 PPM for 1926; whichever method is deemed most feasible during the assessment.

3.4. When the H2S alarms sound, Whiting-Turner employees and Trade Partners shall immediately evacuate the area.

3.5. Where the possibility of this hazard exists, it shall be document on the emergency action plan for that project and employees shall be training and aware of site-specific contingency /emergency plans.

3.6. Hydrogen sulfide gas causes a wide range of health effects. Workers are primarily exposed to hydrogen sulfide by breathing it. The effects depend on how much hydrogen sulfide you breathe and for how long. Exposure to very high concentrations can quickly lead to death. Some of the acute symptoms and health effects are:

3.6.1. Prolonged exposure may cause nausea, tearing of the eyes, headaches or loss of sleep. Airway problems (bronchial constriction) in some asthma patients. (2-5 PPM)
3.6.2. Possible fatigue, loss of appetite, headache, irritability, poor memory, dizziness. (20 PPM)
3.6.3. Slight conjunctivitis ("gas eye") and respiratory tract irritation after 1 hour. May cause digestive upset and loss of appetite. (50-100 PPM)
3.6.4. Coughing, eye irritation, loss of smell after 2-15 minutes (olfactory fatigue). Altered breathing, drowsiness after 15-30 minutes. Throat irritation after 1 hour. Gradual increase in severity of symptoms over several hours. Death may occur after 48 hours. (100 PPM)
3.6.5. Loss of smell (olfactory fatigue or paralysis). (100-150 PPM)
3.6.6. Marked conjunctivitis and respiratory tract irritation after 1 hour. Pulmonary edema may occur from prolonged exposure. (200-300 PPM)
3.6.7. Staggering, collapsing in 5 minutes. Serious damage to the eyes in 30 minutes. Death after 30-60 minutes. (500-700 PPM)
3.6.8. Rapid unconsciousness, "knockdown" or immediate collapse within 1 to 2 breaths, breathing stops, death within minutes. (700-1000 PPM)
3.6.9. Nearly instant death. (1000-2000 PPM)

E.7 Silica Exposure Prevention

Introduction

Exposure to silica can lead to silicosis, a serious and sometimes fatal respiratory disease. Silicosis develops from being exposed to and breathing in silica dust. Excessive amounts of silica dust may be generated during activities such as, but not limited to: sandblasting, rock drilling, roof bolting, foundry work, stonecutting, drilling, quarrying, brick/block/concrete cutting, gunite operations, lead-based paint encapsulate applications, asphalt paving, cement products manufacturing, demolition operations, hammering and chipping and sweeping concrete or masonry, drywall sanding and concrete saw cutting.

Trade Partners are required to submit their company's Silica Exposure Control Plan to Whiting-Turner prior to start of work. Their plan shall be in compliance with the requirements set forth in 29 CFR §1926.1153.

Releases of plumes of silica laden dust are prohibited. Observation of such activities will require cessation of the task and a meeting with the creating Trade Partner to determine which engineer or work practices need to be put in place to ensure that these releases are reduced/eliminated as prescribed in their silica exposure plan.

Procedures

1. **General Requirements.**
   - 1.1. To determine whether a product contains silica, the Safety Data Sheet shall be obtained and evaluated. In the event silica is present in products on-site, the following safe working procedures shall be followed to eliminate or control silica dust exposure:
     - 1.1.1. Any Trade Partner producing silica dust shall follow Table 1 of the 29CFR1926.1153, if the activity is listed. If the activity is not listed, an exposure determination shall be completed, or historical or objective data must be submitted to the project team for review.
     - 1.1.2. Whiting-Turner project teams shall not allow Trade Partners to only utilize PPE as a means of protecting their employees. All feasible engineering and work practices shall be deployed first.
     - 1.1.3. If PPE is required, refer to the Trade Partner's Respiratory Protection Program for specific guidelines.
     - 1.1.4. After working with products that contain silica, everyone will be required to thoroughly wash their hands before eating, drinking, or smoking. Eating, drinking, or smoking near silica or in a silica-regulated area is prohibited.
     - 1.1.5. Whiting-Turner will not allow the use of any compound used for abrasive cleaning that contains more than 1% silica. Employee sampling shall be conducted to verify that concentrations released from the media being finished does not exceed allowable OSHA PEL's. For abrasive blasting, replace silica sand with less toxic materials. The National Institute for Occupational Safety and Health highly discourages the use of sand or any abrasive with more than 1% crystalline silica in it. As an alternative, garnet, slag and steel grit and shot may be suitable substitutes.

1.1.6. All Trade Partners are to supply any exposure monitoring, testing, or engineering information regarding silica exposure in their operations prior to beginning work. An example may be the masonry Trade Partner using brick/block saws and associated experience data that the Trade Partner has obtained.

1.1.7. Releases of plumes of silica laden dust will be prohibited on Whiting-Turner projects. Project team members that observe such activities will stop the activity and have a meeting with that Trade Partner to determine what engineering or work practices need to be put in place to ensure these releases are reduced and/or eliminated.

1.1.8. Activities that produce high levels of crystalline silica, i.e. walk behind saws, shall be considered for off-hours completion to help reduce potential exposure.

2. **Additional Information, Requirements and Considerations.**

2.1. Assessment of the written silica exposure control program's effectiveness shall take place at least annually.

2.2. A copy of the written exposure control plan shall be available to all respective employees.

2.3. All applicable records (e.g. air sampling, medical surveillance) shall be kept in accordance with OSHA's medical and exposure record requirements.

E.8 Respiratory Protection

Introduction

Trade Partners are required to submit and have accepted by Whiting-Turner, their company's respiratory protection program prior to start of work. Compliance with this policy applies to filtering face-piece respirators (N95 Respirators) as well. All programs shall meet or exceed Whiting-Turner policies as well as federal, state and local regulatory requirements.

Procedures

1. **Program Administration.**
   1.1. A formal annual audit of the respiratory program is required for all companies who are actively using respirators. The evaluation shall be documented and recommended changes shall be recorded.

2. **Workplace exposure assessment & ongoing surveillance.**
   2.1. Exposure assessment is critical in identifying harmful airborne contaminants, their extent and magnitude and how to control them.
   2.2. Supervision shall make every effort through evaluations and training to ensure that employee exposure does not exceed permissible concentrations
   2.3. Results of these evaluations will be summarized and a record maintained in the jobsite project files. Additional evaluations are necessary if exposures change due to new materials, process changes or other conditions, increasing the degree of employee exposure.
   2.4. Trade Partners shall provide proof of exposure assessments, training, fit-testing and medical surveillance for their employees prior to performing work with any material that may require respiratory protection.
   2.5. Trade Partners may not perform any work with chemicals or materials that may cause a respiratory hazard or nuisance odor for Whiting-Turner employees, other Trade Partners, or the general public without scheduling the work with Whiting-Turner. Examples of such activities include, but may not be limited to, applying hazardous paints or coatings; saw-cutting or grinding concrete and applying spray-on fireproofing.

3. **Respirator selection.**
   3.1. In those instances where engineering and administrative means do not achieve the desired control, respirators shall be worn. Distinct types of respirators are available for a variety of applications. Trade Partner shall ensure that the proper NIOSH/MSHA approved respirator is selected and used for the kind of work being performed and the hazards involved.
   3.2. Respirator selection information shall be completed to document the selection process.

3.3. In cases where dust-mask like respirators (N95 – two strap) is being used voluntarily at the discretion of the employee, but is not required for the task, a copy Appendix D to OSHA Respirator Standard 1910.134 shall be provided.

3.4. In cases where tight-fitting facepiece type respirators are being worn voluntarily, at the discretion of the employee, but is not required for the task, a copy of Appendix D to the OSHA Respirator Standard 1910.134 shall be provided. The Trade Partner shall also provide the employee with a medical evaluation and training on how to properly clean, store and maintain that respirator.

4. Evaluating respirator wearer health status.

4.1. Even with appropriate equipment and adequate training provided, an employee's health status shall be considered before allowing respirator use. The wearer's physical and medical condition, duration and difficulty of the tasks, toxicity of the contaminant and type of respirator all affect an employee's ability to wear a respirator while working. Therefore, the Trade Partner shall ensure that each employee's physical ability to wear a respirator is evaluated.

4.2. Each respirator wearer will be given a medical evaluation. The project will make appropriate arrangements with a proper medical organization to perform the evaluation. A medical evaluation and work restriction report shall be completed for each person.

5. Respirator fit testing & assignment.

5.1. After selecting the appropriate type of respirator and verifying the employee's ability to work while wearing a respirator, the Trade Partner will ensure that a qualitative fit test is conducted to choose the best fitting face piece and determine the specific brand, model and size for each employee.

5.2. Quantitative fit is the preferred alternative to qualitative fitting. Although it requires specialized equipment and trained personnel, some exposures require a quantitative fit test.

6. Training.

6.1. Once the employee is fitted with the correct respirator for the task, that employee shall be thoroughly trained in the need, use, limitations, inspection, fit checks, maintenance and storage of the equipment. This training may be initiated during the fit test.

6.2. The manufacturer of the equipment provides detailed instructions for use and care of the respirator and this information is to be used in the training.

6.3. Proof of training shall be made available upon request.

7. Additional Information, Requirements and Considerations

7.1. An effective face seal of respiratory protection is required and shall be maintained; facial hair is permitted if it does not interfere with the effective seal of the face piece or valve function and was existing during a passed fit test.

7.2. Employees shall leave the area that requires the use of a respirator if a vapor/gas breakthrough, changes in breathing resistance and/or leakage of the facepiece occurs.
7.3. Whiting-Turner does not permit employees to work in immediately dangerous to life or health conditions.
7.4. Respirators shall be properly cleaned and stored in accordance with the manufacturer's recommendation.
7.5. All respirators shall be inspected prior to use.

E.9 Asbestos Exposure Prevention Policy

Introduction

Where asbestos is found, best practice is for the Owner to contract for its removal. Prior to the commencement of construction activities, Whiting-Turner shall obtain documentation records that the asbestos has been removed and the area is safe to work.

Procedures – Whiting-Turner Contracts for Asbestos Abatement

1.1. Each Trade Partner engaged in asbestos abatement on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart Z – Toxic and Hazardous Substances in addition to the following guidelines.
1.2. Prior to the start of work, Whiting-Turner will request that the Owner has a survey conducted by certified testing company, industrial hygienist, or asbestos removal Trade Partner to determine the presence of asbestos.
1.3. Whiting-Turner shall request written documentation from the Owner detailing the presence, location and quantity of asbestos containing material and presumed asbestos containing material.
1.4. Whiting-Turner will ensure that the documentation detailing the presence, location and quantity of ACM and PACM is shared with all Trade Partners that will be working with or around said materials.
1.5. The Whiting-Turner project team will ensure that the asbestos abatement Trade Partner chosen to perform the abatement is licensed and trained to perform such activities in the applicable authority.
1.6. Prior to the start of work, the Whiting-Turner project team will request that the Trade Partner provide the name of the competent person on Whiting-Turner's Competent Person Acknowledgement Form, in addition to an alternate competent person.
1.7. The Whiting-Turner project team shall contract a 3rd party industrial hygiene consultant who will act as a supervisor to properly oversee the work and ensure compliance with 29 CFR 1926, Construction Industry Regulations, Subpart Z – Toxic and Hazardous Substances.
1.8. Whiting-Turner and Trade Partner performing the abatement shall advise all other Trade Partners to stay out of the regulated areas until clearances are provided.
1.9. The Trade Partner performing the abatement's competent person shall assure that containment/barriers/ventilation and other safeguards are maintained.
1.10. All individuals shall receive asbestos hazard awareness training prior to beginning work in areas that have asbestos containing materials that will remain in place, undisturbed.
1.11. Whiting-Turner should never agree to be the originators or sign any documents indicating that it is the originator of any hazardous materials.
1.12. All individuals shall receive asbestos hazard awareness training prior to beginning work in areas that have asbestos containing materials that will remain in place, undisturbed.

1.13. Whiting-Turner should never agree to be the originators or sign any documents indicating that it is the originator of any hazardous materials.

2. **Roles and Responsibilities**

2.1. The Trade Partner management personnel shall:

2.1.1. Comply with and furnish materials necessary to comply with Whiting-Turner policy.

2.1.2. Provide asbestos hazard awareness training to employees working in or around material containing asbestos.

2.2. The Trade Partner employees shall:

2.2.1. Attend and participate in project orientations and asbestos hazard awareness training

2.2.2. Immediately report disturbance or discovery of asbestos containing material by completing the asbestos encounter form.

E.10 Lead Exposure Prevention

Introduction

Where lead is found and it is determined that the lead must be abated, the owner shall contract directly with an abatement Trade Partner for its removal. Whiting-Turner shall obtain certification that the lead has been removed and the area is safe to work.

Procedures – Whiting-Turner Contracts for Lead Abatement

1.1. Each Trade Partner engaged in lead abatement on a Whiting-Turner project will comply with 29 CFR 1926.62 -Lead, in addition to the following guidelines.
1.2. Prior to the start of work, Whiting-Turner will request that the Owner has a survey conducted by certified testing company, industrial hygienist, or lead removal Trade Partner to determine the presence of lead.
1.3. Whiting-Turner shall request written documentation from the Owner detailing the presence and location of lead containing materials.
1.4. Whiting-Turner will ensure that the documentation detailing the presence and location of lead containing materials is shared with all Trade Partners that will be working with or around said materials.
1.5. The Whiting-Turner project team will ensure that the lead abatement Trade Partner chosen to perform the abatement is licensed and trained to perform such activities in applicable jurisdiction.
1.6. Prior to the start of work, the Whiting-Turner project team will request that the Trade Partner provide the name of the competent person on Whiting-Turner's Competent Person Acknowledgement Form, in addition to an alternate competent person.
1.7. The Whiting-Turner project team shall have an individual, on the team, who will act as a supervisor to properly oversee the work and ensure compliance with 29 CFR 1926.62 -Lead.
1.8. Whiting-Turner and Trade Partner performing the abatement shall advise all other Trade Partners to stay out of the regulated areas until clearances are provided.
1.9. The Trade Partner performing the abatement's competent person shall assure that containment/barriers/ventilation and other safeguards are maintained.
1.10. All individuals shall receive lead hazard awareness training prior to beginning work in areas that have lead containing materials that will remain in place undisturbed.
1.11. Whiting-Turner should never agree to be the originators or sign any documents indicating that it is the originator of any hazardous materials.

2. **Procedures – Trade Partner employees who will disturb lead containing material that will not be abated, or during the installation of lead containing material.**
   2.1. An exposure assessment shall be conducted to determine exposure levels.

2.2. If the exposure assessment shows exposure levels above the action level, monitoring shall be conducted every six months until two consecutive results are below the action level.
2.3. If the exposure assessment shows exposure levels above the PEL, engineering and administrative controls shall be put in place and additional exposure assessments will be completed until levels are below the PEL.
2.4. If an exposure assessment shows exposure levels over the PEL after all feasible engineering and administrative controls have been put in place, employees shall be protected with respiratory protection. Whiting-Turner and Trade Partners shall follow the Whiting-Turner Respiratory Protection policy in this manual, OSHA Lead Standard 29CFR 1926.62 and OSHA Respiratory Protection Standard, 29CFR 1910.134.
2.5. Objective data may be used in lieu of an exposure assessment, as long at it meets the requirements of 29CFR 1926.62.

3. **General Requirements**
3.1. Employees shall be notified in writing of the air monitoring results and corrective actions taken.
3.2. A written (site specific) compliance program shall be developed & implemented to reduce exposures to or below the permissible limits.
3.3. Respirators are to be used during the time period necessary to install or implement engineering or work practice controls, where engineering and work practice controls are insufficient and in emergencies. Employees shall be provided with a) full face-piece respirators instead of half-mask respirators for protection against lead aerosols that cause eye or skin irritation at the use concentrations. b) HEPA filters for powered and non-powered air-purifying respirators. c) Powered air-purifying respirator (PAPR) instead of a negative pressure respirator when an employee chooses to use a PAPR and it provides adequate protection to the employee.
3.4. Based on the hazard assessment and fit-testing, specific types of PPE shall be provided to the employees for their protection.
3.5. A medical surveillance program shall be in for all employees who are or may be exposed above the action level for more than 30 days.
3.6. Medical monitoring is required for workers exposed to lead.
3.7. Lunchrooms, changing, shower & hygiene facilities shall be provided when exposure to lead is above the PEL.
3.8. Signs shall be posted in and around the regulated work area.

4. **Roles and Responsibilities**
4.1. Whiting-Turner management shall:
4.1.1. Conduct inspections of the workplace for compliance with this policy.
4.1.2. Discuss policy applications during project orientation with Trade Partners; and
4.1.3. Provide lead hazard awareness training to all Whiting-Turner employees working in or around material containing lead.

4.2. Trade Partner management shall:
   4.2.1. Comply with and furnish materials necessary to comply with Whiting-Turner's policy and
   4.2.2. Provide lead hazard awareness training to employees working in or around material containing lead.
4.3. Trade Partner employees shall:
   4.3.1. Attend and participate in project orientations and lead hazard awareness training and immediately report the discovery or disturbance of lead containing material.

E.11 Access to Employee Medical and Exposure Records - U.S.

Introduction

This policy covers employee access to exposure and medical records and analyses that concern their employment. Trade Partners shall permit employee to access exposure and medical records relevant to the employee, free of charge, within a reasonable period of time.

Procedures

1. Employee medical record means a record concerning the health status of an employee which is made or maintained by a physician, nurse, or other health care personnel or technician.
2. Medical records shall be retained for the duration of employment plus 30 years.
3. Environmental (workplace) monitoring or measuring of a toxic substance or harmful physical agent, including personal, area, grab, wipe, or other form of sampling, as well as related collection and analytical methodologies, calculations and other background data relevant to interpretation of the results obtained.
4. Biological monitoring results which directly assess the absorption of a toxic substance or harmful physical agent by body systems (e.g., the level of a chemical in the blood, urine, breath, hair, fingernails, etc.) but not including results which assess the biological effect of a substance or agent, or which assess an employee's use of alcohol or drugs.
5. Employee exposure records shall be retained for 30 years.
6. Access to records shall be provided in a reasonable time, place and manner. If access to records cannot reasonably be provided within fifteen (15) working days, the employer shall within the fifteen (15) working days apprise the employee or designated representative requesting the record of the reason for the delay and the earliest date when the record can be made available.
7. Whenever an employee or designated representative requests a copy of a record, that record shall be provided at no cost.
8. Whenever access is requested to an analysis which reports the contents of employee medical records by either direct identifier (name, address, social security number, payroll number, etc.) or by information which could reasonably be used under the circumstances indirectly to identify specific employees (exact age, height, weight, race, sex, date of initial employment, job title, etc.), personal identifiers shall be removed before access is provided.
9. Upon an employee's first entering employment and at least annually thereafter, information shall be given to current employees of the existence, location, availability and the person responsible for maintaining and providing access to records and each employee's rights of access to these records.
10. Whenever an employer is ceasing to do business, the employer shall transfer all records subject to this section to the successor employer. Whenever an employer is ceasing to do business and there is no successor employer to receive and maintain the records, or intends to dispose of any records required to be preserved for at least thirty (30) years, the employer shall notify affected current employees of their rights of access to records at least three (3) months prior to the cessation of the employer's business.

E.12 Bloodborne Pathogen Exposure Prevention

Introduction

This baseline expectation applies to all Trade Partner employees who could be *reasonably anticipated*, as a result of performing job duties, to come in contact with blood and other potentially infectious bodily fluids. Trade Partner employees trained and certified in first aid and cardiopulmonary resuscitation (CPR), who might be *reasonably anticipated* to come in contact with bodily fluids also shall follow the guidelines set forth in this section.

Procedures

1. General Requirements.
   - 1.1. When dealing with blood or other bodily fluids, Whiting-Turner and Trade Partner employees are required to follow universal precautions. Accordingly, all human blood and other human body fluids are treated as if known to be contaminated with bloodborne infectious diseases.
   - 1.2. All project sites and offices are required to provide employees with disposable latex free gloves and one-way resuscitation masks for use in the event of an emergency.
   - 1.3. All certified first aid providers are required to wear disposable latex free gloves and eye protection while performing first aid on an injured individual. If rescue breathing or CPR is performed, a one-way resuscitation mask shall be provided for the protection of the injured and the provider.
   - 1.4. Whiting-Turner and each Trade Partner shall have, at all times, a trained and certified FA/CPR representative.
   - 1.5. All blood spills shall be immediately contained and cleaned with an anti-viral solution, or by a solution of 5:1 water to bleach. In the event of a serious accident, the Whiting-Turner project team shall contract an outside biohazardous materials firm.
   - 1.6. Any material saturated with blood or other bodily fluid shall be considered regulated waste. Discarded band-aids and gauze containing small amounts of blood products are not considered regulated waste. Disposal of all regulated waste shall be conducted by an outside biohazardous waste firm or emergency medical personnel.

2. **Additional Requirements and Considerations.**
   - 2.1. Employers whose workers can be reasonably anticipated to contact blood, or other potentially infectious materials, shall ensure the following:
     - 2.1.1. Proper training is provided on Bloodborne Pathogens
     - 2.1.2. An Exposure Control Plan is readily available to employees
     - 2.1.3. A Hepatitis B Vaccine is provided to all employees with occupational exposure, unless an employee signs a declination statement and
     - 2.1.4. Records of employee exposure to Bloodborne Pathogens shall be accessible for the duration of employment plus 30 years
   - 2.2. Training records shall be retained for 3 years or more.

E.13 Carbon Monoxide Exposure Prevention

Introduction

Carbon monoxide (CO) is a highly-toxic, flammable, non-irritating, tasteless, odorless gas that is slightly lighter than air. CO is one of the most common asphyxiants encountered in the construction industry. CO is produced in copious amounts by internal combustion engines such as automobiles, diesel powered compressors, welding machines, concrete machines and forklifts. Some of the common symptoms of carbon monoxide poisoning are shortness of breath, headache, dizziness, muscular weakness and nausea.

Procedures

1. The use of fuel powered engines and tools—which could potentially increase the likelihood of carbon monoxide exposures beyond the permissible exposure limit—are prohibited in poorly ventilated areas on Whiting-Turner projects.
2. In enclosed or poorly ventilated spaces, tools and equipment shall be powered by electricity, batteries, or compressed air.
3. All fuel driven equipment being used indoors or in partially enclosed spaces shall have scrubbers where carbon monoxide exposure exists, regardless of the engine tier.
4. When using gasoline powered generators and compressors, place them outside away from air intakes to ensure that the exhaust is not being drawn back indoors.
5. Where the potential for exposure beyond the permissible exposure limit is probable—using any device that discharges the products of combustion into a work area—the following testing requirements shall be followed:
   - 5.1. Monitor the work area continuously for the concentration of carbon monoxide,
   - 5.2. Monitor several different points within the area at the working/breathing heights of employees.
   - 5.3. Remove the employees from the area when the concentration of carbon monoxide reaches 35 PPM. Supplemental ventilation and reduction or elimination of the source shall be provided to reduce the concentration below 35 PPM before the employees are allowed to resume work in the area.

E.14 Pandemic Preparedness

Introduction

This section is to be applied in the event of a pandemic.

Procedures

1. Whiting-Turner and its Trade Partners will uphold federal and local policies and requirements.
2. In addition to all other requirements in our Pandemic Policy and Statement, following a pandemic event, the person responsible for implementation of the plan should identify learning opportunities and take action to implement any corrective actions.

E.15 Heat Illness Prevention

Introduction

Heat-related illnesses can manifest in various ways, potentially causing work-related injuries and fatalities. These include:

- Heat stroke
- Heat exhaustion
- Fainting or dizziness; and
- Heat cramps.

Trade Partner Supervisors and field personnel shall take the following precautions to minimize the risk of heat illness when working in temperatures of 80 degrees and above. Supervisors shall receive training in the prevention of heat-related illnesses prior to supervising employees working in heat.

Procedures

1. **Written Heat Illness Prevention and Management Plan**
    1.1. Throughout employees' work shifts, supervision is required to monitor the heat index where employees are working. This can be done by:
        1.1.1. Directly measuring the temperature and humidity at the same time and location in the areas where employees are performing work;
        1.1.2. Using local weather data reported by the National Weather service or other recognized source to determine the heat index; or
        1.1.3. Using the National Institute for Occupational Safety and Health's (NIOSH) Heat Safety Tool application to determine the heat index
    1.2. Trade Partners shall measure the temperature and humidity at the same time and location in areas where employees perform work in building and structures that do not have a functioning mechanical ventilation system.

2. **Acclimatization Requirements**
    2.1. Trade Partners shall provide an acclimatization period for affected employees for up to 14 days. This shall be done with employees newly exposed to heat and those employees who have been away from work for seven or more consecutive days.
    2.2. Trade Partners shall monitor employees during the acclimatization period for signs of heat-related illness through regular communication by using:
        2.2.1. A phone or radio
        2.2.2. A buddy system, or
        2.2.3. Other effective means of observation
    2.3. Trade Partners shall develop and implement an acclimatization schedule that complies with one of the following:
        2.3.1. A schedule that gradually increases an employee's exposure time over a five-to 14-day period, with a maximum 20% increase each day;
        2.3.2. A schedule that uses the current NIOSH recommendations for acclimatization; or

2.3.3. A schedule that uses a combination of gradual introduction and alternative cooling and control measures that acclimate an employee to the heat.

2.4. Alternative cooling and control measures include engineering, work practices, administrative or other controls to manage heat such as:
   2.4.1. Job rotation;
   2.4.2. Mechanical ventilation systems;
   2.4.3. Misting equipment;
   2.4.4. Cooling vests;
   2.4.5. Air-cooled or water-cooled garments; and
   2.4.6. Access to recreational water.

2.5. The acclimatization schedule shall be in writing and consider the following elements:
   2.5.1. Acclimated and unacclimated employees;
   2.5.2. The environmental conditions and anticipated workload;
   2.5.3. The impact of required clothing and personal protection equipment on employees' heat burden;
   2.5.4. The personal risk factors that put an employee at a higher risk of heat-related illness;
   2.5.5. Reacclimatizing employees as necessary and;
   2.5.6. The use of alternative cooling and control measures.

3. **Shade Access Requirements**

3.1. Trade Partners shall provide shaded areas for their exposed employees as close to the work area as practical. Shaded areas shall:
   3.1.1. Be outside, open and exposed to air on at least three sides;
   3.1.2. Prevent contributing heat sources from reducing effectiveness;
   3.1.3. Be sufficiently sized for the number of employees utilizing the shaded area;
   3.1.4. Be arranged in a configuration that allows employees to sit in normal posture; and
   3.1.5. Accommodate the removal and storage of personal protective equipment during periods of use.

3.2. If providing or creating shade in outdoor work areas is not feasible or unsafe, there should be alternative cooling control measures that provide protections similar to shade. Trade Partners may provide cooling with an indoor mechanical ventilation system as an alternative to outdoor shade provided that the indoor space satisfies the following:
   3.2.1. Prevents contributing heat sources from reducing effectiveness;
   3.2.2. Is sufficiently sized for the number of employees utilizing the shaded area;
   3.2.3. Is arranged in a configuration that allows employees to sit in normal posture; and
   3.2.4. Accommodates the removal and storage of personal protective equipment during periods of use.

4. **Drinking Water Requirements**

4.1. Trade Partners shall provide employees with at least 32 ounces of drinking water per hour to each exposed employee per day at no cost to the employee. The water should be provided as close to the work area as possible.

4.2. Trade Partners are not required to provide the entire supply of drinking water at the beginning of an employee's shift bust shall always make drinking water available while work is being performed.

5. **High-heat Procedures**
   5.1. Trade Partners shall implement high-heat procedures when the index reaches or exceeds 90 degrees Fahrenheit in the area where the work is being performed. The high-heat procedures shall include a work and rest schedule to protect employees from heat-related illness that is adjusted for:
      5.1.1. Environmental conditions;
      5.1.2. Workload; and
      5.1.3. Impact of required clothing or personal protective equipment.
   5.2. Unless Trade Partners can demonstrate effective heat management protection from heat-related illness through alternative cooling and control measures, high-heat procedures shall include the following:
      5.2.1. A minimum rest period of 10 minutes for every two-hour worked where employees are exposed to a heat index above 90 and below 100 degrees Fahrenheit; and
      5.2.2. A minimum rest period of 15 minutes for every hour worked where employees are exposed to a heat index above 100 degrees Fahrenheit; or
      5.2.3. A rest period, as provided in the current NIOSH recommendations for work and rest schedules, to manage heat exposures.
   5.3. If Trade Partners can demonstrate effective heat management and protection from heat-related illness through alternative cooling and control measures, the work and rest schedules set forth above may not be required.
   5.4. If Trade Partners utilize alternative cooling and control measures, the measures:
      5.4.1. Shall be readily available and accessible to all employees at all times work is being performed;
      5.4.2. Shall be documented in writing; and
      5.4.3. May not supersede any other requirements of this standard.
   5.5. Trade Partners may coincide rest periods with a scheduled rest or meal period. Rest periods shall be taken in the shade in accordance with the shade access section of this standard. In addition, Whiting-Turner supervision and its Trade Partners may not discourage employees from taking rest breaks as needed to prevent heat-related illness.
   5.6. Trade Partners shall monitor exposed employees from signs of heat-related illness with regular communication when high-heat procedures are in effect by:
      5.6.1. A phone or radio
      5.6.2. A buddy system, or
      5.6.3. Other effective means of observation.
   5.7. Trade Partners shall make high-heat procedures available in writing in a language and manner that all employees understand.
6. **Emergency Response Procedures**
   6.1. Trade Partners shall have an emergency response plan in place that provides procedures for:

6.1.1. Ensuring effective and accessible means of communication at all times at the worksite to enable an employee to contact a supervisor or emergency medical services;
6.1.2. Responding to signs and symptoms of possible heat-related illness in employees;
6.1.3. Monitoring and providing care to employees who are exhibiting symptoms of heat-related illness; and
6.1.4. Contacting emergency medical services and, if necessary, transporting employees to a location accessible to emergency medical services.

7. **Training Requirements**

7.1. Trade Partners shall provide initial heat stress training to employees and supervisors prior to an employee's first exposure to heat and retrain employees and supervisors. Retraining of employees and supervisors shall occur annual prior to exposure and immediately following any incident at the worksite involving a suspected or confirmed heat-related illness.
7.2. The heat stress training shall be presented in a language and manner that all employees and supervisors can understand. Training shall include, at a minimum, the following:
7.2.1. The work and environmental conditions that affect heat-related illness;
7.2.2. The personal risk factors that affect heat-related illness;
7.2.3. The concept, importance and methods of acclimatization;
7.2.4. The importance of frequent consumption of water and rest breaks in preventing heat-related illness;
7.2.5. The types of heat-related illness, signs and symptoms of heat-related illness and the appropriate first aid and emergency response measures;
7.2.6. The importance of and procedures for employees immediately reporting to the supervisor signs and symptoms of heat-related illness; and
7.2.7. Trade Partner procedures and requirements.

8. **Recordkeeping Requirements**

8.1. Trade Partners shall retain training for one year from the date on which the training occurred. Training records shall include the following:
8.1.1. The names of the persons trained;
8.1.2. The dates of the training sessions; and
8.1.3. A summary or outline of the content of the training sessions.
8.2. Training records shall be made available to regulatory authorities upon request.

E.16 Cold Weather Safety/Cold Stress Prevention

Introduction

The purpose of this program is to address jobs, tasks or employees who are at risk for cold exposure.

Procedures

1. Employees who are required to work in cold weather conditions receive initial and annual training regarding the health effects of cold exposure.
2. Proper cold weather protection shall be worn by employees when working in cold, wet and windy conditions.
3. Protective Clothing is the most important way to avoid cold stress. The type of fabric also makes a difference. Cotton loses its insulation value when it becomes wet. Wool, silk and most synthetics, on the other hand, retain their insulation even when wet. The following are recommendations for working in cold environments:
   3.1. Wear at least three layers of clothing. An inner layer of wool, silk or synthetic to wick moisture away from the body. A middle layer of wool or synthetic to provide insulation even when wet. An outer wind and rain protection layer that allows some ventilation to prevent overheating.
   3.2. Wear a hat or hood. Up to 40% of body heat can be lost when the head is left exposed.
   3.3. Wear insulated, sturdy, leather boots.
   3.4. Wear insulated/moisture wicking socks.
   3.5. Keep a change of dry clothing available in case work clothes become wet.
   3.6. Except for the wicking layer do not wear tight clothing. Loose clothing allows better ventilation of heat away from the body.
   3.7. Do not underestimate the wetting effects of perspiration. Oftentimes wicking and venting of the body's sweat and heat are more important than protecting from rain or snow.
4. Employees are familiar with the signs and symptoms of cold weather induced health problems such as hypothermia, frostbite and trench foot.
   4.1. Hypothermia occurs when body heat is lost faster than it can be replaced. When the core body temperature drops below the normal 98.6° F to around 95° F, the onset of symptoms normally begins. The person may begin to shiver and stomp their feet in order to generate heat. Workers may lose coordination, have slurred speech and fumble with items in the hand. The skin will likely be pale and cold.
   4.2. Frostbite occurs when the skin actually freezes and loses water. In severe cases, amputation of the frostbitten area may be required. While frostbite usually occurs when the temperatures are 30° F or lower, wind chill factors can allow frostbite to occur in above freezing temperatures. Frostbite typically affects extremities, particularly the feet and hands. The affected body part will be cold, tingling, stinging or aching followed by

numbness. Skin color turns red, then purple, then white and is cold to the touch. There may be blisters in severe cases.

4.3. Trench Foot or immersion foot is caused by having feet immersed in cold water at temperatures above freezing for long periods of time. It is similar to frostbite but considered less severe. Symptoms usually consist of tingling, itching or burning sensation. Blisters may be present.

5. Workers working in cold weather environments shall be under constant protective observation by a co-worker or supervisor; a "Buddy System" shall be implemented to ensure that no employee is working alone in cold work environments.
6. Preventative measures are implemented to avoid cold induced injuries. Some preventive measures include drinking plenty of liquids, avoiding caffeine and alcohol. It is easy to become dehydrated in cold weather. If possible, heavy work should be scheduled during the warmer parts of the day. Take breaks out of the cold. Try to work in pairs to keep an eye on each other and watch for signs of cold stress. Avoid fatigue since energy is needed to keep muscles warm. Take frequent breaks and consume warm, high calorie food such as pasta to maintain energy reserves.
7. All employees who are required to perform work in cold conditions should be knowledgeable on how to administer first aid treatment on cold induced injuries or illnesses.
8. Regularly used walkways and travel paths shall be sanded, salted, or cleared of snow and ice as soon as practicable.
9. All employees should be informed of the dangers and destructive potential caused by unstable snow buildup, sharp icicles and ice dams and know how to prevent accidents caused by them.
10. Regular inspections on cold weather supplies (e.g. hand warmers, jackets, shovels, etc.) should be carried out to ensure that supplies are always in stock. Items shall be restocked as necessary.

E.17 Fatigue Management

Introduction

Construction work that includes fatigue stressors such as night work or extended duty periods is especially vulnerable to increased risk of safety incidents. Higher fatigue levels may also contribute to errors in the construction process that impact quality and schedule.

Procedures

1. Training shall be provided on how to recognize fatigue, how to control fatigue through appropriate work and personal habits and reporting of fatigue to supervision.
2. Each employer will set work hour limitations and will control job rotation schedules to control fatigue, allow for sufficient sleep and increase mental fitness to help control employee turnover and absenteeism.
3. Ergonomic friendly equipment will be used to improve workstation conditions such as anti-fatigue mats for standing, lift assist devices for repetitive lifting, proper lighting and control of temperature and other ergonomic devices as deemed appropriate.
4. Work tasks to control fatigue shall be analyzed and evaluated periodically.
5. Chairs will be provided in break areas for workers to sit periodically and will provide periodic rest breaks for personnel.
6. It is the role and responsibility of each employee to report fatigue/tiredness to supervisors, especially those in safety critical positions, to report fatigue/tiredness and lack of mental acuity to supervision; as well as supervisory personnel to make safety critical decisions and take appropriate actions to prevent loss.
7. Employees shall not chronically use over the counter or prescription drugs to increase mental alertness. Employees should be discouraged from taking any substance known to increase fatigue in that employee, including fatigue that sets in after the effects of the drug wear off.

E18. Housekeeping

Introduction

These requirements apply to all work performed by Trade Partners and vendors. Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart C – General Safety and Health Provisions.

Procedures

1. **General Housekeeping**
   1.1. Work areas and paths of ingress/egress shall be kept clear and free of obstructions by material/debris.
   1.2. Clean-as-you-go practices are required.
   1.3. Require Trade Partners to sort and organize material, sweep daily and standardize activities to aid in the elimination of storage of excess/unused material in active work areas
   1.4. Materials shall not be stored in a manner that will block, restrict, impede, or prevent access to an egress path or emergency equipment, such as fire extinguishers, emergency eyewash or shower, emergency shutoff buttons or emergency disconnect devices.
   1.5. Work that may temporarily block emergency exits, safety showers, elevators, corridors and hallways will require prior Whiting-Turner approval.

2. **Power Cord Management**
   2.1. All cords shall be inspected before use.
   2.2. At no time shall cords be strung across exits or in front of emergency equipment.
   2.3. Run cords overhead in a supported fashion with nonconductive material, when feasible.
   2.4. Tape cords down or use cord covers if they present a tripping hazard.
   2.5. Support all cords that run through floors or ceilings with appropriate means.
   2.6. All cords shall be stored and put away after use. (i.e. not coiled up on floor).
   2.7. All extension cords shall be equipped with GFCI protection or be plugged into an outlet equipped with GFCI protection.

NOTE: If the above-listed safety requirements cannot be met, temporary wiring shall be installed to facilitate proper cord management.

3. **Material Storage**
   3.1. Materials stored near the area where work is performed should be limited to only those materials that will be used in the same shift.
   3.2. Any material stored in a work area longer than 24 hours shall be approved by Whiting-Turner.
   3.3. Store all items neatly in cabinets or on shelves.
   3.4. Gang boxes, toolboxes and sea containers/conex boxes shall not have materials stored on top of them.

4. **Chemical Storage**
    4.1. All chemicals brought on site shall have been specified for use on the project.
    4.2. The user of the chemical shall provide Whiting-Turner an SDS prior to bringing the substance on site.
    4.3. All chemicals and equipment containing chemicals shall be stored in approved areas (i.e. chemical cabinet).
    4.4. Trade Partners are responsible for removing all unused chemicals from the Whiting-Turner project site at the completion of their contract.
    4.5. All chemical containers shall be properly labeled.
    4.6. Chemical/gas cylinders (welding, purging, leak detection cylinders, etc.) shall be secured at all times.
    4.7. All dedicated chemical storage areas shall have safety data sheets (SDS) available at the storage location.

NOTE: If you are unsure of appropriate storage areas, contact Whiting-Turner Superintendent for direction.

5. **Material/Waste Disposal**
    5.1. Waste disposal methods shall be specified within each Trade Partner's Activity Hazard Analysis (AHA) and Work task Plan (PTP).
    5.2. All hazardous waste shall be disposed of in accordance with Federal, State and Local regulations and shall comply with applicable Whiting-Turner hazardous waste programs.
    5.3. All hazardous waste shall be properly labeled.
    5.4. Hazardous waste materials shall be discarded into proper disposal containers
    5.5. Non-hazardous waste shall be disposed of into appropriate recycling or disposal containers.
    5.6. Whiting-Turner employees will be made aware of the proper method to dispose of wastes.

E.19 Signs, Signals and Barricades

Introduction

All employees of the Whiting-Turner and its Trade Partner will comply with 29 CFR 1926, Construction Industry Regulations, Subpart G, Signs, Signals and Barricades.

Procedures

1. Signs shall be posted to warn others that a temporary hazard exist in the area where the Trade Partner is working.
2. Barricades shall be utilized to deter the passage of persons or vehicles when necessary.
3. Required signs will comply with the OSHA standards described in 1926.200.
4. Where areas may require additional awareness or present unique danger, the use of warning tape may be necessary.
   - 4.1. For areas that require additional caution, (e.g. uneven surfaces, wet surfaces) yellow caution tape should be used. Caution tape does not prohibit access.
   - 4.2. For areas where entry and travel are prohibited, (e.g. areas where fall protection is being erected or areas with overhead work being performed) red danger tape should be used. Danger tape is intended to prohibit access.
   - 4.3. Caution/danger tape has limitations; therefore, it shall not be used in lieu of physical barricades when an occurrence calls for a physical barricade.
   - 4.4. The intent of the warning tape is to notify of hazards that may arise during construction activities. Every effort shall be made to correct these situations with permanent solutions in a timely fashion.
   - 4.5. All caution and danger tape used on Whiting-Turner project sites shall be of the reinforced type and shall be supplemented with a tag/label affixed with the responsible party's name, company, contact number and potential hazard.
5. All flag persons shall be trained on appropriate procedures before controlling traffic, as required by the Manual on Uniform Traffic Control Devices (MUTCD) and any municipal or state guidelines.
6. All flag persons shall utilize sign paddles and shall be outfitted with high visibility garments, as required by current ANSI standards. All PPE and traffic control equipment shall be outfitted with reflectorized material for night work as required by current ANSI standards.
7. All crane and hoist signals shall comply with applicable ANSI standards.
8. All traffic control devices shall comply with the MUTCD and any applicable Municipal or State guidelines.
9. All signs shall be secured in such a way that would allow venting in the event of high winds.

E.20 Hand and Power Tools

Introduction

All Whiting-Turner Employees and Trade Partners working on a Whiting-Turner project shall comply with 29 CFR 1926, Construction Industry Regulations, Subpart I – Tools – Hand and Power, the manufacturer's safety recommendations and the following guidelines.

Procedures

1. **General**
    1.1. Hand and power tools shall be maintained in a safe condition, per manufacturers' guidelines. All defective tools shall be removed from service with a visible tag that states 'DO NOT USE' until the tool is repaired or removed from the project.
    1.2. If the tool is designed to accommodate a guard, the guard shall be in place while the tool is being used.
    1.3. Tools manufactured with a handle shall be used with the handle in place.
    1.4. All two-handed tools shall be used with two hands.
    1.5. Additional personal protective equipment (PPE), such as a face shield or hearing protection, may be required while operating a tool. If so, they shall be in use.
    1.6. Any person found making a tool guard inoperable will be immediately removed from the site.
2. **Hand tools.**
    2.1. Drift pins, wedges, chisels and other impact tools shall be kept free of mushroomed heads.
    2.2. Wrenches shall not be issued or used when the jaws are sprung and slippage is probable.
3. **Electric powered tools.**
    3.1. All power tools shall be double insulated or provided with a three-wire, grounded connection.
4. **Pneumatic power.**
    4.1. Each connection on a pneumatic tool and air hose shall be secured with a "whip-check" or similar device.
    4.2. All air hoses, with an inside diameter exceeding ½ inch, shall have a flow reduction device (OSHA valve) at the supply source to reduce pressure in case of hose failure.
    4.3. Compressed air shall not be used for cleaning unless the pressure is reduced to less than 30 psi and appropriate guarding and PPE are in place.
    4.4. Compressed air equipment shall be inspected and safety valves shall be tested.
    4.5. Every air receiver shall be equipped with an indicating pressure gauge.
    4.6. Compressed air receivers shall be drained frequently.
5. **Fuel powered tools.**
    5.1. Fuel powered tools shall be stopped and turned off while being refueled, serviced, or maintained.
6. **Powder-actuated tools.**

- 6.1. Users of powder actuated tools shall be properly trained and certified to operate the equipment. Certification shall always be kept on the user's person while the tool is in use.
- 6.2. The tool shall be tested each day, according to manufacturer's recommendations, before loading to see that safety devices are in proper working condition.
- 6.3. Tools shall not be loaded until just prior to the intended firing time.
- 6.4. Loaded tools shall not be left unattended.
- 6.5. All tools shall be used with the correct shield, guard or attachment recommended by the manufacturer.
- 6.6. Unspent shots shall be disposed of according to the manufacturer's recommendation and removed from site by the user each day.

7. **Abrasive wheels and tools.**
   - 7.1. The RPM rating on all grinding machine motors shall not exceed the speed rating of the grinding wheel attachment.
   - 7.2. The grinding wheel shall be compatible with the grinder for which it is being used (i.e. proper size and type).
   - 7.3. All abrasive wheels shall be closely inspected and ring tested before mounting to ensure they are free from cracks or defects.
   - 7.4. The gap between the work rest and abrasive wheel of a bench or floor mounted grinder shall not exceed 1/8-inch.
   - 7.5. Proper wheel selection is required and is contingent upon the intended work activity.
8. **Woodworking tools.**
   - 8.1. All portable, power-driven circular saws shall be equipped with guards above and below the base plate or shoe.
   - 8.2. When the tool is withdrawn from the wood, the lower guard shall automatically and instantly return to the covering position.
   - 8.3. Any worker who is found disabling a blade guard will be immediately removed from the site.
   - 8.4. Saws shall have the electrical cord unplugged when making blade adjustments or changing blades.

E.21 Welding

Introduction

All Whiting-Turner employees and all Trade Partner employees working on a Whiting-Turner project shall comply with 29 CFR 1926, Construction Industry Regulations, Subpart J – Welding and Cutting, in addition to the following guidelines.

Procedures

1. **General Requirements.**
   - 1.1. The Trade Partner will assure that adequate precautionary measures have been taken to protect all personnel, members of the public and property from contact with flash or contact with falling slag and sparks.
   - 1.2. Welders, cutters and their supervisor shall be trained in the safe operation of their equipment, safe welding/cutting practices and welding/cutting respiratory and fire protection.
   - 1.3. All Trade Partners performing welding, burning/cutting operations shall submit an AHA to the Superintendent prior to the commencement of work.
   - 1.4. Trade Partners are required to use welding shields between the welding operation and the public/other trades for protection against arc flash.
   - 1.5. All welding lead connection lugs are required to have non-conductive boot covers installed.
   - 1.6. Local fire department fire details may be required and all costs associated with the detail will be the responsibility of the Trade Partner doing the hot work requiring the detail.
   - 1.7. Trade Partners may be required to provide adequate engineering controls (ventilation and/or smoke eaters) for welding and burning operations.
   - 1.8. Trade Partners performing welding and/or burning operations may be required to provide air monitoring in areas with inadequate ventilation.
   - 1.9. All employees engaged in hot work activities shall wear the appropriate personal protective equipment (PPE); shields shall be compatible with a head protection.
   - 1.10. Oxy-fuel gas welding shall have hoses that are readily distinguishable.
   - 1.11. Welding and cutting systems using cylinder-hose-torch shall have a reverse-flow check valve and a flash arrestor.
   - 1.12. Oxygen cylinders in storage shall be separated from fuel gas cylinders or combustible material by 20 feet or by a non-combustible barrier at least five (5) feet high having a fire-resistance rating of at least ½ hour.
   - 1.13. All cylinders shall be considered in storage at the end of each shift; cylinders shall have gauges removed and caps in place.
   - 1.14. All compressed gas cylinders shall be secured against displacement by cinch straps or chains.

2. **Hot Work Permit.**
    2.1. A Hot Work Permit shall be completed daily by each Trade Partner performing all welding, burning/cutting operations.
    2.2. The hot work permit will designate the location of the operation, date of the operation, type of work to be done and the time commenced and completed.
    2.3. A hot work permit is required to be submitted by the Trade Partner prior to beginning the hot work activity.
    2.4. The hot work permit shall be conspicuously located near the area where the activity is taking place.
    2.5. Trade Partners are responsible for providing a charged, 20lb ABC dry chemical fire extinguisher for each welding and burning activity.
    2.6. Other hot work activities besides welding and burning may require the use of a fire watch, as designated by Whiting-Turner.
    2.7. Whiting-Turner life safety fire extinguishers shall not be used for fire watch duties.
    2.8. A fire watch is to be positioned at each location where sparks/slag and/or molten metal can fall/drop/strike combustible material, potentially resulting in a fire.
    2.9. A fire watch is required to remain in place at all times during the hot work activity and for a minimum of one half (1/2) hour after the welding or burning operation has been completed or longer periods of time as determined by the project-specific rules.
    2.10. The fire watch is to be replaced by another fire watch if the person needs to leave the area for any reason.
    2.11. Additional permits may be required by the local Fire Department and will be at the Trade Partner’s expense.
    2.12. Copies of these additional permits are to be submitted to Whiting-Turner upon request.
    2.13. Upon completion of the work, the completed hot work permit shall be returned to Whiting-Turner for file retention.

3. **Additional Information, Requirements and Considerations.**
    3.1. A fire extinguisher shall be readily available while hot work is being performed.
    3.2. Persons with fire watch duties shall be provided training to be able to recognize and abate the fire hazard and to understand the primary and secondary functions of their role.
    3.3. Combustible materials shall be removed from the area where hot work is being performed.
    3.4. In areas where it is infeasible to remove combustible materials, guards/shields shall be used to separate hot work activities from combustible materials.
    3.5. A hot work permit is required prior to the commencement of hot work activities.
    3.6. Hot work activities shall not be performed if it is unsafe to do so.
    3.7. During hot work activities, proper ventilation and/or respiratory equipment shall be used when hazardous fumes/gases or dust may be present.

3.8. Defective hot work equipment shall be tagged out and removed from service immediately.

E.22 Hexavalent Chromium

Introduction

Hexavalent chromium (Cr(VI)) compounds are widely used in the chemical industry as ingredients and catalysts in pigments, metal plating and chemical synthesis. Hexavalent chromium can also be found in the construction industry through welding on stainless steel or on hexavalent chromium painted surfaces. The major health effects include lung cancer, nasal and sinus cancer, skin ulcerations and contact dermatitis. The purpose of this policy is to prevent employee exposure to hexavalent chromium compounds during construction activity. Each Trade Partner working on a Whiting-Turner project shall comply with 29 CFR §1926.1126, Chromium (VI), in addition to the following guidelines. Training shall be provided on chromium hazards, control methods and medical surveillance.

Procedures

1. **Permissible Exposure Limit.**
   - 1.1. Since this construction activity is limited to specialty work, Whiting-Turner will direct the Trade Partner to provide specific Activity Hazard Analysis (AHA) meetings to address potential exposure.
   - 1.2. The Trade Partner shall ensure that no employee is exposed to an airborne concentration Cr(VI) in excess of 5 micrograms per cubic meter of air (5 ug/m^3) calculated as an 8-hour time-weighted average (TWA).
   - 1.3. Engineering controls are the preferred method to achieve the Permissible Exposure Limit (PEL). Engineering and work practice controls should be provided to reduce exposure to the lowest feasible level. If employees can demonstrate that such controls are not feasible, The Trade Partner shall use engineering/work controls to reduce employee exposure to the lowest levels achievable and shall supplement them by the use of respiratory protection.
   - 1.4. Regulated areas shall be established when an employee's exposure is or is expected to be in excess of the PEL. Regulated areas shall be marked with warning signs to alert employees. Access is restricted to "authorized persons".

2. **Exposure Determination.**
   - 2.1. The Trade Partner shall determine the 8-hour TWA exposure for each employee exposed to Cr(VI). This may be accomplished using two options; scheduled or performance-oriented monitoring.
   - 2.2. Scheduled Monitoring
     - 2.2.1. The Trade Partner shall perform initial monitoring to determine the 8-hour TWA for each employee based on a sufficient number of personal breathing zone samples.
     - 2.2.2. If the Trade Partner does representative sampling, it shall be conducted on the employee(s) expected to receive the highest exposure.

- 2.2.3. If the monitoring indicates that employee exposures are below the action level (1/2 the PEL or 2.5 ug/m3), the employee may discontinue monitoring.
- 2.2.4. If the monitoring indicates that employee exposures are at or above the action level, the Trade Partner shall perform periodic monitoring at least every six months.

2.3. Performance-Oriented Monitoring

- 2.3.1. If this option is chosen, the Trade Partner shall determine the 8-hour TWA for each employee based on any combination of air monitoring, historical data, or objective data sufficient to accurately characterize employee exposure to Cr(VI).

3. **Methods of compliance.**

3.1. As stated previously, engineering and work practice controls shall be used to reduce and maintain employee exposure to Cr(VI) to or below the PEL.

3.2. If feasible engineering and work practice controls are insufficient to reduce exposure below the PEL, then respiratory protection shall be used.

3.3. The Trade Partner will not be allowed to rotate employees to different jobs to achieve compliance with the PEL.

4. **Respiratory protection.**

4.1. Respiratory protection use shall comply with OSHA's respiratory protection requirements.

4.2. Respirators shall be used when engineering controls and work practices cannot reduce employee exposure below the PEL during work operations where engineering controls and work practices are not feasible and emergencies. The Trade Partner shall provide respiratory protection in the following circumstances:

- 4.2.1. Periods necessary to install or implement feasible engineering or work practice controls.
- 4.2.2. Work operations where an employer has implemented all feasible engineering and work practice controls and such controls are not sufficient to reduce the PEL.
- 4.2.3. Emergencies

5. **Protective work clothing and equipment.**

5.1. Where there may be a hazard to the skin or eyes from exposure to Cr(VI) the Trade Partner shall provide protective clothing or equipment to the employee. PPE shall be provided when there is a hazard from skin or eye contact. Gloves, aprons, coveralls, goggles, foot covers etc.

5.2. The Trade Partner shall ensure that the employees remove all clothing and equipment that may be contaminated with Cr(VI) when the work is complete or at the end of the shift.

5.3. The Trade Partner shall ensure that chromium-contaminated clothing is not removed from the workplace.

5.4. When contaminated protective clothing or equipment is removed for laundering or cleaning, the Trade Partner shall ensure that it is stored and transported in impermeable bags or containers.
5.5. The Trade Partner shall inform any person who launders or cleans clothing or equipment of the potential effects of exposure to Cr(VI) and that the clothing or equipment should be laundered or cleaned in a manner that minimizes skin or eye contact.

6. **Hygiene areas and practices.**
   6.1. The Trade Partner shall provide change rooms for decontamination and ensure facilities prevent cross-contamination. Washing facilities shall be readily accessible for removing chromium from the skin. Workers shall wash their hands and face or any other potentially exposed skin before eating, drinking or smoking.
   6.2. Where protective clothing and equipment is required, the Trade Partner shall provide change rooms that comply with 29 CFR 1926.51.
   6.3. Where skin contact may occur, the Trade Partner shall provide hand-washing facilities that comply with the previously noted standard.

7. **Medical surveillance.**
   7.1. Medical surveillance shall be provided when an employee experiences signs or symptoms of the adverse health effects of Hexavalent Chromium (dermatitis, asthma, bronchitis, etc.).
   7.2. The Trade Partner shall make medical surveillance available, at no cost, to employees who meet the following criteria:
      7.2.1. Those who are or may be occupationally exposed to Cr(VI) at or above the action level for 30 or more days a year
      7.2.2. Those who are experiencing signs or symptoms of adverse health effects associated with Cr(VI) exposure
      7.2.3. Those exposed in an emergency
      7.2.4. Examinations will be performed by or under the supervision of a physician or other licensed health care professional.

8. **Communication.**
   8.1. Shall follow the same communication of hazardous chemicals highlighted in Whiting-Turner's Hazard Communication Program.

9. **Recordkeeping.**
   9.1. Trade Partners are required to maintain and make available an accurate record of all employee exposure monitoring, medical surveillance and training records.
   9.2. Trade Partner shall maintain the following data records:
      9.2.1. Air monitoring
      9.2.2. Historical monitoring
      9.2.3. Objective data

9.2.4. Medical surveillance

10. **Additional Requirements and Considerations**

10.1. Surfaces shall be maintained as free as practicable of accumulation of chromium. All spills and releases of chromium shall be cleaned promptly. Methods of cleaning include HEPA filtered vacuums, dry or wet sweeping, shoveling or other methods to minimize exposure.

E.23 Confined Spaces

Introduction

No Whiting-Turner or Trade Partner employee shall enter into any type of confined space until it has been identified and labeled by their respective employer's competent person and all applicable safety requirements contained in 29 CFR 1926 Subpart AA – Confined Spaces in Construction and this section have been met. All Trade Partners involved in confined space activities shall submit a program—which meets or exceeds all federal, state and local regulations as well as the directives contained herein—to Whiting-Turner prior to the commencement of work in said spaces. All confined space entry programs shall go through annual review.

Procedures

1. **General.**
   1.1. All confined spaces, regardless of classification, shall have continuous multi-gas/4-gas air monitoring while the space is occupied by tradespersons. Atmospheric testing shall also be performed prior to persons entering the space.
2. **Pre-entry assessment.**
   2.1. Prior to confined space entry, each employer shall ensure that a competent person identifies all confined spaces in which one or more of the employees it directs may work, identifies each space that is a permit space, through consideration and evaluation of the elements of that space, including testing.
   2.2. It is Whiting-Turner's position that all confined spaces are permit required until proven otherwise [in writing] by the Trade Partner's competent person. Depending upon the type of confined space identified, specific criteria shall be satisfied before entry.
3. **Signage.**
   3.1. All confined spaces shall be labeled. If the workplace contains permit spaces, the entry supervisor shall inform employees by posting danger signs at all entrances of confined spaces. The signs will be legible in English and in the predominant language of non-English reading tradespersons. At a minimum, the following information will be included:

DANGER
PERMIT-REQUIRED CONFINED SPACE
DO NOT ENTER

4. **Authorized entry.**
   4.1. If permit spaces exist in the workplace, only authorized employees may enter the spaces. The entry supervisor shall take effective measures to prevent unauthorized employees from entering permit spaces. No work will be permitted in confined

spaces until the Trade Partner has initiated, maintained, completed and posted a permit accepted by a designated Whiting-Turner employee.

5. **Modification of non-permit spaces.**
    5.1. If non-permit spaces are modified or experience any change that causes an increased hazard to entrants, the supervisor of the exposed employees, shall ensure that the space is reevaluated by the competent person.
6. **Permit required spaces.**
    6.1. If permit spaces are identified, the following program elements, at a minimum, shall be addressed in a written project specific confined space procedure. Whiting-Turner does not permit work in confined spaces where IDLH conditions are present.
    6.2. Environmental Controls – to ensure that pre-entry precautions (i.e. hazard evaluations, operating procedures, isolation methods, safety equipment, etc.) have been implemented.
    6.3. Atmospheric Testing – for oxygen content, explosive vapors, toxic substances and carbon monoxide to ensure that acceptable entry conditions exist.
    6.4. Assigned Duties – of each participant shall be established and clearly communicated.
    6.5. The entry employer shall ensure that all **authorized entrants**:
        6.5.1. Are familiar with and understand the hazards that may be faced during entry, including information on the mode, signs or symptoms and consequences of the exposure;
        6.5.2. Properly use equipment as required by §1926.1204(d);
        6.5.3. Communicate with the attendant as necessary to enable the attendant to assess entrant status and to enable the attendant to alert entrants of the need to evacuate the space as required by §1926.1209(f).
        6.5.4. Alert the attendant whenever:
            6.5.4.1. There is any warning sign or symptom of exposure to a dangerous situation; or
            6.5.4.2. The entrant detects a prohibited condition;
        6.5.5. Exit from the permit space as quickly as possible whenever:
            6.5.5.1. An order to evacuate is given by the attendant or the entry supervisor;
            6.5.5.2. There is any warning sign or symptom of exposure to a dangerous situation;
            6.5.5.3. The entrant detects a prohibited condition; or
            6.5.5.4. An evacuation alarm is activated.
    6.6. An attendant shall be stationed outside a confined space while it is occupied by workers. A single attendant may not monitor multiple confined spaces.
    6.7. The entry employer shall ensure that each **attendant**:
        6.7.1. Is familiar with and understands the hazards that may be faced during entry, including information on the mode, signs or symptoms and consequences of the exposure;

6.7.2. Is aware of possible behavioral effects of hazard exposure in authorized entrants;
6.7.3. Continuously maintains an accurate count of authorized entrants in the permit space and ensures that the means used to identify authorized entrants under § 1926.1206(d) accurately identifies who is in the permit space;
6.7.4. Remains outside the permit space during entry operations until relieved by another attendant;
6.7.5. Communicates with authorized entrants as necessary to assess entrant status and to alert entrants of the need to evacuate the space under § 1926.1208(e);
6.7.6. Assesses activities and conditions inside and outside the space to determine if it is safe for entrants to remain in the space and orders the authorized entrants to evacuate the permit space immediately under any of the following conditions:
6.7.6.1. If there is a prohibited condition;
6.7.6.2. If the behavioral effects of hazard exposure are apparent in an authorized entrant;
6.7.6.3. If there is a situation outside the space that could endanger the authorized entrants; or
6.7.6.4. If the attendant cannot effectively and safely perform all the duties required under this section;
6.7.7. Summons rescue and other emergency services as soon as the attendant determines that authorized entrants may need assistance to escape from permit space hazards;
6.7.8. Takes the following actions when unauthorized persons approach or enter a permit space while entry is underway:
6.7.9. Warns the unauthorized persons that they shall stay away from the permit space;
6.7.10. Advises the unauthorized persons that they shall exit immediately if they have entered the permit space; and
6.7.11. Informs the authorized entrants and the entry supervisor if unauthorized persons have entered the permit space;
6.7.12. Performs non-entry rescues as specified by the employer's rescue procedure; and
6.7.13. Performs no duties that might interfere with the attendant's primary duty to assess and protect the authorized entrants.

6.8. The entry employer shall ensure that each **entry supervisor**:
6.8.1. Is familiar with and understands the hazards that may be faced during entry, including information on the mode, signs or symptoms and consequences of the exposure;
6.8.2. Verifies, by checking that the appropriate entries have been made on the permit, that all tests specified by the permit have been conducted and that all

procedures and equipment specified by the permit are in place before endorsing the permit and allowing entry to begin;

- 6.8.3. Terminates the entry and cancels or suspends the permit as required by §1926.1205(e);
- 6.8.4. Verifies that rescue services are available and that the means for summoning them are operable and that the employer will be notified as soon as the services become unavailable;
- 6.8.5. Removes unauthorized individuals who enter or who attempt to enter the permit space during entry operations; and
- 6.8.6. Determines, whenever responsibility for a permit space entry operation is transferred and at intervals dictated by the hazards and operations performed within the space, that entry operations remain consistent with terms of the entry permit and that acceptable entry conditions are maintained.

6.9. Rescue Equipment and Emergency Services – develop and implement procedures for summoning rescue and emergency services, for rescuing entrants from permit spaces, for providing necessary emergency services to rescued employees and for preventing unauthorized personnel from attempting a rescue. Only those trained in proper rescue procedures are permitted to perform rescue services in the event of a confined space emergency. The means of communication to summon rescue from the entrant to the attendant shall be facilitated by a two-way communication system.

6.10. Entry Permit System – used to record critical data and serve as official entry authorization shall be implemented and managed accordingly following the completion of permit space work. Barriers/barricades shall be used to prevent unauthorized entry into a confined space.

6.11. Training – of employees expected to enter permit required confined spaces shall be provided to ensure that understanding of assigned duties and the requirements of 29 CFR §1926.1207 and training shall be documented.

6.12. Medical Surveillance Program – for all employees who shall enter permit spaces shall be established to ensure that they have been medically evaluated and cleared to work in such spaces.

6.13. Coordination of operations shall take place prior to multiple employers working in the same confined space.

6.14. The entry supervisor shall ensure that the following procedures for terminating and closing out a confined space permit are followed:

- 6.14.1. Verify that all authorized entrants have vacated the space by signing that they have safely exited the space;
- 6.14.2. Replace all covers, doors and barricade that were breached for entry;
- 6.14.3. Replace permit required confined space signage if it was removed or destroyed;
- 6.14.4. Sign the permit indicating if the job was completed or the permit was terminated and return it to a Whiting-Turner team member.

E.24 Concrete and Masonry
Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart Q – Concrete and Masonry Construction in addition to the following guidelines.

Procedures

1. **General requirements.**
   1.1. No construction load may be placed on a concrete structure unless a qualified person, knowledgeable in structural design, determines that the structure is capable of supporting the load.
   1.2. Protruding reinforced steel—and similar protrusions—onto or into which employees could fall, shall be protected and maintained by the creating Trade Partner to eliminate the hazard of impalement.
   1.3. Trade Partners whose work involves working at heights 6 feet or above a lower level shall utilize conventional fall protection. In addition, the Trade Partner using a personal fall arrest system shall submit a fall protection plan and rescue procedures to Whiting-Turner, including the name and qualifications of the designated competent person.
   1.4. Eyewash stations and washing facilities shall be provided per 29 CFR 1926.50(g) and 29 CFR 1926.51(f)(1), respectively.
   1.5. Ensure that proper concrete washout stations are provided and maintained to prevent runoff of liquids and to consolidate solids for disposal.

2. **Equipment and tool requirements.**
   2.1. Powered and rotating concrete troweling machines shall have a “dead man” switch that automatically shuts off power whenever the hands of the operator are removed from the machine.
   2.2. Masonry saws shall be provided with a semi-circular guard.
   2.3. Machines shall be locked and tagged out of service before employees can perform any maintenance or repair work.

3. **Cast-in-place concrete requirements.**
   3.1. Formwork shall be designed, fabricated, erected, supported, braced and maintained so it can support all anticipated lateral and vertical loads.
   3.2. All shoring equipment shall be inspected prior to erection to determine if it meets the requirements specified in the formwork drawings.
   3.3. Erected shoring equipment shall be inspected immediately prior to, during and after concrete placement.
   3.4. Wherever single post shores are used on top of one another a qualified designer shall prepare the design of the shoring and an engineer qualified in structural design shall

inspect the erected shoring prior to concrete placement. Written documentation shall be provided to Whiting-Turner prior to the start of the pour.

3.5. Forms and shores shall not be removed until the employer determines that the concrete has gained sufficient strength.

3.6. Fall protection shall be maintained while employees are climbing rebar including point to point movement anytime, they are exposed to falls six (6) feet or greater.

3.7. Fall protection is required on all decks where gaps or voids exist in the decking 10” or greater. At the building’s perimeter where the decking steps down to allow for a beam pour, the height of the rails shall be increased accordingly to ensure that guardrail protection remains adequate for persons working on the deck.

3.8. Areas shall be properly barricaded with tape or fence where form stripping is to be performed and signage shall be posted on all sides. This should include areas below stripping.

3.9. Protruding nails in stripped lumber shall be removed or bent immediately.

3.10. Outrigger platforms used for material movement in and out of the building via a crane or forklift shall be designed by an engineer and incorporate 100% fall protection systems. Workers using the platform shall utilize PFAS if any guardrail is removed.

4. **Concrete forms, falsework and vertical shoring.**

4.1. The concrete Trade Partner shall install guardrails along all perimeter edges and interior floor openings.

4.2. The installation of the guardrail system shall progress along the leading-edge formwork.

4.3. Proper guardrail heights shall be maintained during the formwork phase and again after the concrete has been placed; the proper top-rail height is between 39”-45” above the walking/working surface and the mid-rail is half that distance.

NOTE: Proper guardrail height in California is required to be between 42”-45” above the walking/working surface.

4.4. If at any time a worker breaches the height of a guardrail system, that worker shall be protected by a personal fall arrest system or a tiered guardrail system that provides adequate protection.

4.5. If the distance from a person’s walking/working surface and the top-rail is less than 39 inches, that person has breached the fall protection system and another method of protection shall be employed.

4.6. Likewise, if the distance from a person’s walking/working surface and the top-rail exceeds 45 inches, that person is considered to have inadequate protection and another method of protection shall be employed.

5. **Masonry requirements.**

5.1. A limited access zone shall be established prior to the start of any masonry work.

5.2. The zone shall be equal to the height of the wall, plus four feet.

5.3. Employees that are working at heights greater than six (6) feet and are reaching more than 10 inches below the level of the walking / working surface on which they are working, shall be protected from falling by personal fall arrest systems.
5.4. For overhand bricklaying from a scaffold, fall protection is required if the working side of the scaffold has a gap greater than 14" between the scaffold and structure.

E.25 Hydroblasting

Introduction

The purpose of this policy is to ensure that proper safeguards are in place for workers who use high velocity water (hydroblasting).

Procedures

1. Employees should be trained on the hazards (including penetration of the skin by high pressure water), operating procedures and maintenance of hydro-blasters prior to performing hydro-blasting work.
2. Employees performing hydro-blasting work should, at a minimum, wear waterproof body protection, eye protection, head protection including a full-face shield, waterproof foot protection with steel toe caps, appropriate hand protection and hearing protection.
3. Training includes a demonstration of the cutting action of the high-pressure water and an explanation of the effects of high-pressure water penetrating the skin. Training also addresses the potential hazard to the human body by cutting through a piece of lumber, concrete block or rubber boot. If an accident should occur and high-pressure water penetrates skin, medical attention shall be given immediately.
4. A pre-operational, operational and post-operational hydro-blasting permit shall be developed by the site or Trade Partner performing the work. At minimum the permit shall include: job description and equipment being cleaned, precautions taken to protect electrical equipment, maximum operating pressure and list of qualified personnel.
5. Adequate barricades and signs should be in place to protect personnel when approaching all ends of the equipment being cleaned.
6. The operator shall inspect the high-pressure unit and hoses for defects, proper fluid levels and filters and properly sized/rated end fittings.
7. The blast cleaning nozzles shall be equipped with an operating valve (on the gun or foot pedal) which shall be held open manually and always under the control of the operator.
8. Objects to be cleaned shall never be held manually.
9. The minimum total length of a hydro-blasting gun (hand-operated control valve, lance and nozzle resembling a gun layout) shall be 66 inches from the shoulder pad to the nozzle.
10. Properly sized anti-reversal device (stinger assembly attached to a nozzle to prevent it from turning around inside a pipe or large tube) shall be used throughout the task. The combined length of the hose connection, stinger and nozzle shall be a minimum of 1.5 times the diameter of the pipe being cleaned unless the pipe being cleaned has a "T" then the combined length shall be 3 times the diameter of the largest pipe.
11. Moleing device or lance shall require a minimum 2 feet end identification when a pipe flange is available. If no flange or other means to secure anti-reversal device is used, the hose/lance shall require a 2 feet end identification marking and a 4 feet end identification marking of a different color or different pattern.

12. The system shall be shut down and depressurized any time: the barricade is violated, the equipment malfunctions (special attention should be given to the dump control valve), repairs need to be made, or the system is left unattended.
13. A system is not to be operated above the lowest working pressure (40% of the burst pressure) of any of its components.
14. At minimum the hydro-blasting team will consist of a pump operator and a nozzle operator.
15. All hydro-blasting shall be completed from a stable work surface.
16. Ladders, step stools, benches, etc. shall not be used. Use only approved scaffolding or platforms that are job specific.

E.26 Steel Erection

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart R – Steel Erection. In addition to the items listed in this section, all Trade Partners are to comply with all federal, state and local requirements and codes including those in other sections of this Manual and those imposed by the owner. All Trade Partners shall be required to comply with all parts of these requirements based on their scope of work. This section set forth requirements to protect tradespersons from hazards associated with steel erection activities involved in construction.

Procedures

1. **Structure steel design.**
   - 1.1. Column anchorage
     - 1.1.1. All columns shall be anchored by a minimum of four (4) anchor bolts.
     - 1.1.2. When two structural members on opposite sides of a column web, or a beam web over a column, are connected sharing common connection holes, at least one bolt with its wrench-tight nut shall remain connected to the first member unless a shop-attached or field-attached seat or equivalent connection device is supplied with the member to secure the first member and prevent the column from being displaced. If a seat or equivalent device is used, the seat (or device) shall be designed to support the load during the double connection process. It shall be bolted or welded to both a supporting member and the first member before the nuts on the shared bolts are removed to make the double connection.
     - 1.1.3. The perimeter columns have holes or other devices in or attached to perimeter columns at 42-45 inches above the finished floor and the midpoint between the finished floor and the top cable to permit installation of perimeter safety cables. Welded nuts are not permitted as part of the perimeter safety fall protection system.

2. **General requirements.**
   - 2.1. Fall protection provided by the steel erector shall remain in the area where steel erection activity has been completed to be used by other trades. The Whiting-Turner project team shall assign the repair and maintenance to another trade; however, this assignment does not in any way release each Trade Partner of their responsibility to provide and maintain fall protection in their work areas for the protection of their employees and the reinstallation of fall protection devices removed by their employees.
   - 2.2. When floors are ready for turnover, the steel erection Trade Partner shall notify Whiting-Turner. Whiting-Turner will evaluate the installation and condition of the

guardrail cable system provided by the steel erector and decide to accept custody of the system or to have the system removed.

- 2.3. The Steel Erector has a responsibility to inspect and control fall protection in their work areas based on their scope of work.
- 2.4. Site specific erection plans are to be posted and shall include the following topics:
  - 2.4.1. Material deliveries, material staging and storage
  - 2.4.2. Coordination with other trades and construction activities.
  - 2.4.3. Path for overhead loads
  - 2.4.4. Critical lifts, including rigging supplies and equipment.
  - 2.4.5. A description of steel erection activities and procedures
  - 2.4.6. Stability considerations requiring temporary bracing and guying
  - 2.4.7. Erection bridging terminus point
  - 2.4.8. Columns and beams connections
  - 2.4.9. Decking installation
  - 2.4.10. Ornamental and miscellaneous iron.
  - 2.4.11. Procedures that will be used to comply with protection from falling objects.
  - 2.4.12. Special procedures required for hazardous non-routine tasks.

3. **Repairs and/or modifications.**
   - 3.1. Prior to the erection of a column, Whiting-Turner shall provide written notification to the steel erector if there has been any repair, replacement, or modification of the anchor bolts of that column.
   - 3.2. Anchor bolts shall not be repaired, replaced or field-modified without the approval of the project structural engineer of record.
   - 3.3. Columns shall be set on level finished floors, pre-grouted leveling plates, leveling nuts, or shim packs which are adequate to transfer the construction loads.

4. **Fall protection.**
   - 4.1. All tradespersons, including connectors, engaged in steel erection activities on a walking/working surface with an unprotected side or edge (6) feet or more above a lower level shall be protected from fall hazards by a conventional fall protection method.
   - 4.2. All hoisting operations in steel erection shall be pre-planned to ensure that the project requirements are met for employee protection.
   - 4.3. Other construction processes below steel erection are prohibited unless overhead protection for the employees below is provided.
   - 4.4. Full body harnesses are required for fall arrest and these harnesses require daily inspections of this equipment.
   - 4.5. Fall protection is required in baskets of aerial lifts.
   - 4.6. On multi-story structures, perimeter safety cables shall be installed at the final interior and exterior perimeters of the floors as soon as the metal decking has been installed.

4.7. All materials, equipment and tools, which are not in use while aloft, shall be secured against accidental displacement.

4.8. All materials, equipment and tools, which are not in use while aloft, shall be secured against accidental displacement.

5. **Cranes**.
    5.1. Each Trade Partner is responsible for complying with and providing documentation based on the following requirements and providing a description of the crane selection and placement procedures.
    5.2. All operators shall submit documentation showing that they meet the requirements to operate the crane provided.
    5.3. A description of the crane and placement procedures shall be submitted in writing to Whiting-Turner prior to delivery of this equipment.
    5.4. Each crane shall have a current annual inspection conducted by a third party and document shall be submitted to Whiting-Turner.
    5.5. Cranes are to have the following in cab areas; load charts, manufacture manual, fire extinguisher and hand signal poster.
    5.6. Operator shall perform daily inspections on the crane that they will operate.
    5.7. The operator shall not operate the crane until the counterweight is properly barricaded as per OSHA standards.
    5.8. Each Trade Partner is required to inspect the rigging equipment that will be used before each lift and prior to placing that equipment in service, such as; grab hooks, spreader bars, extension devises, slings and wire ropes, etc.

6. **Structural steel erection.**
    6.1. Beams and columns
        6.1.1. During the final placing of solid web structural members, the load shall not be released from the hoisting line until the members are secured with at least two bolts per connection, of the same size and strength as shown in the erection drawings, drawn up wrench-tight as specified by the project structural engineer of record.
        6.1.2. A competent person shall determine if more than two bolts are necessary to ensure the stability of cantilevered members; if additional bolts are needed, they shall be installed.
        6.1.3. Solid web structural members used as diagonal bracing shall be secured by at least one bolt per connection drawn up wrench-tight or the equivalent as specified by the project structural engineer of record.
    6.2. Hoisting and rigging
        6.2.1. The steel erection Trade Partner shall ensure that all the following provisions are complied with as applicable to hoisting and rigging.
        6.2.2. A competent person shall visually inspect cranes being used in steel erection activities prior to each shift; the inspection shall include observation for

deficiencies during operation. At a minimum this inspection shall include the following:

6.2.2.1. All control mechanisms for maladjustments

6.2.2.2. Control and drive mechanism for excessive wear of components and contamination by lubricants, water, or other foreign matter.

6.2.2.3. Safety devices, including but not limited to boom angle indicators, boom stops, boom kick out devices, anti-two block devices and load moment indicators where required.

6.2.2.4. Air, hydraulic and other pressurized lines for deterioration or leakage, particularly those that flex in normal operation.

6.2.2.5. Hooks and latches for deformation, chemical damage, cracks, or wear.

6.2.2.6. Wire rope reeving for compliance with hoisting equipment manufacturer's specifications.

6.2.2.7. Electrical apparatus for malfunctioning, signs of excessive deterioration, dirt, or moisture accumulation.

6.2.2.8. Hydraulic system for proper fluid level

6.2.2.9. Ground conditions around the hoisting equipment for proper support, including ground settling under and around outriggers, ground water accumulation, or similar conditions.

6.2.2.10. The hoisting equipment for level position; and the hoisting equipment for level position after each move and setup.

6.2.3. If any deficiency is identified, the competent person shall make an immediate determination as to whether the deficiency constitutes a hazard.

6.2.4. If the deficiency is determined to constitute a hazard, the hoisting equipment shall be removed from service until the deficiency has been corrected.

6.2.5. The operator shall be responsible for those operations under the operator's direct control. Whenever there is any doubt as to safety, the operator shall have the authority to stop and refuse to handle loads until safety has been assured.

6.2.6. A qualified rigger (a rigger who is also a qualified person) shall inspect the rigging prior to each lift.

6.2.7. The headache ball, hook or load shall not be used to transport personnel except unless all the OSHA standards are being met to hoist employees on a personnel platform and written approval has been granted by executive management (See Cranes).

6.2.8. Safety latches on hooks shall not be deactivated or made inoperable.

6.3. Working under loads.

6.3.1. Routes for suspended loads shall be pre-planned to ensure that no employee is required to work directly below a suspended load except for:

6.3.1.1. Employees engaged in the initial connection of the steel; or

6.3.1.2. Employees necessary for the hooking or unhooking of the load.

6.3.2. When working under suspended loads, the following criteria shall be met:

6.3.2.1. Materials being hoisted shall be rigged to prevent unintentional displacement;

6.3.2.2. Hooks with self-closing safety latches or their equivalent shall be used to prevent components from slipping out of the hook; and

6.3.2.3. A qualified rigger shall rig all loads.

6.4. Structural steel assembly

6.4.1. Structural stability shall be maintained during the erection process.

6.4.2. The following additional requirements shall apply for multi-story structures:

6.4.2.1. The permanent floors shall be installed as the erection of structural members progress and there shall be not more than eight stories between the erection floor and the uppermost permanent floor, except where the structural integrity is maintained because of the design.

6.4.2.2. At no time shall there be more than four floors or 48 feet, whichever is less, of unfinished bolting or welding above the foundation or uppermost permanently secured floor, except where the structural integrity is maintained because of the design.

6.4.2.3. A fully planked or decked floor or nets shall be maintained within two stories or 30 feet, whichever is less, directly under any erection work being performed.

6.4.3. Plumbing-up

6.4.3.1. The steel erector shall evaluate to determine whether guying or bracing is needed; if guying or bracing is needed and if determined, it shall be installed.

6.4.3.2. When deemed necessary by a competent person, plumbing-up equipment shall be installed in conjunction with the steel erection process to ensure the stability of the structure.

6.4.3.3. When used, plumbing-up equipment shall be in place and properly installed before the structure is loaded with construction material such as loads of joists, bundles of decking or bundles of bridging.

6.4.3.4. Plumbing-up equipment shall be removed only with the approval of a competent person.

6.5. Hoisting, landing and placing of metal decking bundles.

6.5.1. Bundle packaging and strapping shall not be used for hoisting unless specifically designed for that purpose.

6.5.2. If loose items such as dunnage, flashing, or other materials are placed on the top of metal decking bundles to be hoisted, such items shall be secured to the bundles.

6.5.3. Bundles of metal decking on joists shall be landed in accordance with OSHA standards.

6.5.4. Metal decking bundles shall be landed on framing members so that enough support is provided to allow removal of banding without dislodging the bundles from the supports.

6.5.5. At the end of the shift or when environmental or jobsite conditions require, metal decking shall be secured against displacement.

6.6. Roof and floor holes and opening

6.6.1. Metal decking at roof and floor holes and openings shall be installed as follows:

6.6.1.1. Metal deck openings shall have structural members turned down to allow continuous deck installation except where not allowed by structural design constraints or constructability.

6.6.1.2. Roof and floor holes and openings shall be decked over. Where large size, configuration or other structural design does not allow openings to be decked over (such as elevator shafts, stair wells, etc.) employees shall be protected in accordance with OSHA standards.

6.6.1.3. Metal decking holes and openings shall not be cut until immediately prior to being permanently filled with the equipment or structure needed or intended to fulfill its specific use and which meets the strength requirements of this section policy or shall be immediately covered.

6.7. Covering roof and floor openings

6.7.1. Covers for roof and floor openings shall be capable of supporting, without failure, twice the weight of the employees, equipment and materials that may be imposed on the cover at any one time.

6.7.2. All covers shall be secured when installed to prevent accidental displacement by the wind, equipment, or employees.

6.7.3. All floor holes greater than 12” x 12” shall have a Whiting-Turner floor hole cover sign in place. Smaller holes shall be labeled “hole” in accordance with the OSHA standard.

6.7.4. Particle board, medium density fiber board (MDF) or similar material is prohibited from being used as floor hole covers on Whiting-Turner projects.

6.7.5. Smoke dome or skylight fixtures that have been installed are not considered covers for the purpose of this section unless they meet the strength requirements.

6.8. Decking gaps around columns

6.8.1. Wire mesh, exterior plywood, or equivalent, shall be installed around columns where planks or metal decking do not fit tightly.

6.8.2. The materials used shall be of sufficient strength to provide fall protection for personnel and prevent objects from falling through.

6.9. Installation of metal decking

6.9.1. Metal decking shall be laid tightly and immediately secured upon placement to prevent accidental movement or displacement.

6.9.2. During initial placement, metal decking panels shall be placed to ensure full support by structural members.

6.10. Erection of steel joists.

6.10.1. Both sides of the seat of one end of each steel joist that requires bridging under OSHA requirements shall meet those requirements set forth in Tables A and B shall be attached to the support structure before hoisting cables are released.

6.10.2. For joists over 60 feet, both ends of the joist shall be attached as specified in the OSHA regulations before the hoisting cables are released.

6.10.3. Connections shall be drawn up, wrench tight with 2 bolts minimum before releasing the load.

6.10.4. On steel joists that do not require erection bridging under Tables A and B, only one employee shall not be allowed on the joist until all bridging is installed and anchored.

6.10.5. Employees shall not be allowed on steel joists where the span of the steel joist is equal to or greater than the span shown in Tables A and B of subpart R.

6.10.6. When permanent bridging terminus points cannot be used during erection, additional temporary bridging terminus points are required to provide stability.

6.10.7. Bar joists unless specified by the engineer of record shall not be used as fall protection anchorage points.

7. **Training.**

7.1. The requirements in this section are mandatory and the Trade Partner shall provide certification for each employee who has received training for performing steel erection operations based upon OSHA standards.

8. **Competent person.**

8.1. Each Trade Partner shall list the qualified and competent persons based on OSHA definition and the steel erector shall have a competent person knowledgeable of current OSHA standards with an OSHA 30-hour certification. This person shall remain on site during the completion of their work activities. If the Trade Partner uses lower tier steel erectors to erect steel onsite the Trade Partner with whom Whiting-Turner holds the contract shall provide an onsite competent person to direct their lower tier Trade Partner.

9. **Fall hazard training.**

9.1. The Trade Partner shall provide a training program for all employees exposed to fall hazards. The program shall include but not limited to the following training and instructions in the following areas:

9.1.1. The recognition and identification of fall hazards in the work area;

9.1.2. The use and operation of guardrail systems (including perimeter safety cable systems), personal fall arrest systems, positioning device systems, fall restraint systems, safety net systems and other protection to be used;

9.1.3. The correct procedures for erecting, maintaining, disassembling and inspecting the fall protection systems to be used;

9.1.4. The procedures to be followed to prevent falls to lower levels and through or into holes and openings in walking/working surfaces and walls; and

9.1.5. The fall protection requirements of this subpart and those found in section E.30 of this manual, Fall Protection and Prevention.

10. **Additional training requirements.**

10.1. In addition to the training requirements above in this section, the Trade Partner shall provide special training to employees engaged in the following activities.

10.1.1. Multiple Lift Rigging Procedure - The Trade Partner shall ensure that each employee who performs multiple lift rigging has been provided training in the following areas:

10.1.1.1. The nature of the hazards associated with multiple lifts; and

10.1.1.2. The proper procedures and equipment to perform multiple lifts.

10.1.1.3. Multiple lifts shall utilize a multiple lift rigging assembly and meet all other criteria as listed in OSHA standards regarding multiple lifts.

10.1.2. Connector Procedures - The Trade Partner shall ensure that each connector has been provided training in the following areas:

10.1.2.1. The nature of the hazards associated with connecting; and

10.1.2.2. The establishment, access, proper connecting techniques and work practices.

E.27 Demolition

Introduction

Each Trade Partner performing demolition on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart T – Demolition, in addition to the following guidelines.

Procedures

1. **Preparatory operations.**
   1.1. Prior to initiating demolition activities, a competent person shall make an engineering survey of the building to determine the condition of the structure and identify areas subject to unplanned collapse. A copy of this inspection shall remain on site.
   1.2. All utilities shall be shut off, capped, or locked out of service beyond the building line before demolition work is initiated.
   1.3. Prior to the start of work, a hazard assessment shall be performed to identify any hazardous chemicals, gases, explosives, flammable materials, or similarly dangerous substances that may have been used on the property.
   1.4. Where employees are exposed to fall hazards, guardrail and personal fall arrest systems shall be used. All hole covers shall be identified and secured against accidental displacement.
   1.5. Any openings cut in a floor for the disposal of materials can be no larger than 25% of the aggregate of the total floor area, unless the lateral supports of the removed flooring remain in place.
   1.6. Trade Partner shall verify that all local ordinances and permitting issues have been addressed as they relate to demolition.
   1.7. Safety data sheets for known demolished material shall be provided by the demolition Trade Partner.
   1.8. Task lighting, which meets or exceeds the requirements of the standard, shall be provided by the demolition Trade Partner.

2. **Stairs, passageways and ladders.**
   2.1. Access to a structure being demolished shall be restricted; tradespersons shall be rerouted to designated stairways, passageways and ladders. All other points of access shall remain closed.
   2.2. All designated access points shall clearly be marked, inspected periodically and maintained in a clean, safe condition.

3. **Chutes**.
   3.1. No material may be dropped to a point outside the building unless that area is delineated with a protective barricade and the distance to any point does not exceed 20 feet.

3.2. All chutes shall be entirely enclosed except for openings at or slightly above the floor level for the insertion of materials.
3.3. At all stories below the top floor chute openings shall remain closed.
3.4. A substantial gate shall be installed in each chute at or near the discharge end. A competent person shall be assigned to control the operation of the gate and the backing and loading of trucks.
3.5. Chutes shall be designed and constructed of such strength as to eliminate failure due to the impact of material and debris loaded into them.
3.6. Fall protection may be required where chute opening may create a fall exposure

4. **Removal of walls, masonry sections and chimneys.**
   4.1. Masonry walls, including sections of walls, will not be permitted to fall onto the floor of the building under demolition unless an engineer has determined that the floor can withstand the imposed load.
   4.2. No wall section, more than one story in height, will be permitted to stand alone without lateral bracing unless it was designed to stand alone.
   4.3. Structural or load-supporting members of any floor will not be cut or removed until all stories above such a floor have been demolished or removed.

5. **Removal of walls, floors and material with equipment.**
   5.1. Mechanical equipment will not be used on floors unless the floors are of sufficient strength to safely support the equipment.
   5.2. Mechanical equipment will only be used for its intended purpose according to the manufacturer's recommendations.
   5.3. Curbs or stop logs shall be installed and maintained by the demo Trade Partner where a possibility exists for equipment going over the edge.

6. **Removal of steel construction.**
   6.1. Steel construction will be dismantled column length by column length, tier by tier.
   6.2. When floors have been removed planking—18" wide by 2" thick—shall be used by employees engaged in razing the steel framing.

7. **Mechanical demolition.**
   7.1. No employees will be permitted in an area where "ball" or "clam" work is being performed. Only employees necessary for the performance of the operation may be permitted in this area.
   7.2. The area shall be identified with warning barricades and signs.
   7.3. During this operation continuous observations, by the competent person, shall be made to identify potential areas of failure.

E.28 Stairways and Ladders

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart X – Stairways and Ladders, in addition to the following guidelines.

Procedures

1. **General Requirements.**
   - 1.1. Ladders shall be utilized as the last resort for performing work at heights.
   - 1.2. All ladders used on Whiting-Turner projects shall meet OSHA/ANSI specifications.
   - 1.3. Workers shall be trained on proper use of ladders and will be required to select ladders that are capable of safely supporting their weight and the weight of their tools.
   - 1.4. Ladders shall be inspected by the competent person daily and prior to use by the persons using the ladder. Upon inspection, defective ladders shall be tagged and removed from service.
   - 1.5. Additionally, at a minimum, only Type 1A Extra Heavy Duty (300 lb. limit) ladders may be used on Whiting-Turner projects.
   - 1.6. Two points of access/egress shall be established and maintained as soon as practical to each level with structurally complete floors. One means of access shall be a stair tower or stairway system.
   - 1.7. A stairway or ladder shall be provided at all personnel points of access where there is a break in elevation of 19” or more.
   - 1.8. A double-cleated ladder or two or more separate ladders shall be used when ladders are the only means of egress from a working area with 25 or more employees.
   - 1.9. All aluminum and commercially manufactured wooden ladders shall not be used on Whiting-Turner projects.
   - 1.10. Upon inspection, all defective ladders are tagged and/or removed from service.

2. **Stairways.**
   - 2.1. When doors from an office or storage trailer open directly onto a stairway, a platform shall be provided and the swing of the door shall allow an additional 20” to prevent the door from striking an employee.
   - 2.2. Workers are not allowed to use metal pan stairs unless they have been fitted with wooden filler blocks or poured with concrete.
   - 2.3. Incomplete/unsafe stairways shall be physically blocked off to prevent unauthorized use; barricade tape would be an insufficient restrictive method.
   - 2.4. Stairways with four or more risers or rising more than 30”, whichever is less, shall have a handrail and a stair rail along each unprotected side or edge.

3. **Ladders**.
    3.1. All hazards shall be evaluated by a competent person when employees are engaged in work from a ladder 6’ or more above an adjacent surface.
    3.2. Fall protection shall be considered by the competent person if employees work from a ladder 6’ or more above a lower level and are exposed to a fall.
    3.3. Employees working on ladders shall be at least one and a half times the height of the ladder away from any perimeter, shaft, stairway and opening where the fall distance exceeds 6’ without employing additional means of fall protection.
    3.4. Trade Partners shall provide ladders with duty ratings that meet the needs of their employees.
    3.5. Workers are required to select ladders that are capable of safely supporting their weight and the weight of their tools.
    3.6. When employees ascend or descend a ladder, they shall maintain three-points of contact (e.g. two hands and a foot or two feet and a hand).
    3.7. Pull ropes should be placed at all access ladders so employees can safely lift tools or equipment to upper levels.
    3.8. Stepladders shall be opened fully and set level when in use, unless manufactured to be used in the closed position, in which the manufacturer’s rules for setup and use shall be followed.
    3.9. When ladders are used to access upper landings, the side rails shall extend at least 3 feet above the landing and be secured from displacement at the bottom and top.
    3.10. When ladders are used to access upper landings, a guardrail system/corral shall be placed around the opening through which the ladder is protruding.
    3.11. All ladders shall be used for the purpose for which they were designed.
    3.12. The base of an extension and or straight ladder is to be placed 1 foot horizontal from the face of the surface for every 4 feet vertical.

4. **Job made ladder requirements.**
    4.1. Job made wooden ladders shall meet ANSI ASC A.14.4-2009 specifications.

5. **Training.**
    5.1. Each worker involved in stair and ladder use shall be trained by their company’s competent person in the recognition and avoidance of stair and ladder hazards.

E.29 Scaffolds

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart L – Scaffolds, in addition to the following guidelines.

Procedures

1. **General Requirements.**
    - 1.1. Scaffolds are required to be designed, erected and inspected by a competent person in accordance with OSHA 1926 – Subpart L Scaffolding Standards.
    - 1.2. The Whiting-Turner project team may contract for scaffold erection/dismantling services directly. In addition, each Trade Partner whose employees will need to access the scaffold system for work shall have a competent person present to inspect and sign off on the scaffold prior to the start of work each day.
    - 1.3. Whiting-Turner may contract directly for stair towers. A competent person shall inspect and sign off on the stair towers each day prior to the start of work.
    - 1.4. Trade Partners erecting scaffolding shall submit a scaffold erection plan/drawing to the Whiting-Turner team showing a section view of necessary components including ties/braces, platforms and means of access.
    - 1.5. All Supported Scaffold types, including Systems Scaffold, over 125 feet in height shall be required to submit Professional Engineer stamped drawings to Whiting-Turner prior to erection.
    - 1.6. Employees erecting or dismantling a scaffold are required to utilize appropriate fall protection at heights six (6) feet or above unless proven to be infeasible or more hazardous as determined by their company's competent person.
    - 1.7. Base plates/screw jacks and mudsills or other adequate firm foundation are required to be used as part of a complete supported scaffold system.
    - 1.8. All scaffolds, including carpenters' bracket scaffolds, over six (6) feet in height shall have guardrails on all open sides, unless otherwise specified by the manufacturer. If guardrails cannot be used on a walking/working platform, Trade Partners are required to use another means to protect employees from a fall.
    - 1.9. Cross-braces are not considered to be an adequate guardrail (fall protection) system and shall not be used as a top or mid rail on Whiting-Turner projects.
    - 1.10. Trade Partners shall comply with Whiting-Turner's scaffold tagging policy. The scaffold tag system shall be color coded and visible. The competent person shall inspect the scaffolding system before each work shift. The competent person shall sign and date the scaffold tag.
        - 1.10.1. Green tags are reserved for complete systems
        - 1.10.2. Red tags are reserved for erection/dismantling activities and for scaffolds with deficiencies in the system.

1.10.3. Yellow tags are reserved for systems that require the use of both PFAS and guardrail systems for incomplete scaffold systems or platforms.

1.11. Scaffold tags are required at each entry point of the scaffold system.

1.12. Each Trade Partner using the scaffold system shall have its competent person place a tag at each point of entry for their employees.

1.12.1. In the case of stair towers, the party's competent person responsible shall put the tags in place and inspect the stair towers daily.

1.13. Fall protection is required when walking/working at six (6) feet or above.

1.14. The scaffold erection competent person will consider/determine if a horizontal, diagonal brace shall be in place to prevent the scaffold from wracking.

1.15. Scaffold working platforms require a fully planked deck (5 planks). Walking platforms for use during the erection/dismantling of a scaffold are required to be at least two (2) planks wide.

1.16. Walking/working surfaces are required to be made of scaffold grade planks. Planking is to be inspected by the competent person erecting the scaffold prior to installation during the scaffold erection. Damaged planks found during the initial inspection and all concurrent inspections are to be immediately removed from service and replaced.

1.17. Walking/working surfaces are required to comply with the following.

1.17.1. Planking that does not extend at least six (6) inches past the bearing surface are required to be secured in such a manner to prevent accidental displacement.

1.17.2. Planking that is less than ten (10) feet in length is not to extend more than twelve (12) inches past the bearing surface and planking greater than ten (10) feet in length is not to extend past the bearing surface more than eighteen (18) inches without adequate barricades to prevent employees from stepping on the cantilevered end.

1.18. Trade Partners are to provide a proper and safe access/egress for employees working on a scaffold. Only approved built-in scaffold stairs or ladders are to be used for access and egress while working on scaffolds.

1.19. Trade Partners are required to load materials (brick, block, stone, etc.) as close to the load bearing surface as possible. Loading in the center of the scaffold bay is not permitted. Scaffolds shall be designed to support 4x the intended load; load capacity ratings shall not be exceeded.

1.20. Trade Partners performing work on scaffolding are required to complete a daily scaffold inspection prior to allowing their employees on the scaffold. The above-mentioned inspection is required to be completed by an employee competent in scaffold safety.

1.21. Trade Partners are responsible for training their employees and shall submit the training documentation to Whiting-Turner upon request.

1.22. Rolling/mobile scaffold systems 30" or less in width (baker-type scaffolds) are required to have guardrails on all sides unless otherwise specified by manufacturer or local jurisdiction having authority; all wheels are required to be locked when an employee is on the working platform. Outriggers are required when the height of scaffold exceeds

3x the base width or according to the manufacturer's recommendation, whichever is more stringent. Self-propelling/skateboarding is not permitted.

1.22.1. Rolling/mobile scaffold systems shall be accessed through swing gates. Removing or climbing over guardrails is not permitted. When guardrails are used, swing gates shall be in place. Removing or climbing over guardrails is not permitted.

1.23. Suspended scaffolds shall be designed, constructed, operated, inspected, tested and maintained as specified in the operating manual for the device; all operating instructions and design specification shall be submitted to Whiting-Turner prior to installation and a copy shall remain on the jobsite.

2. **Additional Information, Requirements and Considerations**.

2.1. All employees required to access scaffold shall be provided training on safe use of scaffolds. The training shall include the following areas, as applicable:

2.1.1. The nature of any electrical hazards, fall hazards and falling object hazards in the work area;

2.1.2. The correct procedures for dealing with electrical hazards and for erecting, maintaining and disassembling the fall protection systems and falling object protection systems being used;

2.1.3. The proper use of the scaffold and the proper handling of materials on the scaffold;

2.1.4. The maximum intended load and the load-carrying capacities of the scaffolds used; and

2.1.5. Any other pertinent requirements of subpart L.

2.2. Retraining is required when Whiting-Turner personnel, or the Trade Partner employer has reason to believe that an employee lacks the skill or understanding needed for safe work involving the erection, use or dismantling of scaffolds, the employer shall retrain each such employee so that the requisite proficiency is regained. Retraining is required in at least the following situations:

2.2.1. Where changes at the worksite present a hazard about which an employee has not been previously trained; or

2.2.2. Where changes in the types of scaffolds, fall protection, falling object protection, or other equipment present a hazard about which an employee has not been previously trained; or

2.2.3. Where inadequacies in an affected employee's work involving scaffolds indicate that the employee has not retained the requisite proficiency.

2.3. Load Rating.

2.3.1. The design working load of ladder stands shall be calculated on the basis of one or more 200-pound persons together with 50 pounds of equipment each.

2.3.2. The design load of all scaffolds shall be calculated on the basis of:

2.3.2.1. Light -Designed and constructed to carry a working load of 25 pounds per square foot.

2.3.2.2. Medium -Designed and constructed to carry a working load of 50 pounds per square foot.

2.3.3. Heavy -Designed and constructed to carry a working load of 75 pounds per square foot.

2.4. Work Levels.

2.4.1. The maximum work level height shall not exceed 3 times the least base dimension below the platform. Where the basic mobile unit does not meet this requirement, outrigger frames shall be employed to achieve this least base dimension, or provisions shall be made to guy or brace the unit against tipping.

2.4.2. The minimum platform width for any work level shall not be less than 20 inches for mobile scaffolds (towers). Ladder stands shall have a minimum step width of 16 inches.

2.4.3. The supporting structure for the work level shall be rigidly braced, using cross bracing or diagonal bracing with rigid platforms at each work level.

2.4.4. The steps of ladder stands shall be slip-resistant.

2.4.5. The work level platform of scaffolds (towers) shall be the full width of the scaffold, except for necessary openings. Work platforms shall be secured in place.

2.4.6. All scaffold work levels 6 feet or higher above the ground or floor shall have a toeboard at locations where persons are required to work or pass under the scaffold.

2.4.7. All scaffold work levels 30 inches or higher above the ground or floor shall have guardrail protection.

2.4.8. A climbing ladder or stairway shall be provided for proper access and egress and shall be affixed or built into the scaffold and so located that its use will not tend to tip the scaffold. A landing platform shall be provided at intervals not to exceed 30 feet.

2.5. Wheels or Casters.

2.5.1. Wheels or casters shall be properly designed for strength and dimensions to support 4 times the design working load.

2.5.2. All scaffold wheels, casters and swivels shall be provided with a positive locking device, or other effective means to prevent movement of the scaffold.

2.5.3. Ladder stands shall have at least 2 locking casters or other means of locking the unit in position. If only 2 casters are used, they shall be of the directional type and if 4 casters are used, at least 2 of the 4 shall be of the swivel type.

2.5.4. Locking devices shall be kept in the locked position when workers are climbing or working on scaffolds and ladder stands.

2.5.5. Where leveling of the elevated work platform is required, screw jacks or other similar means for adjusting the height shall be provided in the base section of each mobile unit. The screw jack shall extend into its leg tube at least 1/3 its length, but in no case shall the exposed portion of the screw jack exceed 12 inches.

E.30 Fall Prevention and Protection

Introduction

Falls continue to be the leading cause of fatalities in the construction industry; in consequence, Whiting-Turner has zero tolerance for failure to comply with all provisions of our fall protection and prevention policy. Any tradespersons found violating said policy shall be retrained. In addition, at the discretion of Whiting-Turner personnel, violators may be permanently removed from the project. Any Trade Partner with any fall exposure shall submit to Whiting-Turner the resume of their competent person trained in fall protection techniques. All Whiting-Turner employees and all Trade Partner employees working on a Whiting-Turner project shall comply with 29 CFR 1926, Construction Industry Regulations, Subpart M - Fall Protection, in addition to the following guidelines.

Procedures

1. General Requirements.

- 1.1. Each Trade Partner, with employees exposed to a fall 6' or greater to a lower level shall ensure that effective fall protection measures and rescue procedures are addressed in their company activity hazard analysis (AHA) prior to beginning work on site. This is to include the name and qualifications of the designated competent person.
  - 1.1.1. Exception: The provisions of this part do not apply when inspectors and competent persons are briefly inspecting, investigating, or assessing workplace conditions.
  - 1.1.2. Exception: Fall protection policy includes any surface, except ladders, vehicles, or trailers.
  - 1.1.3. Under the terms of 1926.500, fall protection is not required for employees who are on vehicles and trailers when the employee shall be on the vehicle or trailer to perform his or her duties. For example, if the employee shall climb on the tractor trailer rigs to connect the rigging for loading and unloading materials and equipment, the employer will utilize the sound judgement of their competent person for Fall Protection activities to determine how to best proceed.
- 1.2. If at any time during work at heights, it is observed that the means as stated in the Trade Partner's AHA for protecting the workers from falls are perceived as ineffective, work will be suspended until the Trade Partner's competent person assesses the situation and makes necessary corrections. If corrections determined by the competent person involve modified work practices listed in the Trade Partner's AHA, then these changes shall be completed and any necessary retraining shall be provided to affected workers prior to allowing the continuation of the Trade Partner's work at heights.
- 1.3. At no time shall a safety monitor system be used as a sole means of fall protection.
- 1.4. The use of controlled access zones (CAZ) and controlled decking zones (CDZ) as a means of fall protection are prohibited on Whiting-Turner projects.

1.5. A Personal Fall Arrest System (PFAS) [comprised of a full body harness, double lanyards, anchorage point and anchorage connector], a personal fall restraint system (PFRS) [comprised of a full body harness, lanyard, anchorage point and anchorage connector], a guardrail, or safety net system shall be in place to protect all trade persons from exposure to falls working at or above 6 feet. All fall protection equipment shall meet industry or regulatory standards.
1.6. Employees shall be protected from falling objects by the installation of toe boards, barricades, or canopy structures.
1.7. Employees working on ladders shall be at least one and a half times the height of the ladder away from any perimeter, shaft, stairway and opening where the fall distance exceeds 6'. If that distance is not feasible, a conventional fall protection method shall be employed.
1.8. Stilts are only permitted in broom swept areas, where there is no change in elevation.
1.9. Non-roofing activities – Trade Partners other than roofers who shall perform work on roofs shall develop a written/drawn plan showing the roof area where work is planned to take place to protect their workers from potential exposure to falls.

2. Floor and wall holes and openings.

2.1. The Whiting-Turner project team shall ensure that hole and perimeter protection inspections are conducted and documented twice a day.
2.2. Prior to creating a hole or opening in any elevated work surfaces, Trade Partners shall submit an elevated work surface modification permit.
2.3. All floor and roof openings into which persons can accidentally walk or fall through shall be guarded by a physical barrier or covered.
2.4. All floor and roof holes through which equipment, materials, or debris can fall shall be covered.
2.5. All floor holes greater than 12" x 12" shall have a Whiting-Turner floor hole cover sign in place. Smaller holes shall be labeled "hole" in accordance with the OSHA standard.
2.6. Coverings for floor and roof openings shall be of sufficient strength to support two times the maximum intended load that may be imposed and shall be secured in place to prevent accidental removal or displacement.
2.7. Conduits, trenches and manhole covers and their supports, when exposed to vehicles or equipment, shall be designed to carry a truck rear axle load of two times the maximum anticipated load.
2.8. Particle board, medium density fiber board (MDF) or similar material is prohibited from being used as a floor hole covers on Whiting-Turner projects.
2.9. Wall openings 18" or greater in width from which there is a drop of 6' feet or greater and the bottom of the opening is less than 39" above the working surface, shall be guarded with a top rail or a top rail and intermediate rail or a standard guardrail. A toe board or enclosing screen shall be provided where the bottom of the wall opening, regardless of width, is less than 3 ½ inches above the working surface.

2.10. A hinged floor-opening cover shall guard every hatchway and chute floor opening. Or opening shall be barricaded with railings to leave only one exposed side. The exposed side shall be provided either with a swinging gate or so offset that a person cannot walk into the opening.

2.11. Wall opening protection shall meet one of the following requirements:

2.11.1. Barriers of such construction and mounting that, when in place at the opening, the barrier can withstand a load of at least 200 pounds applied in any direction (except upward) with a minimum of deflection at any point on the top rail or corresponding member.

2.11.2. An extension platform outside a wall opening onto which materials can be hoisted for handling shall have a standard railing that meets handrail standards. However, the inside of an extension platform may have removable railings to facilitate handling materials. Workers removing guardrails shall utilize personal fall prevention equipment to tie off while working on or near platforms that are not 100% protected by guardrail.

3. Fall protection systems.

3.1. Guardrail Systems

3.1.1. The top rail height of a guardrail system shall be 42”, + or – 3”. Midrail heights shall be half of that distance.

3.1.2. Perimeter cable shall not be less than 3/8” steel cable.

3.1.3. Corner uprights shall be braced so that the required tension may be maintained.

3.1.4. The cable shall be terminated with three U-bolt wire rope clips that maintain an efficiency rating of at least 80% of the wire rope’s breaking strength as proven through product documentation (e.g. Crosby clips).

3.1.5. The U-bolt clips shall have the U-bolt section on the dead or short end of the rope and the saddle on the live or running end of the rope.

3.1.6. The use of any combination of hook turnbuckles as part of the perimeter cable system is prohibited.

3.1.7. A Personal Fall Arrest System (PFAS) shall not be attached to a guardrail system unless a registered professional engineer designs the system to accommodate the PFAS; documentation of that design shall be maintained on site.

3.1.8. Guardrail systems shall be able to withstand a force of 200 lbs. in all directions, without failure and be smooth surfaced to prevent hand injuries.

3.1.9. The Trade Partner installing the perimeter cable guardrail system shall submit a design with details on how the system will be installed and maintained.

3.1.10. All guardrail systems [with the exception of scaffolds systems or where it can be proven to create a greater hazard] shall be equipped with orange perimeter screening or mesh to prevent the ability to breach the system by climbing through rails. The installation of the screening shall be compliant with Whiting-Turner’s orange perimeter screening guidelines. The orange perimeter screening policy satisfies the requirement of flagging a wire rope guardrail system every 6' for high visibility.

3.2. Personal Fall Arrest Systems and Fall Restraint Systems.

3.2.1. A PFAS shall be used when working from suspended scaffolds, when breaching any guardrail systems and when working near unprotected floor openings and perimeter edges.

3.2.2. A fall restraint system shall be employed when working from an articulating boom man lift and shall also be employed when working from a scissor lift. A competent person shall ensure that fall-distance calculations have been evaluated in each circumstance where a PFAS is used.

3.2.3. A competent person shall ensure that fall-distance calculations have been evaluated in each circumstance where a PFAS is used.

3.2.4. A PFAS is not required when climbing up or down a ladder. Fall protection shall be considered by the competent person if employees work from a ladder 6' or more above a lower level and are exposed to a fall.

3.2.5. Employees shall use positive fall protection devices when working in proximity to any leading edge.

3.2.6. Retractable lanyards shall incorporate either a 3/16" steel wire cable or a nylon strap with a minimum width of 1".

3.2.7. All anchorage points utilized in a personal fall arrest system shall be capable of supporting a load of no less than 5000 lbs.

3.2.8. Steel erectors and metal decking installers shall utilize 100% fall protection devices at all times when working at 6' or above.

3.2.9. Horizontal lifelines shall be designed by a registered engineer and installed under the supervision of a qualified person. A safety factor of two shall be maintained.

3.2.10. Adequate fall protection devices shall be provided, installed and used at all loading platforms by the Trade Partner wishing to remove existing perimeter protection prior to its removal.

3.2.11. Rescue procedures shall be provided in writing when using a personal fall arrest system.

3.3. Training Requirements.

3.3.1. Each employee exposed to a fall hazard shall be trained by a competent person in the recognition and avoidance of such a hazard. Proof of training shall be made available to Whiting-Turner upon request.

3.3.2. Specific training courses include, but is not limited to the following:

3.3.2.1. The type of fall exposures expected

3.3.2.2. The correct procedures for erecting, maintaining, dismantling and inspecting of any fall protection system used by the employee

3.3.3. When employee lacks the understanding and demonstrative skill required for the proper application of fall protection systems, the employer shall provide retraining of such employee(s).

3.3.4. Retraining documentation—to include instructor's name and qualifications, training literature and sign-in sheet—shall be submitted to Whiting-Turner on company letterhead.

E.31 Mobile Elevated Work Platforms

Introduction

Each Trade Partner using an aerial lift on a Whiting-Turner project shall comply with 29 CFR §1926.453, in addition to the following guidelines.

Procedures

1. **General Requirements.**
   - 1.1. Only authorized and trained individuals may operate aerial lifts. Companies will make verification of training available to Whiting-Turner upon request.
   - 1.2. Aerial lifts shall be inspected by the trained operator each day prior to use, in accordance with the manufacturer's requirements.
   - 1.3. All aerial and scissor lifts shall be tagged according to Whiting-Turner's Scaffold tagging policy.
   - 1.4. Fall protection requirements for aerial lift use shall comply with the manufacturer's recommendations; load limits shall not be exceeded. The minimum clearance between electrical lines and any part of the equipment shall be at least 20 feet. Please refer to power line safety requirements.
   - 1.5. Worker shall use personal fall arrest systems (PFAS) or personal fall restraint systems (PFRS) whichever the manufacturer requires—when working from articulating boom platforms.
   - 1.6. Workers shall keep both feet on the floor of the basket; use of guardrails or toe boards to gain additional height is prohibited on Whiting-Turner projects. Persons observed committing this unsafe act will be removed from site for a minimum of 3 days; their employer shall retrain them prior to their return to the site.
   - 1.7. Where aerial and scissor lifts are used on concrete slabs, any floor depressions or grade changes are required to be barricaded with a fixed barricade to restrict travel onto that area.
   - 1.8. The area(s) below the basket or platform of aerial lifts shall be cordoned off using reinforced danger tape—or something of equivalent or greater tensile strength—and by using signage to identify the overhead hazard if/when the possibility of falling objects exist. Spotters may be used in lieu of danger tape; they shall stand clear of overhead hazards.
   - 1.9. Modifications to aerial lifts shall not be made without written approval from the manufacturer. Aerial lifts shall not be used to hoist, raise, or position material outside of the platform or basket unless manufactured to do so.
   - 1.10. Load limits shall not be exceeded. The minimum clearance between electrical lines and any part of the equipment shall be at least 20 feet. If a Trade Partner has determined that this is infeasible, a detailed plan shall be developed and submitted prior to the commencement of work.

1.11. Modifications to aerial lifts shall not be made without written approval from the manufacturer.

E.32 Cranes and Derricks

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926 Subpart CC – Cranes & Derricks in Construction in addition to the following guidelines.

Procedures

1. **General Requirements**
    - 1.1. No crane or hoist shall be placed in service on a Whiting-Turner project until an annual, third-party inspection and supplemental reports are submitted to Whiting-Turner indicating that the crane or hoist meets the manufacturer's inspection criteria.
    - 1.2. If the manufacturer's inspection criterion does not exist, a structural engineer, familiar with crane or hoist's design and dynamics, may develop or use existing inspection criteria.
    - 1.3. Written approval from the manufacturer (or a Professional Engineer) shall be obtained before modifying equipment if the changes may impact safe operation.
    - 1.4. Whiting-Turner requires all crane operators to be certified or qualified in accordance with the requirements of Subpart CC.
        - 1.4.1. Copies of all certifications shall be submitted to Whiting-Turner supervision prior to the start of work.
        - 1.4.2. A certified operator license shall be provided where state or local ordinances require additional certifications.
    - 1.5. The crane operator's manual shall be located in the cab at all times.
    - 1.6. Mobile crane movement on site shall be in accordance with manufacturer's recommendations.
    - 1.7. The swing radius of cranes shall be properly barricaded at all times while working on site.
    - 1.8. Wire rope, its attachments, fittings, sheaves and safety devices shall be inspected according to the manufacturer's recommendations. Copies of the inspections shall be submitted to Whiting-Turner.
    - 1.9. Wedge sockets and fittings shall be the proper size to match the wire rope and shall not move when holding the wire rope under load. The dead end shall be terminated according to ANSI B30.5 and shall not be attached, in any manner, to the live side of the load line.
    - 1.10. An anti-two-block or warning device is required on all cranes except those engaged in driving piles.
    - 1.11. A qualified rigger shall inspect the rigging prior to each lift.
    - 1.12. All windows in cabs shall be safety glass that produces no visible distortion that will interfere with the safe operation of the machine.

1.13. Outriggers shall be fully extended unless proven to be a hindrance to the safe operation of the crane or infeasible by the operator. In all cases, the use of intermediate outriggers may only be considered if it is permitted by the manufacturer.
1.14. A crane checklist shall be completed prior to each initial lift.

2. **Lift plans.**
    2.1. All pick & place lifts requires a formal lift plan.
        2.1.1. This plan shall be submitted to the Whiting-Turner project team for record and review prior to the commencement of the lift.
        2.1.2. Whiting-Turner personnel shall not approve lift plans; however, Whiting-Turner reserves the right to reject lift plans or require additional information if deemed necessary.
    2.2. All completed documentation with reference to the lift—including Whiting-Turner Crane Authorization Form—shall be onsite prior to beginning and for the duration of the activity.

3. **Ground conditions.**
    3.1. Unless otherwise agreed upon, the Whiting-Turner Superintendent shall ensure that appropriate ground preparations are provided—firm, drained and graded.
    3.2. The agreement of acceptable ground conditions and notification of known underground hazards shall be considered completed once the competent persons have signed the Whiting-Turner Crane Authorization Form.

4. **Controlling loads.**
    4.1. All loads to be lifted at Whiting-Turner project sites shall have a tag line attached.
        4.1.1. The competent person shall determine the size, rope materials and length of the tag line.
        4.1.2. The line shall be attached in a way that maintains control of the load to reduce the risk of caught-in/-between and struck-by hazards to employees and surroundings during any lift.
        4.1.3. Where encroachment precautions are required, taglines shall be of nonconductive material.

5. **Assembly/disassembly.**
    5.1. Manufacturer
        5.1.1. When assembling or disassembling equipment (or attachments), the Trade Partner shall comply with all applicable manufacturer exclusions and procedures for assembly or disassembly.
        5.1.2. Rated capacity limits for loads imposed on the equipment, equipment components (including rigging), lifting lugs and equipment accessories, shall not be exceeded for the equipment being assembled/disassembled during all phases of assembly/disassembly.

5.1.3. The post assembly inspection shall ensure the selection of components and the configuration of equipment that affect the safe operation of the crane are in accordance with manufacturer instructions, prohibitions, limitations and specifications.

5.2. General Requirements

5.2.1. Assembly/disassembly (A/D) shall be directed by a person who meets the criteria for both a competent person and a qualified person, or by a competent person who is assisted by one or more qualified persons.

5.2.2. The A/D director shall be knowledgeable in the A/D procedures prior to the commencement of assembly/disassembly.

5.2.3. The A/D director is responsible for ensuring crew members are knowledgeable of their tasks prior to commencing work.

5.2.4. Prior to starting work, the A/D director is responsible for ensuring crew members are knowledgeable of the hazards associated with their tasks.

5.2.5. The A/D director is responsible for ensuring crew members are knowledgeable before commencing work of the hazardous positions/locations they need to avoid.

5.2.6. The A/D director shall be responsible for managing specific hazards associated with assembly/disassembly operations. These include the following hazards:

5.2.6.1. Site and ground bearing conditions
5.2.6.2. Blocking material
5.2.6.3. Proper location of blocking
5.2.6.4. Verifying assist crane loads
5.2.6.5. Boom and jib pick points
5.2.6.6. Center of gravity
5.2.6.7. Stability upon pin removal
5.2.6.8. Snagging
5.2.6.9. Struck by counterweights
5.2.6.10. Boom hoist brake failure
5.2.6.11. Loss of backward stability
5.2.6.12. Wind speed and weather

6. **Power line safety. (Up to 350 kV)**

6.1. Equipment Operations

6.1.1. Before beginning equipment operations, the employer shall identify the work zone.

6.1.2. If during the operations any part of the equipment will get closer than 20 feet to a power line, the employer shall ensure that (a) The power line is de-energized and visibly grounded at the worksite, (b) No part of the equipment gets closer than 20 feet to the power line; or (c) A table in the standard is used to determine the minimum safe distance based on the line's voltage.

7. **Inspections**.
   7.1. Copies of all required inspections completed shall be submitted to the Whiting-Turner project team after completion of the inspection for record at the project site.
   7.2. Inspections are required during the following intervals and specifications
      7.2.1. Modified equipment – Modified equipment shall be inspected by a qualified person after the modifications have been completed, but prior to initial use.
      7.2.2. Repaired and Adjusted Equipment – Repaired and adjusted equipment shall be inspected by a qualified person after the repairs or adjustments have been completed, but prior to initial use.
      7.2.3. Post Assembly – Upon completion of assembly, a qualified person shall inspect the equipment to assure that it is configured in accordance with manufacturer equipment criteria.
      7.2.4. Post Assembly – a post assembly inspection is required for all Crawlers and Tower Cranes by a person properly trained and qualified to inspect such equipment.
      7.2.5. Each Shift – A competent person shall begin a visual inspection prior to each shift the equipment will be used, which shall be completed before or during that shift.
      7.2.6. Monthly – Each month that the equipment is in service it shall be inspected in accordance with the criteria established in the standard.
      7.2.7. Annual Comprehensive – At least every 12 months the equipment shall be inspected by a qualified person in accordance with the criteria established in the standard.
      7.2.8. Severe Service – When the equipment is used enough that there is a reasonable possibility of damage or excessive wear the employer shall stop using the equipment and have it inspected by a qualified person using the inspection criteria established in the standard.
      7.2.9. Equipment Not in Regular Use – Equipment that has been idle for three (3) months or more shall be inspected by a qualified person using the inspection criteria established in the standard.
   7.3. Any crane with a deficiency that could potentially affect the safe operation of the equipment shall be taken out of service immediately and remain so until documented repairs have been made.

8. **Safety devices.**
   8.1. The following safety devices are required on all equipment unless otherwise specified in 29 CFR 1926 Subpart CC:
      8.1.1. Crane level indicator
      8.1.2. Boom stops (except for derricks and hydraulic booms)
      8.1.3. Jib stops (if jib is attached)
      8.1.4. Locks on equipment with foot pedal brakes
      8.1.5. Integral holding device/check valves on hydraulic outrigger jacks and hydraulic stabilizer jacks
      8.1.6. Rail clamps and rail stops on equipment on rails (except for portal cranes)

8.1.7. Horn
8.1.8. Boom-tip anemometer or equivalent device

9. **Operations**.
   9.1. Trade Partners shall comply with all manufacturer procedures applicable to the operational functions of equipment, including its use with attachments.

10. **Authority to stop operation.**
   10.1. Whenever there is a concern about safety, the operator has the authority to stop and refuse to handle loads until a qualified person has determined that the safety concern has been resolved.

11. **Signals.**
   11.1. Each signal person shall:
      11.1.1. Know and understand the type(s) of signals used:
      11.1.2. Be <u>competent</u> in the application of the type of signals used
      11.1.3. Have a basic understanding of equipment operation and limitations, including the crane dynamics involved in swinging and stopping loads and boom deflection from hoisting loads:
      11.1.4. Demonstrate that he/she meets the qualification requirements through an oral or written test and through a practical test.
   11.2. A signal person shall be provided in each of the following situations:
      11.2.1. When the point of operation is not in full view of the operator
      11.2.2. When the view in the direction of travel is obstructed when the equipment is traveling
      11.2.3. When site specific safety concerns are an issue because either the operator or the person handling the load determines that it is necessary.
   11.3. Types of Signals
      11.3.1. Signals to operators shall be by hand, voice, audible, or new signals.
      11.3.2. Whiting-Turner prohibits the use of cell phones for signaling of cranes and equipment
   11.4. Suitability – The signals used and means of transmitting them shall be appropriate for the site conditions.
   11.5. During Operations – During the operations the ability to transmit signals shall be maintained. If that ability is interrupted the operator shall safely stop all operations until the ability to transmit is re-established and a proper signal is given and understood.
   11.6. Safety Problems - If the operator becomes aware of a safety problem and needs to communicate with the signal person, the operator shall safely stop all operations.
   11.7. The operations shall not resume until the operator and signal person agree that the problem has been resolved.
   11.8. Only One Signaler - Only one person may give signals to a crane or derrick at a time, except for those giving the emergency stop signal.

11.9. Safety Problems Alert - Anyone who becomes aware of a safety problem shall alert the operator or signal person by giving the stop or emergency stop signal.

11.10. Communication with Multiple Cranes/Derricks - Where a signal person is in communication with more than one crane/derrick, a system shall be used for identifying the crane/derrick each signal is as follows:

11.10.1. For each signal, prior to giving the function/direction, the signal person shall identify the crane/derrick the signal is for or shall use an equally effective method of identifying which crane/derrick for which the signal is intended.

11.10.2. Testing Signal Transmission Devices - The devices used to transmit signals shall be tested on site before beginning operations to ensure that the signal transmission is effective, clear and reliable.

11.11. Dedicated Channels - Signal transmission shall be through a dedicated channel, except:

11.11.1. Multiple cranes/derricks or more than one signal person may share a dedicated channel for coordinating operations; and

11.11.2. Where a crane is being operated on or adjacent to railroad tracks and the actions of the crane operator need to be coordinated with the movement of other equipment or trains on the same/or adjacent tracks.

11.12. Language - The operator, signal person and lift director (if applicable), shall be able to effectively communicate in the language used.

12. **Work area control.**

12.1. Swing radius hazards – Workers shall be protected from foreseeable risks of being struck by and/or pinched or crushed by the equipment's rotating superstructure.

12.2. Training – Affected tradespersons shall be trained to recognize struck by and pinch/crush hazard areas posed by the rotating superstructure.

12.3. Barriers – Control lines, warning lines, railings or similar barriers shall be erected to mark the boundaries of the hazardous areas. All barriers shall be equipped with a warning sign (such as “danger-swing/crush zone”).

12.4. Protecting tradespersons in the hazard area – Before a worker goes to a location in the hazard area that is out of the view of the operator, the worker (or someone instructed by the worker) shall ensure that the operator is informed that he/she is going to that location.

13. **Keeping clear of the load.**

13.1. Hoisting routes - Where available, hoisting routes that minimize the exposure of tradespersons to hoisted loads shall be used.

13.2. Workers in fall zone - While the operator is not moving a suspended load, no worker shall be within the fall zone except for the following:

13.2.1. Workers engaged in hooking, unhooking, or guiding a load

13.2.2. Workers engaged in the initial attachment of the load to a component or structure

13.2.3. Workers operating a concrete hopper or concrete bucket

14. **Safety criteria for tradespersons in fall zone.**
    14.1. When affected tradespersons shall be in the fall zone the following criteria shall be met:
        14.1.1. The materials being hoisted shall be rigged to prevent unintentional displacement.
        14.1.2. Hooks with self-closing latches or their equivalent shall be used.
        14.1.3. A qualified rigger shall do the rigging.

15. **Safety criteria for lifting over occupied buildings.**
    15.1. Every reasonable effort should be made to avoid lifting over an occupied building. In cases where the competent person for this operation has determined that a lift over an occupied building is necessary and other options are infeasible, the following criteria shall be met:
        15.1.1. Prior to the lift, the area of the top two floors that is within the fall zone shall be evacuated
            15.1.1.1. If the Trade Partner's qualified, competent person determines it—after a thorough hazard assessment—that evacuation is not necessary, this decision shall be submitted to Whiting-Turner be in writing. In addition, the Trade Partner's competent person shall ensure that the critical lift protocols are followed.
        15.1.2. All affected paths of access and egress shall be barricaded to prevent access to the restricted area or fall zone
        15.1.3. Signage notifying occupants of the activity and their new path of travel shall be conspicuously posted.

16. **Safety criteria for receiving a load.**
    16.1. Only tradespersons receiving the load can be within the fall zone when the load is being landed.

17. **Safety criteria for tilt up or tilt down operations.**
    17.1. During tilt up or tilt down operations the following criteria apply:
        17.1.1. No worker may be directly under the load.
        17.1.2. Only tradespersons essential to the operation can be in the fall zone but may not be directly under the load.

18. **Workers essential to the operation.**
    18.1. Workers are essential to the operation only if the following apply:
        18.1.1. It is infeasible for the worker to perform the operation from outside the fall zone and;
        18.1.2. The worker is physically guiding the load; or
        18.1.3. The worker is closely monitoring and giving instructions regarding the load's movement; or

18.1.4. The worker shall detach the load or initially attach the load to another component or structure.

19. **Operator qualification and certification.**
    19.1. Qualification or certification – The Trade Partner shall ensure that the operator is qualified or certified in accordance with the standard. Trade Partner options for getting affected operators qualified or certified follow:
        19.1.1. Certification by an accredited crane operator testing organization;
        19.1.2. Qualification by an audited employer program;
        19.1.3. Qualification by the U.S. military; or
        19.1.4. Licensing by a government entity.
    19.2. Exceptions to qualification and certification requirements - Operators of derricks, side-boom cranes, or equipment with a maximum manufacturer-rated hoisting/lifting capacity of 2,000 pounds or less are exempt from qualification and certification requirements.

20. **Signal person qualifications.**
    20.1. Qualification Requirements - The Trade Partner shall ensure that signal persons meet the following qualification requirements before giving any signals to operators:
        20.1.1. Obtain documentation from a third-party qualified evaluator showing that the signal person meets the qualification requirements established in the standard; or
        20.1.2. Obtain documentation from the Trade Partner's qualified evaluator (not a third party) showing that the signal person meets the qualification requirements established in the standard.
    20.2. Documentation Availability - The employer shall make signaler qualification documentation available at the site where the signal person is employed.
    20.3. Knowledge Requirements - Each signal person shall demonstrate the following:
        20.3.1. Knowledge and understanding of the type of signals used. If hand signals are used, the signal person shall know and understand the Standard Method for hand signals;
        20.3.2. Competence in the application of the types of signals used;
        20.3.3. Basic understanding of the equipment operation and limitations, including the crane dynamics involved in swinging and stopping loads and boom deflection from hoisting loads;
        20.3.4. Knowledge and understanding of the relevant requirements of the standard covered in the sections on Signals-General Requirements, Signals - Hand Signal Chart and Signal Person Qualifications; and
        20.3.5. That he or she meets the qualification requirements through successful completion of an oral or written test and a practical test.

21. **Qualifications of maintenance & repair employees.**

21.1. Maintenance, inspection and repair personnel are permitted to operate the equipment only where specified requirements are met as established in the standard.

22. **Hoisting personnel.**

22.1. Personnel hoisting requirements - The use of equipment to hoist personnel is prohibited unless the employer can demonstrate that other methods would be more hazardous and is able to comply with the personnel hoisting requirements that are established in the standard.

22.2. Hoisting personnel on Whiting-Turner project sites shall be done only after review of the plan and deemed feasible by Vice President and the EH&S Manager assigned to the project.

22.3. Hoisting personnel on Whiting-Turner projects shall be considered a critical lift or activity and therefore, shall meet all requirements of a critical lift before the lift may begin.

E.33 Critical Lifts

Introduction

The goal of this section is to ensure that crane lifts which meet the criteria stated below are done with understanding of the regulatory requirements as well as the stated guidelines contained herein.

The Whiting-Turner Contracting Company identifies a critical or special lift as:

- Any lift where the total weight of the load and the deductions for the equipment combined exceeds 75% of the capacity of the crane capacity chart at the specific boom length and radius of the load,
- Any lift where there will be more than one (1) crane or piece of load handling equipment attached to the load at a time.
- Any lift that involves the lifting of personnel.
- Any lift where the contents of the lift are considered hazardous to health or environment and an accidental release could result harm to either.
- Any lift where encroachments precautions are required for power lines.

Procedures – Lift Plan Development.

1. **The load.**
   1.1. Identify loads weight, center of gravity and dimensions and the sources of that information.
   1.2. Identify any components that could shift during the lift and develop a method to secure any items identified if required.
   1.3. Identify load attachment points and ensure they are suitable for the load to be handled, while maintaining load integrity.
   1.4. Identify the requirements to be met for the load's orientation and securement prior to the release of the load handling equipment and rigging.

2. **Load handling equipment.**
   2.1. Identify the load handling equipment and the anticipated configuration(s).
   2.2. Assure the load handling equipment can handle the total anticipated load, including the rigging, accessories and attachments in the intended configuration(s).
   2.3. Ensure the load handling equipment is following the requirements of the site, the manufacturer or qualified person, industry-recognized standard, local, state and federal regulations.
   2.4. Establish the process to set up, erect, or install and dismantle the load handling equipment using the information provided by the following:
      2.4.1. The manufacturer
      2.4.2. A qualified person
      2.4.3. Site specific recommendations

2.4.4. Applicable regulatory requirements

2.5. Identify all required inspections and tests on the load handling equipment that need to be performed. Additional inspections may be necessary for repetitive critical lifts.

3. **Rigging.**
   3.1. Establish rigging method that will support and secure the load and is suitable for the activity.
   3.2. Ensure that the rigging method and the equipment have the capacity to support the load in the configuration or geometry required by considering the following:
      3.2.1. Dynamic Effects
      3.2.2. Adverse environmental conditions, temperature, wind, lightning etc.
      3.2.3. Position of the center of gravity relative to rigging support points
      3.2.4. D/d ratio
   3.3. Identify the weight of the rigging, accessories and attachments, as well the sources of that information
   3.4. Establish the process to ensure that rigging equipment meets the manufacturer's specifications, regulations and industry-recognized standards (e.g. ASME B30.9, B30.20 and B30.26) and onsite requirements.
   3.5. Identify all necessary inspections and tests for the rigging equipment and that they are carried out prior to the lift
   3.6. For repetitive lifts ensure a more frequent inspection process is in place.
   3.7. Establish the process to install and disassemble the rigging equipment using the manufacturers' information and other applicable standards.
   3.8. Ensure that the rigging will be protected from damage during the activity.

4. **The travel path.**
   4.1. Identify the path of travel for the load and the load handling equipment
   4.2. Establish a process to minimize the exposure to overhead loads by way of barricades, spotters, or lookouts to ensure only authorized personnel are allowed into the fall zone
   4.3. Ensure the load and the load handling equipment has adequate clearance to prevent contact with site-specific hazards or obstructions during the activity
   4.4. Ensure a tagline of proper type and length is obtained for load control
   4.5. Ensure that the path taken if pick and carry operation is instituted that level and substantial support are established.

5. **Personnel**.
   5.1. Identify the tasks to be completed prior to, during and after the load handling activity and the personnel required to complete each task.
   5.2. Identify the specialized training required for these tasks and ensure copies of these training completion documents are attached to the plan.

6. **Site, services and ancillary equipment.**

6.1. Identify the equipment necessary to assist the load handling operation, (high reach, personnel lift, boom lift etc.)
6.2. Identify and ensure specialized training is completed and copies of the completion are attached to the lift plan if required.
6.3. Ensure that ground conditions for equipment support and that proper all underground hazards and placement have been identified and marked for adequate clearance.
6.4. Ensure that all equipment is used and operated in accordance with manufacturer, regulatory and industry standards.

7. **Communication system.**
   7.1. Identify a suitable communication system to be used during the activity. Acceptable methods for Whiting-Turner project sites are:
      7.1.1. Voice or radio signals
      7.1.2. Hand signals
      7.1.3. Other methods of signals are available but would require review prior to being accepted in respect to effectiveness and applicability.
   7.2. Identify back up system to be used in the event of interruption from initial communication system plans.

8. **Site control.**
   8.1. Identify vehicular and pedestrian access and traffic controls to be used.
   8.2. Ensure the plan addresses the following:
      8.2.1. Vehicular and pedestrian traffic in and around the site that could potentially be affected by the load handling activity
      8.2.2. Potential interference for other site activities and the controls to use
      8.2.3. Location of barricades or other measures which may be put in place to restrict traffic or prohibit interference during the load handling activity.

9. **Contingency considerations.**
   9.1. The plan should address, at minimum, the following potential events that could cause deviation from the original lift plan:
      9.1.1. Equipment malfunction or loss of power (e.g. loss of power, fouled rigging, radio communication failure etc.)
      9.1.2. Adverse changes to the environment (weather, visibility)
      9.1.3. Deviation from the planned load characteristics identified
      9.1.4. Adverse changes to the site conditions (surrounding activities, change in ground conditions and unauthorized entry to the fall zone).

10. **Emergency action plan.**
   10.1. Review the existing Emergency Action Plan and coordinate modifications created by the load handling activity if applicable.

11. **Pre-lift meeting.**
    11.1. The lift director shall conduct the meeting and the Whiting-Turner supervisor in charge of the operation or other designated person from Whiting-Turner shall attend the meeting along with all associated personnel to the lift.
        11.1.1. At a minimum, the following elements should be reviewed with all load handling activity personnel:
            11.1.1.1. Overview of the load handling activity
            11.1.1.2. LHE, rigging and other equipment involved in the load handling activity
            11.1.1.3. The sequence of events and step-by-step procedures for the entire load handling activity
            11.1.1.4. Safety measures, as required (e.g. work task plan action items)
            11.1.1.5. Load handling activity personnel assignments, addressing
                11.1.1.5.1. Individual responsibilities (e.g. location, time, task)
                11.1.1.5.2. Work location hazards (e.g. pinch points)
                11.1.1.5.3. Communication methods
                11.1.1.5.4. Personal protective equipment requirements
                11.1.1.5.5. Qualification(s) of assigned personnel
        11.1.2. Concern raised during this meeting shall be addressed prior to proceeding with the load handling activity
        11.1.3. At the completion of the pre-lift meeting, the lift director should confirm that the attendees understand the plan and respective roles and responsibilities during the load handling activity
        11.1.4. For repetitive lifts, the lift director should decide the frequency of pre-lift meetings. Pre-lift meetings are not required prior to each repetition of the load handling activity.

12. **Executing the critical lift plan.**
    12.1. Preparation for the load handling activity - the lift director should confirm that all setup and preparation requirements of the plan are in place and all required inspections and tests on the LHE(s) and rigging equipment have been completed.
    12.2. Initiating the Load Handling Activity - immediately prior to performing the load handling activity, the lift director should ensure that either:
        12.2.1. All requirements of the plan continue to be met and no conditions exist that would preclude implementation of the plan; or
        12.2.2. If a deviation exists, the load handling activity is not initiated until the deviation is addressed by a qualified person or the lift director determines that conditions are acceptable to allow the activity to begin.
    12.3. During the Load Handling Activity - the lift director should ensure that the load handling activity continues to comply with the plan.
        12.3.1. If the operation deviates from the plan, the load handling activity should be stopped and evaluated to determine if
            12.3.1.1. The load handling activity can resume according to plan;

12.3.1.2. The contingency measures can be implemented

12.3.1.3. The plan can be readily modified at the site to accommodate an unexpected condition or event; or

12.3.1.4. The load handling activity can no longer be implemented a planned, requiring a modified plan to be prepared. In such case, the load and the LHE shall be secured, if possible, until a new plan can be developed

12.3.1.5. Changes or modifications to the plan should be communicated to all affected load handling personnel prior to initiating the change.

12.3.1.6. If the load handling activity is stopped for any reason, only the lift director may initiate restart.

12.4. After the completion of the load handling activity, the lift director should

12.4.1. Review the development, planning and execution of the load handling activity with the load handling personnel. Items for review should include, but not be limited to, the requirements of all previous sections.

12.4.2. Identify potential measures to improve future load handling activity

12.4.3. Communicate any recommendations identified during the activity to the appropriate personnel for future considerations

12.5. For repetitive lifts, decide the frequency of post-lift reviews and evaluation of the lift plan. Post-lift reviews may not be required after each repetition of the load handling activity.

E.34 Tower Cranes

Introduction

Certain projects, due to size and scope, require the use of tower cranes. The quantity and location of towers cranes will vary from project to project. In some cases, due to project requirements, multiple tower cranes may be in a manner resulting in an overlapping operating zone (i.e. area where the operating zone or radius of working jib intersects the operating zone of another tower crane). While this will increase the hook access across the working deck and enhance the effectiveness of each crane, it brings with it significant safety concerns. For that reason, Whiting-Turner has established the following guidelines. In addition, notice to the Federal Aviation Administration (FAA) guidelines set forth in 14 CFR—Aeronautics and Space, Part 77—Safe, Efficient Use and Preservation of Navigable Airspace, shall be adhered to.

Procedures

1. **General Requirements.**
   1.1. Employers using tower cranes shall comply with specific provisions in the standard that cover the following subjects:
      1.1.1. Erecting, Climbing and Dismantling
      1.1.2. Signs
      1.1.3. Safety Devices
      1.1.4. Operational Aids
      1.1.5. Inspections

2. **Overlapping operating zones.**
   2.1. Multiple tower cranes operating with an overlapping operating zone have the potential to come into contact with each other. Such an occurrence has the potential to cause damage to the crane and/or its components; cause the crane or cranes to fail and topple; and/or cause severe injury and even death to tradespersons and/or the public. The safe operation of multiple tower cranes with an overlapping operating zone is determined by the communication and coordination of experienced and skilled operators.
   2.2. Written operating procedures shall be developed and implemented to coordinate lifting tasks in the overlapping operating zone to prevent contact, collision, or interference between a component or suspended load of one crane with a component or suspended load of another crane. The following safety procedures / protocols should be considered if a project has multiple tower cranes with an overlapping operating zone to minimize, if not eliminate, the potential for contact/collision between cranes.
   2.3. Establish a means and protocol for communicating between the crane operators when a crane operates in the overlapping zone. Communicating protocols may include the following:

- 2.3.1. Crane operators shall have the ability to communicate to each other (via radio - same channel)
- 2.3.2. Reduce distracting background noise on radios by ensuring crane operators, monitor and limited riggers have separate radio channels
- 2.3.3. Include protocols that all crane operators shall follow before entering another crane's operating zone. Consider including the following:
  - 2.3.3.1. Visually ensure it is safe to enter the other crane's operating zone
  - 2.3.3.2. Communicate entry intent into that zone to the other operator
  - 2.3.3.3. Receive an acknowledgement response from the other operator. If the entering operator does not receive acknowledgement from the other operator than the entering operator shall not proceed with entering the other operator's operating zone

2.4. Standardize the language used for all communication between crane operators (English).

2.5. Consider limiting the number of individuals that can communicate directly with the operators (i.e. individuals on the working deck or other requesting use or support of the crane, etc.)

2.6. Establish clearance requirements for loads passing over or near another tower crane

2.7. Proximity alarms shall be utilized to decrease the potential for tower crane contact/collision. Various zone protection and boundary protection safety systems exist in the marketplace (e.g. TAC-3000 Tower Crane Collision / Accident Avoidance Safety System)

2.8. Consider creating a 'right-of-way' procedure giving one crane priority for working in the overlapping operating zone.

2.9. Consider assigning a monitor to observe / supervise all crane activity and provide that monitor with the ability to stop all crane activity should violations of protocol occur, or potentially hazardous situations be observed. The following elements / activities should be considered related to use of said monitor:

- 2.9.1. Monitor shall be equipped with radio to communicate with all crane operators
- 2.9.2. Demonstrate bright, highly visible outerwear unique to the monitor so to be easily distinguishable to the operators and other tradespersons (e.g. bright colored vest and/or hardhat, etc.).
- 2.9.3. Monitor shall be solely dedicated to its role as monitor and shall not participate in any other activities on the project while acting as monitor
- 2.9.4. Monitor shall observe crane activity to ensure that proper protocol is followed
- 2.9.5. Monitor shall listen to communication between operators to ensure that protocol is followed:
- 2.9.6. If operators fail to communicate with each other as protocol requires, the monitor shall issue a reminder to the operators requiring compliance to protocol
- 2.9.7. Monitor shall record instances when operators fail to comply with protocol and inquire why the violation took place

2.9.8. Violations in protocol should be copied to the appropriate supervisor / Superintendent and safety personnel of the responsible trade. Documentation shall also be made available to Whiting-Turner

2.9.9. Monitor should have the ability to stop crane activity if required to maintain safe crane operations. Methods for delivering an "ALL STOP" directive may include the following:

2.9.9.1. Outfit monitor with air horn to be used to alert a stop directive to operators. Test procedure in field to identify most effective method for communicating signal to operators.

2.9.9.2. Outfit monitor and operators with a separate and dedicated radio to deliver the stop directive. The separate radio is reserved for stop directive between monitor and operators only and would ensure that operators receive directive.

2.9.10. Monitor shall be a competent person and subject to Whiting-Turner approval

2.10. Help to better identify the counter deck of the lower crane by installing bright flags or other markings on the back of the counter deck.

2.11. Identify counter deck radius of lower crane on the current working deck by means of flags, cones, or other clear delineators to be conspicuously visible. Confirm that delineators are clearly visible to operators and other crane related personnel.

2.12. Formal overlapping operating zone plan shall be distributed to and acknowledged by the responsible Trade Partner's Superintendents, foremen, operators, monitor(s) and safety personnel – at a minimum.

2.13. Disciplinary procedures to address violations in protocol, etc. should be included.

2.13.1. Appropriate disciplinary procedures should be identified and implemented on the party responsible if crane contact or associated accident, etc. takes place (e.g. removal of operator or responsible party from the project).

2.13.2. Responsible trade shall identify how it will determine who the responsible party is should an occurrence take place requiring disciplinary action.

2.13.3. Whiting-Turner shall be copied on all recorded non-conformances with protocol as well as related disciplinary procedures/actions.

2.14. If it is determined that only one tower crane is needed for a given activity or shift of work, the overlapping work zone plan shall address the crane that is not in use to avoid interference with the working crane. Procedures for this situation may include the following:

2.14.1. If the non-working crane is the higher crane, it can be allowed to weathervane and the working crane (lower) can work unobstructed.

2.14.2. If the non-working crane is the lower crane:

2.14.2.1. Non-working crane shall be operated, but not used, to keep non-working crane from *weathervaning* or movement of the non-working crane into the working crane's operating zone.

2.14.2.2. Investigate to determine whether the non-working crane is outfitted with a locking mechanism to keep the crane from *weathervaning* except under extreme weather conditions. If this is the case, the non-working crane can be locked to prevent *weathervaning* into the working crane's operating zone. At certain wind speeds (determined by crane manufacturer) the lockdown function will be overridden and the non-working crane will be allowed to weathervane. Weather conditions and wind speed causing this override will require all crane activity to stop in this scenario.

E.35 Materials Handling and Rigging

Introduction

Material handling and rigging incidents account for many tradespersons' compensation claims annually. Each Trade Partner working on a Whiting-Turner project shall comply with 29 CFR 1926, Construction Industry Regulations, Subpart H – Materials Handling, Storage, Use and Disposal, in addition to the following guidelines.

Procedures

1. **General Material Storage.**
    - 1.1. Aisles and passageways shall be kept clear at all times for the safe movement of material handling equipment and employees.
    - 1.2. Materials shall not be stored within 6' of any hoist way or interior floor opening.
    - 1.3. Materials shall not be stored within 10' of an exterior wall which does not extend above the material.

2. **Rigging**.
    - 2.1. General Requirements
        - 2.1.1. Before each use rigging equipment, including its fastenings and attachments, shall be inspected by a competent person/qualified rigger.
        - 2.1.2. Inspections shall also be conducted during use and where additional service conditions warrant.
        - 2.1.3. Defective or damaged slings shall be removed from service immediately and destroyed or tagged out of service.
        - 2.1.4. Taglines shall be utilized to minimize worker exposure to falling and swinging loads.
        - 2.1.5. Rigging equipment shall be removed from the work area when not in use.
        - 2.1.6. All rigging shall have an affixed and legible label that identifies the capacity, size and manufacturer.
    - 2.2. Lifting Chains
        - 2.2.1. Alloy steel lifting chains shall have a permanently affixed, durable identification tag stating size, grade, rated capacity and sling manufacturer.
        - 2.2.2. Attachments, including, but not limited to hooks, rings, oblong links, pear-shaped links or other welded or mechanical links, shall have a rated capacity at least equal to the lifting chain.
        - 2.2.3. Job made shop hooks or links, makeshift fasteners formed from rebar or bolts or other such attachments are not allowed on Whiting-Turner projects.
        - 2.2.4. Additional lifting chain inspection criteria is based upon the frequency of use, the severity of the service conditions, the nature of the lifts being made and the experience gained on the service life of slings used in similar circumstances.
        - 2.2.5. Lifting chains shall be inspected, prior to each use.

2.3. Wire Rope Slings
   2.3.1. The manufacturer's safe working loads shall be followed at all times.
   2.3.2. Wire rope shall not be used if, in any length of eight diameters, the total number of visible broken wires exceeds 10% of the total number of wires.
   2.3.3. Wire rope shall not be used if it shows signs of excessive wear, corrosion, or defects.
   2.3.4. Slings shall not be shortened with knots, bolts, or other makeshift devices.
   2.3.5. Slings shall be protected from sharp edges with padding, softeners, or similar devices.
   2.3.6. Shock loading of a sling is prohibited and slings shall not be pulled from under a load when the load is resting on the sling.

2.4. Synthetic Slings
   2.4.1. Each synthetic sling shall be identified with the name of the manufacturer, rated capacities and type of material.
   2.4.2. Synthetic slings shall be immediately removed from service if any of the following conditions are present; acid or caustic burns, melting or charring of any of the sling surface, snag, puncture, tear or cut, broken, or worn stitches or distorted fittings.
   2.4.3. Slings shall be protected from sharp edges with padding, softeners, or similar devices.

E.36 Manual Lifting

Introduction

Manual lifting safety is critical because improper techniques can lead to serious injuries, including musculoskeletal disorders (MSDs), back pain and strain, which highlights the importance of proper techniques to minimize workplace hazards and exposures.

Procedures

1. Before manual lifting is performed, a hazard assessment shall be completed. The assessment shall consider size, bulk and weight of the object(s), if mechanical lifting equipment is required, if two-man lift is required, whether vision is obscured while carrying and the walking surface and path where the object is to be carried.
2. Training should include general principles of ergonomics, recognition of hazards and injuries, procedures for reporting hazardous conditions and methods and procedures for early reporting of injuries. Additionally, job specific training should be given on safe lifting and work practices, hazards and controls.
3. Musculoskeletal injuries caused by improper lifting shall be investigated and documented. Incorporation of investigation findings into work procedures shall be accomplished to prevent future injuries.
4. Where use of lifting equipment is impractical or not possible, two-man lifts shall be used.
5. Supervision shall periodically evaluate work areas and employees' work techniques to assess the potential for and prevention of injuries. New operations should be evaluated to engineer out hazards before work processes are implemented.
6. Manual lifting equipment such as dollies, hand trucks, lift-assist devices, jacks, carts, hoists shall be provided for employees. Other engineering controls such as conveyors, lift tables and workstation design should be considered.
7. Manual lifting equipment should be used instead of manual lifting where possible. Supervisors should enforce the use of lifting equipment.

E.37 Electrical Hazards Prevention

Introduction

Use of electricity on the jobsite poses serious hazards, with employees potentially becoming exposed to such dangers as electric shock, electrocution, fires and explosions. All Whiting-Turner employees and Trade Partners working on a Whiting-Turner project will comply with NFPA 70E Electrical Safety Practices and 29 CFR 1926, Construction Industry Regulations, Subpart K – Electrical, in addition to the following guidelines.

Procedures

1. **Working on or near exposed energized parts.**
   - 1.1. It is Whiting-Turner's policy that no one works on live electrical circuits. If a situation arises where it is impossible to perform a task with the circuit de-energized, the Whiting-Turner Superintendent or Project Manager shall contact the Vice President and submit a formal request detailing why the circuits cannot be deenergized with a live electrical work plan prior to performing the work. An Area EH&S Manager shall review the plan. A formal pre-construction meeting shall occur prior to any such work occurring.
   - 1.2. Only qualified electricians may work on electric circuit parts that have not been de-energized under the procedures of 1910.333.
   - 1.3. Such persons shall be capable of working safely on energized circuits and shall be familiar with the proper use of special precautionary techniques, personal protective equipment, insulating and shielding materials and insulated tools.
   - 1.4. All work shall be completed with strict compliance to NFPA 70-E requirements and guidelines.
   - 1.5. The Trade Partner shall provide proof of training for all tradespersons when requested by Whiting-Turner.
   - 1.6. Light switches and receptacles shall be protected by permanent or temporary cover plates prior to energizing the circuit.
   - 1.7. When working near overhead lines, a clearance distance of at least 20 feet shall be maintained, or the lines will be de-energized and grounded.
   - 1.8. When working near overhead lines, unqualified persons shall maintain a clearance distance of 20 feet (or greater).
   - 1.9. Vehicles and/or mechanical equipment shall maintain a clearance distance of 20 feet (or greater) from energized overhead lines.

2. **Ground fault circuit interrupters.**
   - 2.1. All 120-volt, single-phase 15 and 20 ampere receptacle outlets which are not part of the permanent wiring of the structure and which are in use by employees shall have approved GFCI's.

2.2. Whiting-Turner requires that all projects are 100% GFCI compliant. An Assured Equipment Grounding Conductor Program may be used in addition to—but not in lieu of—the GFCI program. If permanent power outlets are to be utilized portable GFCI units shall be provided and utilized.

2.3. All GFCI receptacles and circuit breakers shall be tested monthly to ensure the GFCI is properly functioning and protecting the worker. It is the responsibility of the electrical Trade Partner to perform inspection and testing. The documentation shall be made available upon request.

3. **Electric tools.**

3.1. All portable electric tools such as saws, hammers, drills, vibrators and float machines shall bear the label of a Certified Testing Agency, such as Underwriters Laboratories, CSA, ETL, or the like.

4. **Extension cords.**

4.1. Only round, heavy-duty (type S, SJO, SJTW, ST, SO, STD) are acceptable for use on a construction site; at least 12 gauges or larger.

4.2. Cords shall be maintained in their original design configuration.

4.3. Any cord which is damaged or has the grounding pin removed shall be removed from service, including power tools.

4.4. Whenever an extension cord is plugged into permanent power, a GFCI is required between the extension cord and the receptacle.

4.5. All electrical cords shall be out of the hallway, corridor, aisle, stairway, doorways and exit areas where a tripping hazard may occur.

4.6. All electrical cords shall be protected from damage by equipment, carts, trucks and other rolling objects.

4.7. Where possible, all extension cords will be suspended (8’) above the floor or working surface.

4.8. Extension cords shall not be fastened with staples, hung from nails, suspended with non-insulated wire, or hang from fire sprinklers.

4.9. Extension cords shall not be run thru or laid over sharp metal objects that could subject the cord to cuts and possible unintentional energization.

5. **Temporary Panel Requirements**

5.1. All breaker spaces shall be filled.

5.2. All circuits shall be identified.

5.3. Inspection stickers shall be on cover.

5.4. Temporary panel shall be properly installed.

5.5. Temporary panel shall be protected from all energized parts.

5.6. Temporary panel shall be UL listed.

6. **Temporary wiring.**

6.1. All temporary wiring and lighting shall meet current NEC codes.

6.2. Temporary lighting shall never be put on the same circuit as temporary receptacles.
6.3. All temporary lighting circuits shall originate from GFCI protected breakers.
6.4. Temporary wiring shall be rated for all conditions to which it may be subjected and installed per the requirements of NEC, OSHA, NFPA and other authorities having jurisdiction.
6.5. Flexible cords or cables such as extension cords shall not be used to substitute for the fixed wiring on any building or structure. They shall not be concealed behind or transition through temporary or permanent walls, ceilings, or floors.
6.6. Wiring systems shall not be installed in ducts used to transport dust or flammable vapors or in ducts used for vapor removal.
6.7. Metal enclosures for conductors shall be metallically joined together into a continuous metallic run that provides electrical continuity.

7. **Temporary lighting.**
7.1. The minimum illumination level 5 foot-candles; this shall be maintained by the electrical Trade Partner at all times.
7.2. Installation of temporary lighting shall be per manufacturer's specifications and in compliance with OSHA, NFPA, NEC and local codes.
7.3. All lamps for general illumination shall be protected from accidental contact or breakage by a suitable fixture or lamp holder with a guard. The fixtures shall be factory assembly.
7.4. All lamps shall be maintained in a safe condition.
7.5. All missing or broken bulbs and bulb guards shall be replaced.
7.6. Temporary lights shall not be suspended by their electric cords unless cords and lights are designed for this means of suspension.

8. **Inspections of Installed Temporary Wiring**
8.1. Temporary wiring installations shall be inspected monthly to verify that the wiring is in proper working order and to discover damage from construction or maintenance activities.
8.2. Monthly inspections shall include/cover, but not limited to:
8.2.1. Inspect housing, cable condition, strain reliefs and plug attachments.
8.2.2. Inspect cords/conductors, which shall be in good repair and show no signs of wear, defects, bulging, exposed wire, or other damage.
8.2.3. Cords/conductors shall not be affixed to walls, run under doors, or run through walls, ceiling, or floors.
8.2.4. Determine that daisy chaining is NOT in use.
8.2.5. All conductors shall have proper overcurrent breaker protection and insulators.
8.2.6. Switches shall be labeled identifying their control and use.
8.2.7. Temporary light fixtures shall be guarded.
8.2.8. Inspect all GFCI's
8.2.9. Ensure breakers are identified
8.2.10. Ensure all openings are closed, breaker blanks, KO seals.

9. **Clear-Space Requirements**
   9.1. For the purpose of safe operation and maintenance of temporary electrical equipment, sufficient access and working space shall be provided and maintained about all temporary electrical equipment.
   9.2. Temporary Electrical Equipment that may need examination, adjustment, servicing, or maintenance while energized shall have a clear working space of not less than 4' on all sides of such equipment. OSHA 1926.403 Table K-1 or NEC Table 110.26(A)(1) may be used to determine working clearance requirements in lieu of 4' minimum clearance.
   9.3. The clear working space shall not be used for storage.

10. **Fish tapes.**
   10.1. Fish tapes or lines made of metal, or any other conductive material are prohibited when the potential for contact with energized circuits exist; non-conductive tapes and lines will be used instead.

11. **Additional Information, Requirements and Considerations**
   11.1. All employees shall have electrical awareness training.
   11.2. Safe work practices outlined in the applicable activity hazard analysis shall be adhered to prevent electric shock.
   11.3. All exposed de-energized parts shall be treated as live when working on or around the equipment.
   11.4. Whiting-Turner does not have qualified employees who shall adhere to the approach distances in Table S5 when working in the vicinity of overhead lines.
   11.5. The use of portable ladders with conductive side rails is not permitted on Whiting-Turner projects.
   11.6. Conductive apparel shall not be worn unless it is rendered non-conductive by covering, wrapping or other insulating means.

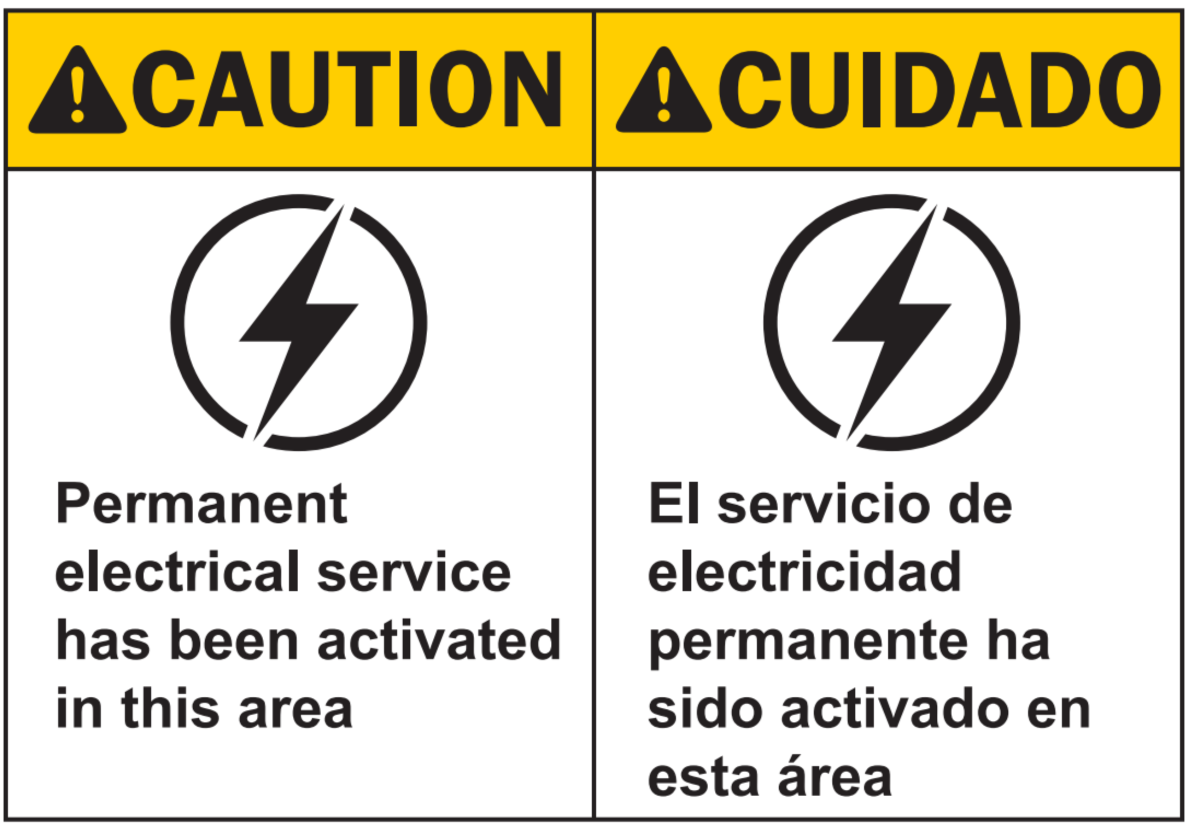
E.38 Project Electrical Transfer Program

Introduction

This procedure is effective on all Whiting-Turner projects and addresses the potential hazards encountered while transferring from temporary electrical service to the permanent building electrical system.

Procedures

1. **General Requirements.**
   1.1. The Whiting-Turner Project Team will provide written notification to all Trade Partners working on site when any portion of the building's permanent electrical system is activated. This notification will be made by memorandum or by email.
   1.2. The project team will review the status of the transfer to permanent power system at all project meetings with all Trade Partners.
   1.3. The project team will make use of signage to alert project workers of the transfer.



E.39 Energy Control Policy

Introduction

The intent and purpose of this procedure is to reduce or eliminate the danger of the unexpected release of stored or residual energy that could cause injury or death to the employee or to the public. Each Trade Partner working on a Whiting-Turner project will comply with NFPA 70E Electrical Safety Practices, 29 CFR §1926.417, "Locking and Tagging of Circuits", in addition to 29 CFR §1910.147, "The Control of Hazardous Energy" along with the following guidelines.

Procedures

1. **General Requirements**
   - 1.1. Lockout/tagout (LOTO) shall not be considered for use until all other avenues of attaining a "zero-energy state" (all potential sources of any type including re-energization or stored) have been exhausted.
   - 1.2. All Trade Partners working with electrical systems are required to have a written lockout/tagout procedure. A competent person shall be responsible to control all aspects of the LOTO procedure. They will ensure coordination with the appropriate tradesmen.
   - 1.3. If a system can be locked out through design or by other means, this will be the preferred method.
   - 1.4. The lockout device shall be substantial enough to prevent removal.
   - 1.5. The lock shall be a separately keyed lock for use only with the lockout system.
   - 1.6. The lockout device shall be tagged with the name of the employee and their company. There shall be one lock for each employee (including Whiting-Turner) exposed to the system.
   - 1.7. If working in a multi-shift environment each worker shall remove their respective locks at the end of their shift. If work on the system is to continue, the next shift worker shall be present to install his lock immediately following removal of the lock from the previous shift. The creating Trade Partner shall have a system in place to ensure that exposure is eliminated by controlling the potential hazardous energy by means of LOTO.
   - 1.8. Workers shall not leave locks on after the completion of their task. This does not apply to the competent person or supervisor. The use of 100% LOTO shall be maintained by the Trade Partner responsible until the completion of the task. Verification by all competent persons in charge of the LOTO shall be completed prior to re-energizing the system.
   - 1.9. If the energy isolation device cannot be locked out and a tag shall be used, authorization from the Area EH&S Manager and Vice President is required prior to start of work.
   - 1.10. Tag out devices, including the means of attachment, shall be substantial enough to prevent accidental removal.

1.11. The tag shall warn against energizing the tagged-out system such as do not start, do not open, do not close, do not energize, do not operate, etc.
1.12. The name of each employee shall be displayed on respective tags.
1.13. The competent person shall be responsible for removing the tag and activating the system after all exposed employees have removed their tags.
1.14. In the event an employee is discovered tampering with or violating the LOTO procedure, the employee will be removed from the project indefinitely.

2. **Training and Documentation.**
   2.1. Each Trade Partner utilizing LOTO shall establish a program and utilize procedures for affixing appropriate lockout or tagout devices to energy isolating devices and to otherwise disable machines, piping, or equipment to prevent unexpected release of stored or residual energy to prevent injury to employees.
   2.2. Each tradesperson affected by the LOTO procedure shall be trained in the procedure. Records of training will be kept on site and be made available to Whiting-Turner upon request.
   2.3. Each Trade Partner involved in LOTO procedures shall maintain a log on site that identifies the following:
      2.3.1. Date of usage
      2.3.2. Number of locks and tags used
      2.3.3. Persons involved
      2.3.4. Time of LOTO initiation
      2.3.5. Time of LOTO removal
      2.3.6. Designated competent persons
      2.3.7. Location of LOTO Devices
   2.4. Electrical or piping and instrumentation drawings identifying specific locations of the LOTO devices shall accompany the log.
   2.5. Locks shall only be removed by the person who applied the lock. In extreme circumstances, after “all reasonable means” to contact the employee who was responsible for the lock were exhausted, the Whiting-Turner Superintendent shall request a meeting with the Area EH&S Manager and Trade Partner’s competent person. This process is to ensure that the Trade Partner’s competent person has considered and documented all the necessary steps to keep the workforce safe prior to removing the lock. The decision to remove a lock rests solely with the Trade Partner’s competent person. However, the decision to remove a lock may be overridden by the EH&S Manager, if deemed necessary.
   2.6. Records of training will be kept on site and be made available to Whiting-Turner upon request.

3. **NPFA 70E and General Industry Specific Requirements and Considerations.**

3.1. Whiting-Turner will advise the host employer of unique hazards in the workplace presented by the Trade Partner's work, unanticipated hazards and any measures taken to correct hazards reported to them by the host employer.
3.2. Additional training shall be conducted for qualified persons who are allowed to work within the Limited Approach Boundary.
3.3. Retraining is required and will be conducted when a tradesperson is not complying with safety-related work practices or when workplace changes necessitate the use of safety-related work practices that are different from those that the tradesperson would normally use.
3.4. Retraining shall be performed at intervals not to exceed 3 years.
3.5. A risk assessment shall be conducted prior to work being done within the Limited Approach Boundary and safe work practices that were determined during the risk assessment shall be implemented when working with the Limited Approach Boundary.
3.6. Job briefings are discussed prior to the commencement of work which includes the steps to perform the tasks, the hazards associated with the task and the methods for abating or mitigating the hazard.
3.7. Only qualified persons shall complete tasks such as testing, troubleshooting and voltage measuring on electrical equipment.
3.8. All test instruments, equipment and their accessories shall be rated for circuits and equipment to which they will be connected.
3.9. All test instruments shall be verified to be in proper working order before and after an absence of voltage test is performed.
3.10. All insulating tools, PPE and other equipment be inspected prior to each day's use and immediately after any incident.
3.11. Electrical protective equipment shall be subjected to periodic electrical tests. Test voltages and the maximum intervals between tests shall be in accordance with Table I-4 (Rubber Insulating Equipment, Voltage Requirements) and Table I-5 (Rubber Insulating Equipment, Test Intervals) of 1910 Subpart I.
3.12. Any time a Trade Partner wishes to perform work on energized equipment, an Energized Electrical Work Permit, shall be completed and submitted to Whiting-Turner. The energized work permit shall include the following:
  3.12.1. Location/Space with time and date of request and estimated completion time
  3.12.2. Description of the work to be performed and by whom along with written justification why the circuit/equipment cannot be de-energized
  3.12.3. Safety Precautions: Shock and arc flash risk assessments; safe work practices; required PPE; and standby person who has required training, adequate PPE, current First Aid/CPR training and has emergency communication capability available
  3.12.4. Confirmation that the work can be completed safely by those performing the activity
  3.12.5. Approvals
  3.12.6. Work Completion

3.13. All work areas shall be properly illuminated.
3.14. All affected tradespersons are trained to understand the specific hazards associated with electrical energy.
3.15. Documentation of training shall be maintained for the duration of the tradesperson's employment.
3.16. PPE shall comply with all applicable laws and regulations.
3.17. Each Trade Partner engaged in this type of work shall perform an audit of field work on an annual basis.

4. **Additional Information, Requirements and Considerations.**
    4.1. The types of energy that may be locked out include—but is not limited to—electrical, mechanical, hydraulic and pneumatic. Chemical, nuclear, thermal and gravitational energy are other types of hazardous energy shall be controlled but may require other means of control instead of LOTO. A risk assessment would help determine the best method to the type of control hazardous energy present in a given situation.
    4.2. All energy control (Lockout/Tagout) procedures shall be inspected annually.
    4.3. The tradesperson who turns off the machine shall be trained and have knowledge of the energy magnitude (preparation for shutdown).
    4.4. After the completion of a risk assessment, the machine/equipment shutdown procedures shall be reviewed with all potentially exposed personnel. Prior to working on any circuit, a live-dead-live test shall be performed to confirm that circuit has been de-energized.
    4.5. Stored energy shall be released after a lockout device is installed.
    4.6. Isolation/zero energy is verified after a lockout device is installed.
    4.7. A qualified, competent person shall be assigned to be in charge of a group or shift change lockout operation.
    4.8. Retraining on Lockout/Tagout is required on an as needed basis—at intervals not exceeding 3 years—to ensure that the tradesperson is well educated and prepared to perform LOTO operations with success.

E.40 Fire Protection and Prevention

Introduction

Each Trade Partner working on a Whiting-Turner project shall comply with 29 CFR 1926, Construction Industry Regulations, Subpart F – Fire Protection and Prevention, in addition to the following guidelines.

Procedures

1. **General requirements.**
    - 1.1. A site-specific fire protection and prevention plan shall be developed at each Whiting-Turner project.
    - 1.2. Client requirements permit procedures, fire watches, shields and blankets shall be considered when developing site-specific fire prevention programs.
    - 1.3. All firefighting equipment shall be clearly visible and access to the equipment shall be maintained at all times.
    - 1.4. At a minimum, a 20 lb. ABC dry chemical fire extinguisher or equivalent shall be provided for each 3,000 square feet of protected building area. An extinguisher shall be placed at every stairwell on each level.
    - 1.5. Travel distance to a fire extinguisher shall not exceed 100 feet.
    - 1.6. Each portable fire extinguisher shall undergo an annual certification & have an inspection tag affixed. These services shall be performed by qualified, authorized, certified and actively licensed fire extinguisher companies.
    - 1.7. Portable fire extinguishers shall be inspected monthly. The documentation shall be a weather resistant tag attached to the extinguisher. A fire extinguisher log is recommended for back-up.
    - 1.8. All tradespersons who are required to use fire extinguishing equipment shall be provided training.
    - 1.9. Re-training on use of fire extinguishing equipment shall be performed annually.
    - 1.10. Residential-like wood framing construction shall have a 20 lb. ABC dry chemical fire extinguisher or equivalent for each 1,500 square feet of protected building area. An extinguisher shall be placed at every stairwell on each level.
    - 1.11. A standpipe, whether temporary or permanent, shall be installed and active as soon as combustibles are present in a structure with a roof.

2. **Compressed Gas Cylinders.**
    - 2.1. Contents of compressed gas cylinders shall be clearly identified.
    - 2.2. If a cylinder cap cannot be removed, do not continue to attempt removal. Tag out the cylinder out, return to storage, notify the vendor or manufacturer and request for it to be picked and replaced.
    - 2.3. Compressed gas cylinders shall be visually inspected for damage and defects prior to being used.

2.4. Regulators, hoses and/or connections shall be inspected prior to using a compressed gas cylinder.
   2.4.1. Look for signs of damage – do not use if gauges are damaged.
   2.4.2. Check that the threads of the regulator and cylinder are clean. If they are not clean, wipe with a lint free cloth.
   2.4.3. Fit the regulator using the correct spanner (not an adjustable one).
   2.4.4. NEVER use oil or grease.
   2.4.5. Undo the captive pressure adjusting screw (fully anticlockwise).
   2.4.6. Open the valve to the cylinder gently.
   2.4.7. Adjust the pressure adjusting screw to get the output you needed.
   2.4.8. Check for leaks using an approved leak detection spray.

2.5. Appropriate tools shall be used to open and close cylinder valves; tool designed for that intended purpose.

2.6. Store cylinders upright and secure them with a chain, strap, or cable to a stationary building support or to a cylinder cart to prevent cylinders from tipping or falling.

2.7. Store cylinders in a well-ventilated area.

2.8. Storage areas shall be designated and labeled for full and empty cylinders.

2.9. Use only approved containers to store and transport liquid nitrogen. Containers should have vented lids to prevent spillage when carried.

2.10. Compressed gas containers shall be separated from each other based on the hazard class of their contents.

2.11. Combustible waste shall be kept a minimum of 10 feet from the compressed gas containers and systems.

2.12. Oxygen cylinders should be kept at a minimum of 20 feet away from flammable gases, liquids, solids, greases and oils.

2.13. Cylinders shall be transported in a vertical, secure position using a cylinder basket or cart.

2.14. Tradespersons who use compressed gas cylinders to perform their tasks shall be trained on the proper use, handling and storage of compressed gas cylinders.

2.15. If a leaking cylinder is encountered, tradespersons are trained not to attempt to repair the leak; it is against Whiting-Turner's safe work practice to attempt to repair equipment under pressure and to tamper with pressure-relief devices. If safe to do so, remove the leaking cylinder to a well-ventilated area, label it appropriately and contact the supplier.

2.16. Cylinders that are no longer needed shall be tagged and marked out of service.

3. Fire prevention.

3.1. Temporary offices or trailers, when located inside of a building under construction, shall be constructed of fire-retardant materials.

3.2. Combustible materials, such as cardboard, wooden pallets, etc., shall be removed from the work area immediately.

4. **Flammable liquids.**
    4.1. All storage, handling or use of flammable liquids shall be under the supervision of qualified persons.
    4.2. All sources of ignition shall be prohibited in areas where flammable liquids are stored, handled and processed. Suitable NO SMOKING OR OPEN FLAMES signs shall be posted in all such areas.
    4.3. Trade Partner shall provide and mount a 20lb ABC fire extinguisher between 25'-75' from flammable storage.
    4.4. Flammable liquids shall not be stored in areas used for exits, stairways or used for safe passage of people.
    4.5. Dispensing systems shall be electrically bonded and grounded.
    4.6. Above ground storage tanks shall be double-walled or provided with a secondary means of containment. Secondary containment shall have a capacity at least equal in volume to that of the largest tank plus 10 percent of all other tanks enclosed. Provision shall be made for draining off accumulations of ground or rainwater or spills. Drain plugs shall remain in place except when draining.
    4.7. A metal cabinet meeting the requirements of NFPA 30, Flammable and Combustible Liquids, shall be provided for the storage of more than a total of 25 gallons of flammable liquids and greases in buildings used for other than storage or processing. Not more than a total of 60 gallons shall be stored in any one cabinet. Individual containers shall be metal, kept tightly closed and shall not exceed five (5) gallons capacity.
    4.8. Smoking or open flames within 50 feet of where flammables are being used or transferred or where equipment is being fueled is prohibited.
    4.9. Areas in which flammable liquids are transferred, in quantities greater than 5 gallons from one tank or container to another tank or container, shall be separated from other operations by 20 feet or by a five-foot partition, having a fire resistance of at least one hour. Drainage or other means shall be provided to control spills; Natural or mechanical ventilation shall be provided to maintain the concentration of flammable vapor at or below 10 percent of the lower flammable limit.
    4.10. Tradespersons shall be required to guard carefully against any part of their clothing becoming contaminated with flammable liquids. They shall not be allowed to continue work when their clothing becomes contaminated and shall remove or wet down the clothing as soon as possible.
    4.11. Equipment using flammable liquid fuel shall be shut down during refueling, servicing or maintenance.
    4.12. Where an automatic extinguishing system is provided, the system shall be designed and installed in accordance with NFPA recommendations.
    4.13. In every inside storage room, there shall be one clear aisle maintained at least three feet wide.
    4.14. Outdoor portable tanks shall be at least 20 feet from buildings. Individual tanks shall be at least 5 feet apart.

4.15. The dispensing units shall be protected against collision damage.

4.16. Handling of all flammable liquids by hand shall be in safety containers with flame arresters. This requirement shall not apply to those flammable liquid materials that are extremely hard to pour, that may be handled in original shipping containers. For quantities of one gallon or less, only the original container or approved metal safety cans shall be used for storage, use and handling of flammable liquids.

4.17. Storage of flammable/combustible liquids on or inside of buildings under construction shall be no more than one-day supply.

5. Paints and painting.

5.1. Paints, varnishes, lacquers, thinners, or other volatile painting materials and containers shall be kept tightly closed when not in use and shall be stored in accordance with NFPA recommendations.

5.2. Unopened containers of paint, varnishes, lacquers, thinners and other flammable paint materials shall be kept in a well-ventilated location, free of excessive heat, smoke, sparks, flame, or direct rays of the sun.

5.3. Paint-soiled clothing and drop cloths, when not in use, shall not be stored on site.

5.4. Paint scraping and paint-saturated debris shall be removed daily from the premises.

5.5. Ventilation adequate to prevent the accumulation of flammable vapors to hazardous levels shall be provided in all areas where painting is done, or paints are mixed.

6. Liquid petroleum gases.

6.1. Storage, handling, installation and use of liquefied petroleum gases and systems shall be in accordance with NFPA 58. LP gas shall not be stored indoors.

6.2. Liquefied petroleum gas containers and equipment shall not be used in unventilated spaces below grade in pits, below decks and other such spaces where dangerous accumulations of heavier-than-air gas may accumulate due to leaks or equipment failure.

6.3. Equipment using liquefied petroleum gas shall be shut down during refueling operations.

6.4. Filling of fuel containers for motor vehicles from bulk storage containers shall be performed not less than 10 feet from the nearest masonry-walled building, or not less than 25 feet from the nearest building or other construction and, in any event, not less than 25 feet from any building opening.

6.5. Filling of portable containers or containers mounted on skids from storage containers shall be performed no less than 50 feet from the nearest building.

7. Gasoline power.

7.1. Fire

7.1.1. Only approved containers are allowed for the storage of flammable liquids. An approved container is one which is constructed of metal, has a spring-loaded top that allows venting of fumes and contains a flash arresting screen and spout cover.

7.1.2. Provide, at a minimum, a 20-pound ABC dry chemical type extinguisher between 25'-75' from area where flammable liquids are being handled

7.1.3. "No Smoking" and "No Open Flame" signs shall be conspicuously posted in service, refueling, or flammable liquid storage areas.

7.1.4. All drums shall be properly labeled in accordance with 29 CFR 1910.1200 Hazard Communication.

7.2. Vapors

7.2.1. Gas engines exhaust carbon dioxide and carbon monoxide. Dioxide is heavier than air; monoxide slightly lighter. A mixture of the gases usually is heavier than air although heat may cause it to rise. Both are without color, taste or smell. Light concentrations cause headache and nausea. Death is swift in heavy concentrations. A few minutes may be too long. Do not discount this hazard. If anyone exhibits symptoms, do not attempt rescue without proper personal protection equipment.

7.2.2. Do not run gas engines in pits, manholes or confined spaces.

7.2.3. If gas engines are to be used inside buildings, excavations, crawl spaces under basement floors, hoist engineers' shanties, then the following shall be done:

7.2.3.1. Ventilation shall be required

7.2.3.2. Testing the space for carbon monoxide shall be done before starting the engines

7.2.3.3. Exhausts shall be equipped with a scrubber

7.2.3.4. Continuous monitoring

8. Temporary heating devices.

8.1. Fresh air shall be supplied in quantities sufficient to maintain the health and safety of all tradespersons. If a competent person deems natural airflow inadequate, then mechanical ventilation shall be provided.

8.2. Heaters used in the vicinity of tarpaulins, canvas or similar coverings shall be located at least 10' from the covering and be secured to prevent ignition due to wind.

8.3. Open fires are not allowed on Whiting-Turner projects.

8.4. All gas piping shall be labeled and flagged or painted; shutoff valves shall remain in place.

8.5. Whiting-Turner permits the use of direct fired and indirect fired heaters on project sites. Clearance from combustibles shall be maintained per the manufacturer's requirements. Direct fired temporary heaters contain a direct flame and require a Firewatch. Direct fired temporary heaters shall be equipped with the following:

8.5.1. High temperature limit switch – monitors operating temperature of equipment; provides automatic shutdown.

8.5.2. Air proving control – continuously monitors operating airflow, provides automatic shutdown.

8.5.3. Low voltage reset – provides protection for insufficient incoming power.

8.5.4. Redundant solenoids – dual gas valves as required, for redundant gas flow protection.

8.5.5. Electronic ignition sequence – flam safety start-up with continuous electronic monitoring of equipment in firing mode of operation.

8.5.6. A fire watch shall be provided by the Trade Partner for the duration of the use of direct fire heaters, plus 30 minutes after the shutdown.

E.41 Motor Vehicles, Mechanized Equipment and Marine Operations

Introduction

Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart O – Motor Vehicles, Mechanized Equipment and Marine Operations in addition to the following guidelines.

Procedures

1. **General Requirements.**
   - 1.1. Operators of mobile equipment shall have sufficient education, training and experience. A tradesperson shall not operate mobile equipment unless the tradesperson: (a) is trained to safely operate the equipment (b) has demonstrated competency in operating the equipment to a designated competent worker (c) is familiar with the equipment's operating instructions (d) is authorized by the Trade Partner to operate the equipment.
   - 1.2. An inspection shall be completed before mobile equipment is used each shift to ensure that it is in safe operating condition and that no one will be endangered by the start-up of the equipment. A checklist should be used for pre-use inspections.
   - 1.3. All operations requiring the use of heavy equipment will require a pre-planning meeting, Trade Partners shall include strategies to prevent injuries to tradespersons and the public.
   - 1.4. All fuel driven equipment being used indoors or in partially enclosed spaces—that does not meet the Final Tier 4 engine requirements—shall have scrubbers where the potential for carbon monoxide overexposure exists.
   - 1.5. Keys are to be removed at the end of the shift and cab doors locked.
   - 1.6. All material handling equipment, with an obstructed view to the rear, shall have a back-up alarm that is functional and distinctly audible. The operator shall make sure the warning signal is operating when the equipment is backing up.
   - 1.7. A "spotter", wearing an ANSI approved high visibility traffic vest, may be used in lieu of an alarm, but only if such devices are not routinely supplied on such a vehicle. Vehicles shall never back "blind" on a Whiting-Turner project.
   - 1.8. If the competent person determines that the back-up alarm is not sufficient, a spotter shall be present to help facilitate the rear operation of the equipment.
   - 1.9. Equipment and dump trucks equipped with faulty back-up alarms shall be repaired within 24 hours or taken out of service until back-up alarms are repaired. During that 24hr period, to remain in service, a spotter shall be present to help facilitate the rear operation of the equipment.
   - 1.10. Forklift operator training records shall be submitted to Whiting-Turner prior to site use.
   - 1.11. A seat belt shall be provided and used when operating equipment on a Whiting-Turner project. Before starting the engine, the operator shall fasten seat belts and adjust them for a proper fit.

1.12. All windows shall be in full working condition. Any equipment with broken glass of any size, including mirrors, will be taken out of service.
1.13. Equipment without a rollover protective structure (ROPS) or seatbelt is not allowed on any Whiting-Turner project unless it is manufactured in such a way and determined to be more hazardous with such protections.
1.14. Use of cell phones and radios is not permitted while operating machinery.
1.15. No one may work within 20' of motorized equipment like an excavator, backhoe, loader etc. unless that person's presence is fundamental to the operation underway and the operator can observe the person at all times.
1.16. All equipment with a rotating superstructure shall have swing radius protection or a designated spotter.
1.17. The operator shall not load mobile equipment beyond its established load limit and shall not move loads because of the length, width, or height that have not been centered and secured for safe transportation.

2. **Hoisting personnel.**
2.1. Personnel hoisting requirements - The use of load handling equipment to hoist personnel is prohibited unless the employer can demonstrate that other methods would be more hazardous and is able to comply with the personnel hoisting requirements that are established in the standard.
2.2. Hoisting personnel on Whiting-Turner project sites shall be done only after review of the plan and deemed feasible by Vice President and the Area EH&S Manager assigned to the project.
2.3. Hoisting personnel on Whiting-Turner projects shall be considered a critical lift or activity and therefore shall meet all requirements of a critical lift before the lift may begin.

E.42 Stop Work Authority

Introduction

Stop Work Authority (SWA) empowers workers to halt work when they perceive an unsafe condition or behavior, preventing potential incidents or injuries by encouraging a safety-conscious workplace.

Procedures

1. Tradespersons shall receive Stop Work Authority training before initial assignment. The training shall be documented including the employee's name, the dates of training and subject.
2. All tradespersons have the authority and obligation to stop any task or operation where concerns or questions regarding the control of EH&S risk exist.
3. No work will resume until all stop work issues and concerns have been adequately addressed.
4. Any form of retribution or intimidation directed at any individual or company for exercising their right to issue a stop work authority will not be tolerated.
5. Tradespersons are responsible to initiate a Stop Work Intervention when warranted and management is responsible to create a culture where Stop Work Authority is exercised freely.
6. When an unsafe condition is identified the Stop Work Intervention will be initiated, coordinated through the supervisor, initiated in a positive manner, notify all affected personnel and supervision of the stop work issue, correct the issue and resume work when safe to do so.
7. All Stop Work Interventions shall be documented for lessons learned and corrective measures to be put into place.
8. Stop Work reports shall be reviewed by a supervisor or manager to measure participation, determine quality of interventions and follow-up, trend common issues, identify opportunities for improvement and facilitate sharing of learnings.
9. It is the desired outcome of any Stop Work Intervention that the identified safety concern(s) have been addressed to the satisfaction of all involved persons prior to the resumption of work. Most issues can be adequately resolved in a timely manner at the job site, occasionally additional investigation and corrective actions may be required to identify and address root causes.

E.43 Excavations

Introduction

The intent of this policy is to limit and/or eliminate the dangers associated with excavation and trenching operations that could expose tradespersons to the possibility of severe injury or death. Each Trade Partner working on a Whiting-Turner project will comply with 29 CFR 1926, Construction Industry Regulations, Subpart P – Excavations, in addition to the following guidelines.

Procedures

1. **General Requirements.**
   1.1. Prior to the commencement of excavation activities where the excavation will be greater than three (3) feet in depth, a pre-excavation checklist shall be completed by the Trade Partner's competent person and submitted to Whiting-Turner.

2. **Specific Excavation Requirements.**
   2.1. Trade Partners whose employees will be entering trenches or excavations shall ensure that a comprehensive training program in the recognition, identification, evaluation and control of excavation hazards shall be provided to all tradespersons prior to working in an excavation or trenching operation. This shall also include a review of the geo-technical report.
   2.2. Underground utility installations shall be identified and marked prior to beginning any excavation. To prevent unintentional contact, all necessary measures shall be employed to locate underground utilities prior to excavating. Acceptable methods include but are not limited to the following: test pitting, ground penetrating radar (GPR), use of as-built drawings and any other obtainable information.
   2.3. When an excavation is performed within 3 feet of any utility line, a non-damaging method of excavation shall be used. The non-damaging method shall be hand digging with nonconductive tools. Other non-damaging methods, such as soft digging, vacuum digging, pneumatic hand tools, may be considered subject to approval by the competent person.
   2.4. A competent person shall be identified on Whiting-Turner's competent person acknowledgment form and their qualifications submitted to Whiting-Turner prior to the start of work.
   2.5. The competent person will be on-site during all excavation work to determine the soil type and its stability by performing one visual and one manual test in accordance with 29 CFR 1926, Subpart P Appendix A.
   2.6. Inspections shall be conducted daily and after every rainstorm or other hazard-increasing occurrence. Daily inspection reports shall be submitted to Whiting-Turner upon request. Daily inspections of excavations, the adjacent areas and protective systems shall be made by a competent person for evidence of a situation that could

result in possible cave-ins, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions. An inspection shall be conducted by the competent person prior to the start of work and as needed throughout the shift. Inspections shall also be made after every rainstorm or other hazard increasing occurrence. These inspections are only required when employee exposure can be reasonably anticipated. Where the competent person finds evidence of a situation that could result in a possible cave-in, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions, exposed employees shall be removed from the hazardous area until the necessary precautions have been taken to ensure their safety.

2.7. All trenches, unless otherwise specified, shall be protected by snow fence, at a minimum.

2.8. A stairway, ladder, ramp or other safe means of egress shall be located in trench excavations that are 4 feet or more in depth so as to require no more than 25 feet of lateral travel for employees.

2.9. Employees exposed to public vehicular traffic shall be provided with and shall wear, warning vests or other suitable garments marked with or made of reflectorized or high-visibility material.

2.10. No employee shall be permitted underneath loads handled by lifting or digging equipment. Employees shall be required to stand away from any vehicle being loaded or unloaded to avoid being struck by any spillage or falling materials. Operators may remain in the cabs of vehicles being loaded or unloaded when the vehicles are equipped to provide adequate protection for the operator during loading and unloading operations.

2.11. Employees shall not work in excavations in which there is accumulated water, or in excavations in which water is accumulating, unless adequate precautions have been taken to protect employees against the hazards posed by water accumulation. The precautions necessary to protect employees adequately vary with each situation but could include special support or shield systems to protect from cave-ins, water removal to control the level of accumulating water, or use of a safety harness and lifeline. If water is controlled or prevented from accumulating by the use of water removal equipment, the water removal equipment and operations shall be monitored by a competent person to ensure proper operation. If excavation work interrupts the natural drainage of surface water (such as streams), diversion ditches, dikes, or other suitable means shall be used to prevent surface water from entering the excavation and to provide adequate drainage of the area adjacent to the excavation. Excavations subject to runoff from heavy rains will require an inspection by a competent person.

3. **Requirements for protective systems.**

3.1. Excavations greater than 5 feet in depth shall be protected by one or more of the following systems as determined by the Trade Partner’s competent person:

3.1.1. Sloping / benching of sides to allowable configurations and slopes as per soil type.
3.1.2. Shielding (i.e. trench boxes) or shoring which shall have the manufacturer's tabulated data and specifications—or a professional engineer stamp—and current annual inspection documentation; all the aforementioned shall be on site.
3.1.3. Using a slope or shield system designed by a registered professional engineer. Refer to 29 CFR Subpart P, Appendix B.

3.2. Temporary and permanent spoils shall not exceed the angle of repose and shall be kept back at least two feet from the edge of an excavation.
3.3. A registered professional engineer shall design sloping or benching systems for excavations greater than 20 feet in depth.
3.4. Soil classifications shall be determined by testing and protective systems designed according to soil classifications.

4. **Requirements for fall protection.**
   4.1. Persons walking or working adjacent to a trench with vertical/shear walls that is equal to or greater than six (6) feet in depth shall be protected from fall hazards unless it has been determined by the Trade Partner's competent person that it is infeasible or creates a greater hazard.
   4.2. Person crossing an excavation that is equal to or greater than six (6) feet in depth shall be protected from fall hazards by means of a guardrail system.
   4.3. Walkways shall be provided where employees or equipment are required or permitted to cross over excavations. Guardrails shall be provided where walkways are 6 feet or more above lower levels.

5. **Training Requirements**
   5.1. Trade Partners whose employees will be entering a trench or excavation shall have received training on recognizing and avoiding hazardous atmospheres, caught-in/between hazards due to cave-in/collapse, entrapment due to access & egress hazards, struck-by falling objects and equipment and falls into excavations.
   5.2. Atmospheric monitoring, if deemed necessary by the competent person or other competent party, shall be documented and conducted by someone trained in the use of atmospheric monitoring equipment.

E.44 Utility Location Avoidance Policy

Introduction

The intent of this policy is to reduce the risk of danger associated with striking underground utilities when performing excavation work tasks. The policy defines the actions to be performed for a "strike free" excavation and the personnel responsible for performing these actions. All Whiting-Turner Project teams, including relevant trade partners, shall ensure that the following process for underground utility location is adhered to on all Whiting-Turner Project Sites.

Procedures

1. **As Built and Existing Condition Documents.**
   - 1.1. Identify Owner Contract provisions that require the owner to provide all as-builts and other such documents that identify location and conditions of all underground utilities and attest to receipt of these documents in writing.
   - 1.2. Request and collect any as-builts and other such documents that identify the location and condition of all underground utilities and obstructions that could be in the pathways of future Whiting-Turner excavations.
     - 1.2.1. These documents shall be obtained before work commences for all excavations and/or changes in excavation locations.
     - 1.2.2. Should these documents be unobtainable or insufficient to identify anticipated utilities, excavation will not proceed without Whiting-Turner Senior Vice President (SVP) and Regional EHS Director approval on how to proceed.

2. **Daily As-Built Development.**
   - 2.1. Owner-provided documentation will be transposed onto a medium (site maps, BIM, as examples) that can incorporate Trade Partner as-built requirements and accuracy to include:
     - 2.1.1. Elevation set control points/ survey point/ GPS/ at all intersections and at two points to recreate a straight line in field.
     - 2.1.2. Intermediate points every 50'
     - 2.1.3. Stubs, Valves and irregularities
     - 2.1.4. Temporary utilities
   - 2.2. The excavating Trade Partner shall Notify utility call centers (811) or private locators to identify and mark utility locations at the excavation locations, including elevation set con

NOTE: Excavating Trade Partner markings shall be maintained during the duration of the excavating work. That responsibility will extend to any other Trade Partner performing excavation work in the area.

   - 2.3. Each Excavating Trade Partner will provide a "Dig Zone" drawing of the work they are about to perform and include the following:

2.3.1. Utility locator utility identification markings
2.3.2. Dig Zone size including grading, over dig, sloping, benching and depth
2.3.3. Comparison of Dig Zone versus as-builts
2.3.4. Daily, each Excavating Trade Partner will update the Project as-built drawing (medium) to include their utility installations performed on that day.

NOTE: Any utilities identified from as-builts that are within 10' radius (vertical/horizontal) of the proposed Dig Zone will require "Utility Verification."

3. Utility Verification.

3.1. Should Dig Zone drawings reveal a utility within a 10' radius of the edge of the planned excavation, the excavating Trade Partner will conduct Utility Verification. Verification will include:
3.1.1. Utility locator markings are in place and legible
3.1.2. Utility Verification has been performed
3.1.3. Validate Dig Zone drawings are up to date and complete
3.2. Non-destructive exploration is performed to physically locate anticipated utilities. At a minimum:
3.2.1. Explorations will be performed for all utilities within a 10' radius of the excavation path and will continue to a depth of at least 5' below the depth of the anticipated utility.
3.2.2. At the point where the anticipated utility is to be intersected by the planned excavation
3.2.3. At two verification points should the excavation run parallel to the utility and the excavation is no greater than 50' in length.
3.2.4. Should a parallel excavation be more than 50' in length, then additional explorations will be performed at 50' intervals.

NOTE: Potholing will be performed regardless of surface conditions (concrete or asphalt, as examples). Should surface conditions prove to be such that routine Non-destructive methods are ineffective, excavation activities will stop and Whiting-Turner Supervision will be informed.

3.3. Update the Dig Zone Drawing with the results and conclusions of the verification efforts.
3.4. Should explorations fail to identify anticipated utilities, excavation will not proceed without Whiting-Turner Senior Vice President (SVP) and Regional EHS Director approval on how to proceed.

4. Excavation Preconstruction Review Meeting.

4.1. Whiting-Turner Project Leadership shall schedule and perform an excavation/utilities Trade Partner Preconstruction Review meeting. Attendees shall include:
4.1.1. Whiting-Turner Project Manager and Lead Superintendent
4.1.2. Excavating Trade Partner authorized competent person
4.1.3. Excavating Trade Partner operator(s)

4.1.4. Representatives of owners/design teams, with direct knowledge of existing underground conditions

4.1.5. Should owners/design teams fail to participate in the Excavation Preconstruction Review Meeting, the excavation will not proceed without Whiting-Turner Vice President (VP) and assigned EHS Manager approval on how to proceed.

4.2. Invite and urge attendance of the following personnel:

4.2.1. Utility company representatives (on public property)

4.2.2. Utility locator

4.3. Perform the Excavation/Utilities Trade Partner Preconstruction Review using Whiting-Turner Trade Partner Preconstruction Review form as the agenda of the meeting. The assigned superintendent or project manager will complete and document the actions on the form.

4.4. Conduct a walk/discussion of the work area with all required parties. Discussions will include:

4.4.1. Dig Zone observations and anticipated excavation safe work actions

4.4.2. Utility Verification methods and locations

4.5. Ensure at completion of the meeting, all parties sign the form where applicable.

5. Pre-Dig Huddle.

5.1. The Excavation Trade Partner will coordinate with Whiting-Turner Project personnel to conduct a Pre-Dig Huddle prior to beginning excavation work for the first time. The Excavating Trade Partner will lead and document the meeting on the Pre-Dig Huddle form. At a minimum, the following personnel will be in attendance:

5.1.1. Whiting-Turner assigned Project Superintendent

5.1.2. Excavating Trade Partner Competent Person

5.1.3. Excavating Trade Partner Foreman

5.1.4. Excavating Trade Partner Operator who will be performing the digging

5.2. The Excavating Trade Partner will provide Whiting-Turner for verification review:

5.2.1. As-Builts have been received

5.2.2. Utility locator markings have been established and are legible

5.2.3. Utilities have been located

5.2.4. SVP/EHS Director approval has been obtained (if applicable)

5.2.5. Process for daily utility installation updates

5.2.6. Existing utility Dig Zone Emergency Contacts are current

5.2.7. Utility shut offs are located and owner contact personnel identified with contact information

5.2.8. Process for daily Dig Zone Drawing reviews are performed

5.2.9. Machine excavation will not be performed within 3-feet of an underground utility. Non-destructive methods will be used up to the utility until it is physically located.

5.3. Should local authorities require distances greater than 3-feet from the utility before machine digging, the Project will follow those requirements.

5.4. The Excavation Trade Partner will perform additional Pre-Dig Huddle should any of the original excavation personnel change (competent person, operator, as examples), work is suspended for 3 weeks or greater or conditions change during the excavation.

5.5. The Whiting-Turner Project representative will ensure that all the talking points reiterating the digging protocols are discussed and acknowledged on the Pre-Dig Meeting form and that the form is signed off by all attendees.

6. Identification for New and Temporary Utility Installations.

6.1. At a minimum, all new permanent and temporary utility will be installed with Detectable Underground Warning Tape, or another method listed in section 7, LIST OF METHODS FOR PHYSICALLY IDENTIFYING NEW AND TEMPORARY UTILITY INSTALLATIONS.

6.2. Detectable underground tape is used to easily identify buried utility lines, pipes, cables and conduits. It is a 4 mil, non-adhesive, polyethylene tape with a printed continuous warning that is resistant to alkalis, acids and other soil components.

6.3. For proper installation, underground tape should be installed 12''-18'' below ground level and 12'' directly above buried pipes, utility lines, cables and conduits before final backfilling. Meets American Public Works Administration (APWA) color code.

E.45 Wildfire Air Quality Management

Introduction

Directives set forth are intended to protect Whiting-Turner employees from adverse health effects when air quality is diminished and to confirm that subcontracted companies understand their role in protecting their personnel as well. This protocol is to be applied to all Whiting-Turner projects working in locations where breathing air quality is reduced, or anticipated to be reduced, due to the wildfire smoke.

Exemptions

The following workplaces and operations are exempt from this protocol:

- Enclosed buildings or structures in which the air is filtered by a mechanical ventilation system and management verifies that windows, doors, bays and other openings are kept closed to minimize contamination by outdoor or unfiltered air.
- Enclosed vehicles in which the air is filtered by a cabin air filter and management confirms that windows, doors and other openings are kept closed to minimize contamination by outdoor or unfiltered air.

Procedures

Reduced Air Quality

Reduced air quality will be determined with the use of the Environmental Protection Agency (EPA) Air Quality Index (AQI) for the region or from measurements taken directly at the Project location.

Air Quality Index

Levels of Health Concern	Numerical Value	Meaning
Good	0 to 50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51 to 100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	100 to 151	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151 to 200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201 to 300	Health warnings of emergency conditions. The entire population is more likely to be affected.
Hazardous	301 to 500	Health alert: everyone may experience more serious health effects.

1. **Air Quality Index** Air Quality Index (AQI) can be obtained from The U.S. EPA AirNow program. AirNow is the national repository of real-time air quality data and forecasts for the United States. AQI is the vehicle for providing timely Air Quality Index (AQI) information to the public.
2. **Action Levels**
   - 2.1. **AQI for PM2.5 of ≥ 151** (Unhealthy) controls will be established to reduce employee exposures as much as feasible and implement the hierarchy of controls.
     - 2.1.1. Reduce exposure through engineering controls as feasible: enclosed buildings, structures or vehicles with filtered air.
     - 2.1.2. Institute administrative controls as feasible by relocating work, changing work schedules, reducing work intensity or providing additional rest periods.
   - 2.2. **AQI for PM2.5 of > 500** (Hazardous):
     - 2.2.1. Whiting-Turner employees will not be assigned outdoor work activities.
     - 2.2.2. Outdoor work activities will require mandatory use of respirators with an assigned protection factor that will reduce exposure levels inside the respirator to an AQI of 151 or less.
3. **Communication**
   - 3.1. All Project workers will be informed of anticipated air quality issues due to wildfires daily as conditions exist (Appendix A provides an example). Whiting-Turner management is encouraged to notify employees and Trade Partner management of:
     - 3.1.1. Current or anticipated AQI levels with the possibility of exceeding Action Levels and required actions, as applicable.
     - 3.1.2. Minimizing outdoor work and reminding personnel of the adverse symptoms of wildfire smoke exposure such as: asthma attacks, difficulty breathing and chest pain.

APPENDIX A - Poor Air Quality Memorandum – (Example)

Recent publication of air quality in our region indicates we will have an Air Quality Index exceeding 150. At this level, we are distributing the following actions to all project site personnel:

- If you suffer from health conditions that can be aggravated by this poor air quality, contact your supervisor and evaluate the suitability of you working from a location with better air quality.
- Recognize signs and symptoms of adverse reaction to poor air quality and report your observations to your supervisor immediately:
- Irritation of the eyes, nose and throat.
- Coughing, chest tightness and shortness of breath—signs of heart problems.
- Reduced lung function.
- Irregular heartbeat.
- Asthma attacks.
- Remain indoors as much as possible and consider changes in work schedule or tasks that require outdoor application.
- Minimize the level of physical exertion and communicate with your supervisor.
- Consider the voluntary use of a “dust mask” when outdoors—see your supervisor.

Reminder to all Trade Partners:

Your Activity Hazard Analysis (AHA) shall reflect and consider this poor air quality. Review and adjust your AHAs as appropriate, taking into consideration the guidance provided above. As always, verify that your AHA developments or revisions are forwarded to Whiting-Turner Safety for review.

We will continue to monitor air quality levels throughout the day and modify actions in accordance with Action Thresholds as changes occur.

Please feel free to discuss this with your supervisor if you have any questions.

Trade Partner Due Diligence – Wildfire Smoke

Trade Partners who may expose their workers to wildfire smoke should identify current AQI levels applicable to the worksite. The employer should consider the following:

- Check the current AQI before and periodically during each shift.
- Provide training to the employees.
- Any necessary methods to lower employee exposures.
- Provide respirators and encourage their use.

Two-Way Communication

Trade Partners should:

- Alert tradespersons when the air quality is harmful and what protective measures are available to the employees.
- Encourage tradespersons to inform them if they notice the air quality is getting worse or if they are suffering from any symptoms due to the air quality without fear of reprisal.

Protection Methods

151 AQI < 500

- Engineering Controls: Stay inside structures or vehicles with filtered ventilation.
- Administrative Controls: change work locations, reduce work time, increase rest breaks, reduce work intensity.
- Adequate PPE based on risk assessment.

> 500 AQI

- Work outdoors will stop, or
- Mandatory use of respirators will be required.
- o Respirators will only be issued in accordance with an OSHA compliant respiratory protection program.
- o Respirators will be issued based on maintaining AQI levels inside the mask >151.

E.46 Severe Weather Protocol

Introduction

Severe weather is a continuous threat to workers on construction project sites because of conductive equipment and the open-air environment. The following procedures aim to ensure worker health and safety in the event of severe weather.

Procedures

1. **Lightning Protocol**
   1.1. Pre-Storm Controls and Monitoring
      1.1.1. Whiting-Turner will monitor for lightning for awareness at a site level and to provide official "Take Shelter" orders. Trade Partners are expected to monitor their activities and employees. If storms are in the forecast, make sure you have a way to monitor the latest weather conditions and have a plan for lightning and suspending outdoor work activities (or reschedule altogether). Watch for rapidly building and darkening cumulus clouds and if you hear thunder, seek shelter. Each company is responsible for the safety of their employees.
   1.2. Shelter Areas
      1.2.1. The buildings under construction are the designated shelter locations for workers. With building(s) closest to turnover being utilized whenever possible as they are roofed, grounded and in some cases have lightning protection equipment already installed. Employees in office trailers may shelter in place. Employees should not take shelter under trees, in a tent area or other outdoor break area. Employees should minimize contact with the ground to only the soles of their boots, no sitting or lying on the ground.
   1.3. Call to Take Shelter
      1.3.1. If lightning is determined to be within 10 miles or 16 kilometers, outdoor work should be suspended and employees are to proceed to shelter areas.
   1.4. "All Clear" Call
      1.4.1. Lightning can strike 30 minutes after rain/thunder ceases. Employees on the site shall wait 30 minutes before resuming work activities after the end of a storm. It is the responsibility of Trade Partners to ensure there is no lightning detected within a 10-mile radius of the jobsite and the radius remains clear of additional storm fronts. The "All Clear" will be given via email, radio, phone, or text. Clear or sunny skies do not necessarily mean it is safe to go back to work.
2. **Tornado Protocol**
   2.1. Upon observing the distinctive signal for an emergency condition, employees of Whiting-Turner and Trade Partners shall follow the specific emergency procedures for that situation. See the site-specific Whiting-Turner Emergency Action Plan for further information.

2.2. Where a specific evacuation plan is mandated, employees shall be educated as to what actions are required of them. Also, diagrams with specific directions will be posted in the buildings and contained in the site safety orientation program.
2.3. The evacuation map shows specific emergency routes, assembly areas and storm shelters for site. The evacuation map is available in the Whiting-Turner's office and posted on-site.
2.4. If a severe weather event that creates an imminent or likely threat is determined, notifications will be made to Whiting-Turner's onsite project management with as much advance notice as possible by phone, radio and/or email. Employees should shut down equipment as completely as time allows and make their way to the authorized shelters, away from high-risk areas. Do not let concern for equipment override the necessity of self-preservation.
2.5. Each Trade Partner is responsible for providing an accurate headcount.
2.6. When a severe weather emergency happens in a vehicle:
2.7. Do not drive during conditions that could produce tornadoes. Never try to outdrive a tornado in a vehicle. Tornados can change direction quickly and lift a vehicle.
2.8. In the occurrence of a tornado immediately get out of your vehicle and seek shelter in a nearby building. If there is not time to seek shelter indoors, get out of the vehicle and lie in a ditch or a low-lying area away from the vehicle. Be aware of the potential for flooding.

3. Training

3.1. Trade Partner employees shall be trained in the specifics of this and the site-specific safety plan when the plan is first implemented, when an employee's responsibilities change, as a new employee arrives and when the plan is revised.